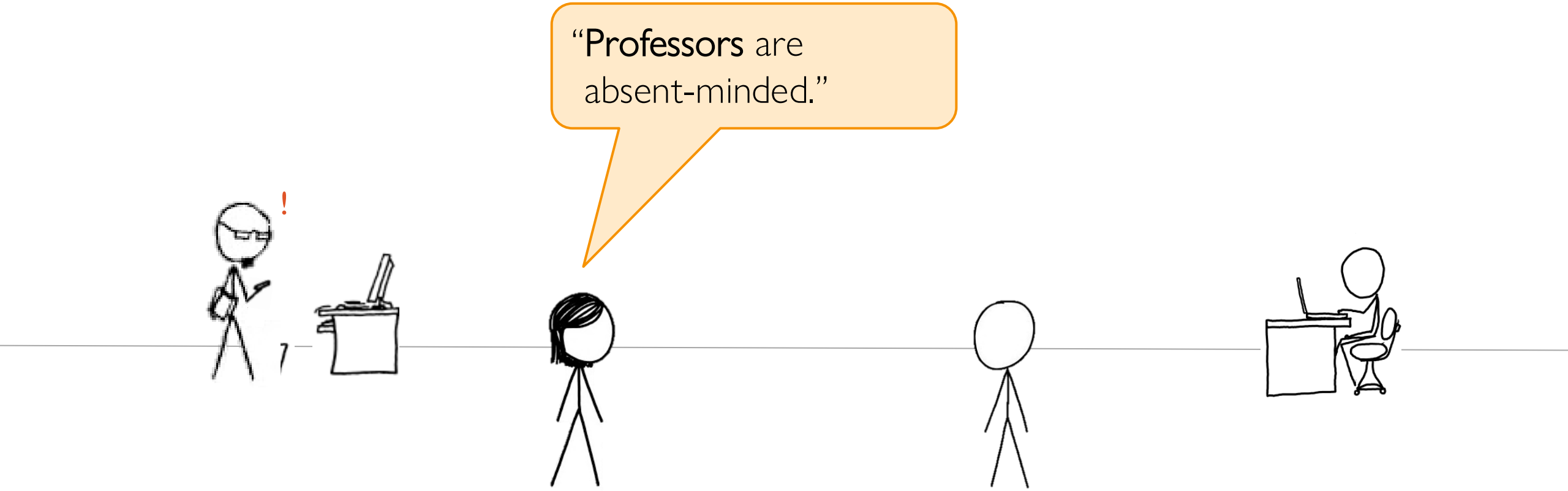
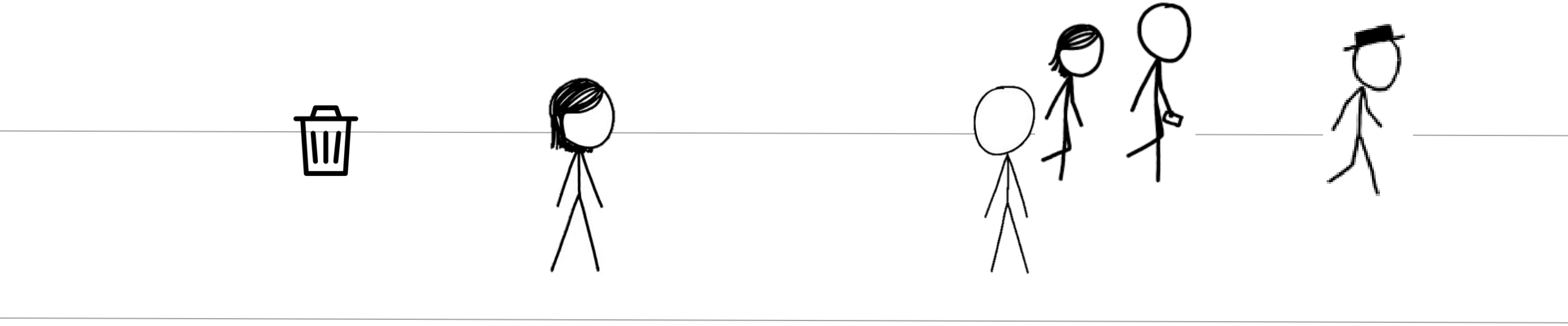
Learning about the inductive potential of categories from language

Marianna Zhang (she/her)
Postdoctoral Researcher in Psychology
New York University

Maya Malaviya
Sarah-Jane Leslie
Marjorie Rhodes
Mark Ho

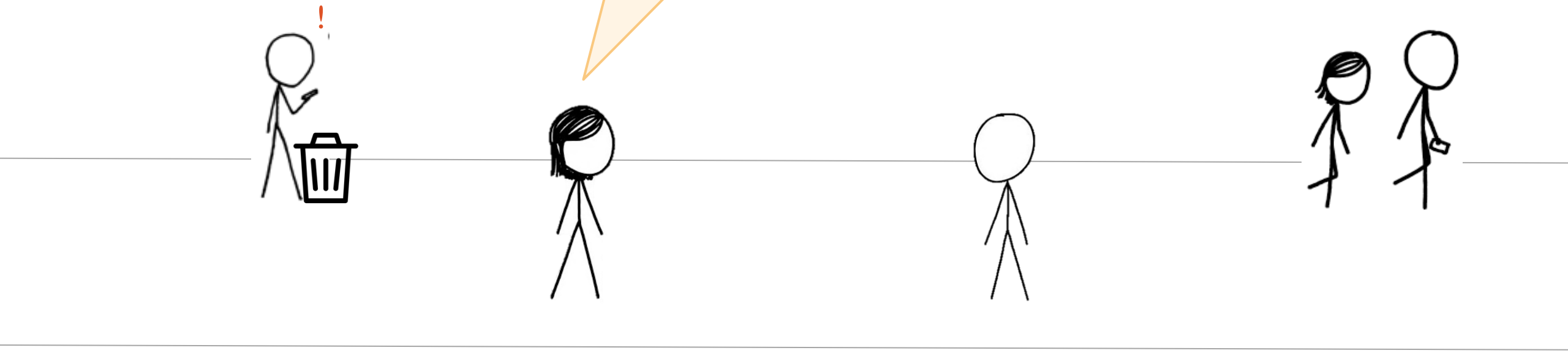
generic statement



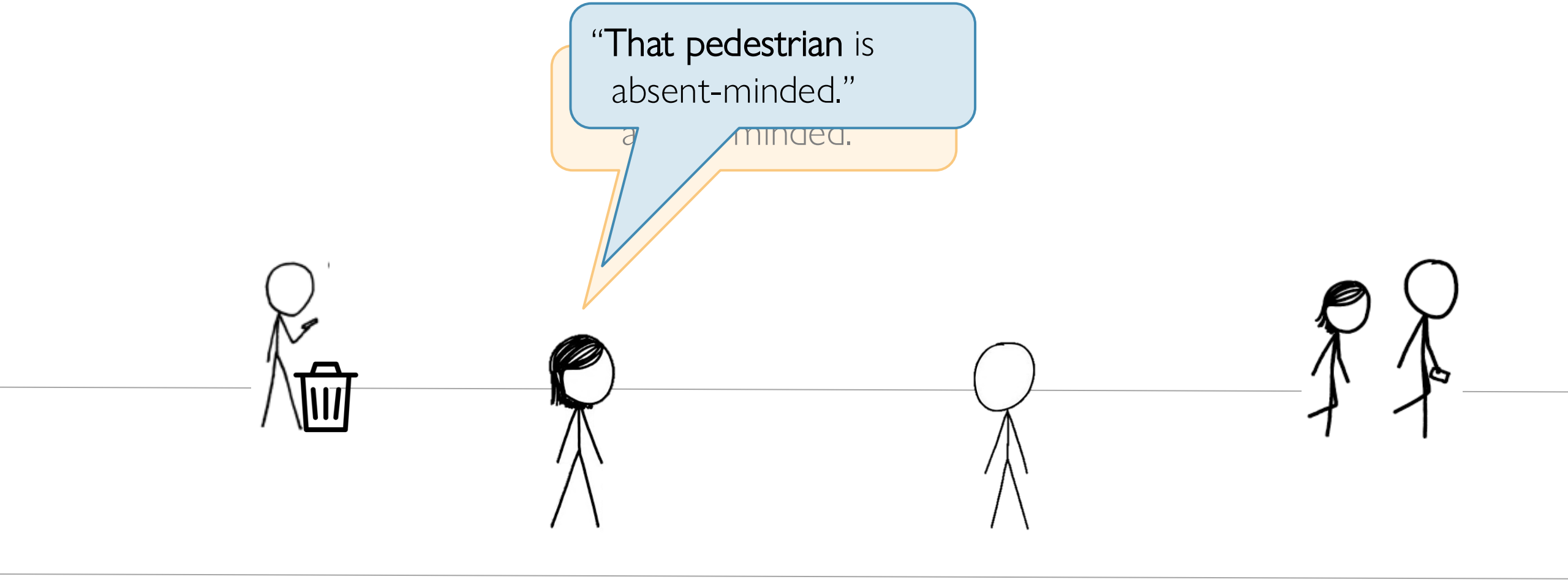


generic statement

"Pedestrians are absent-minded."



specific statement

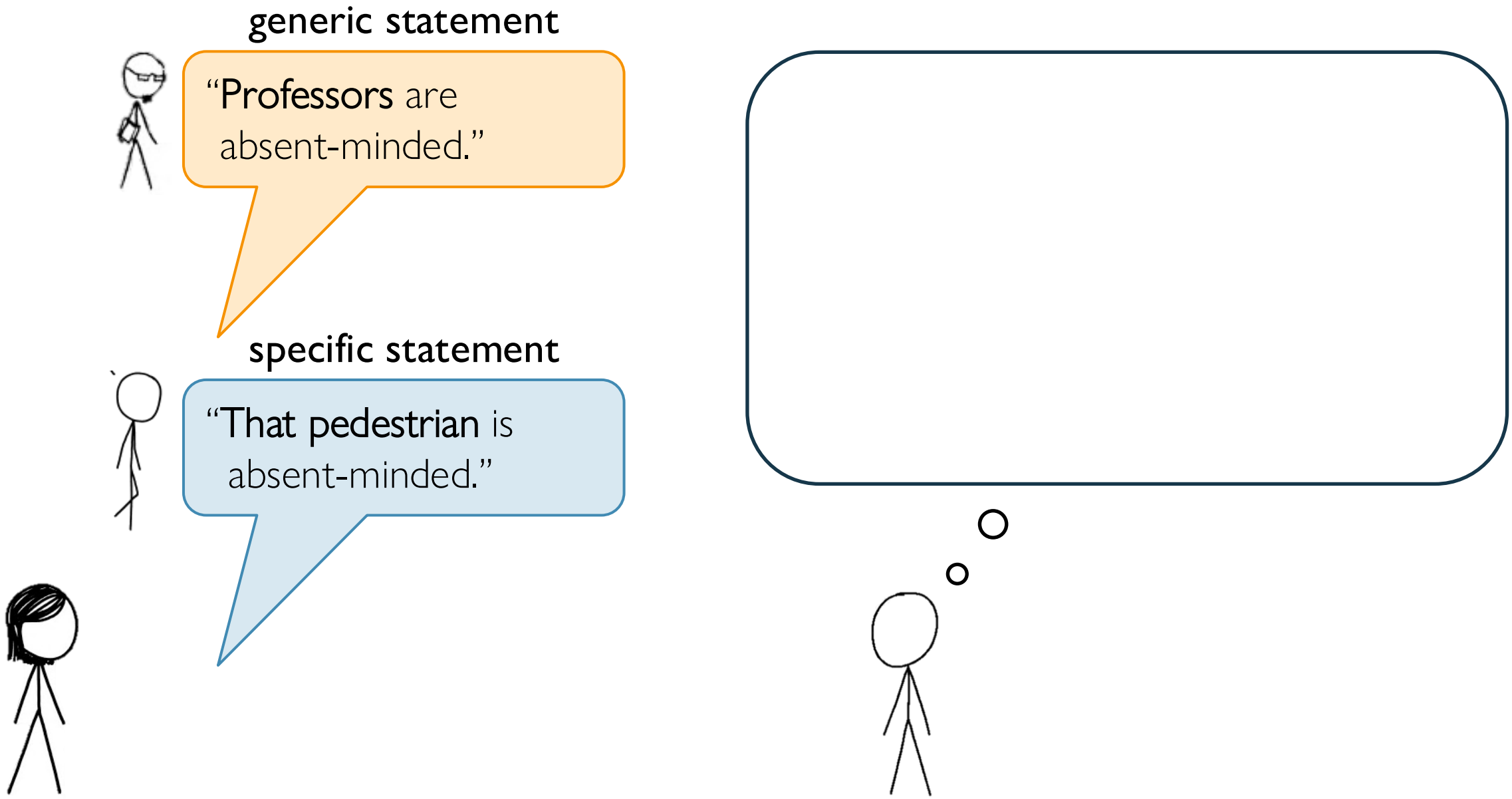


generic statement

"Professors are absent-minded."

specific statement

"That pedestrian is absent-minded."


generic statement
"Professors are absent-minded."
specific statement
"That pedestrian is absent-minded."

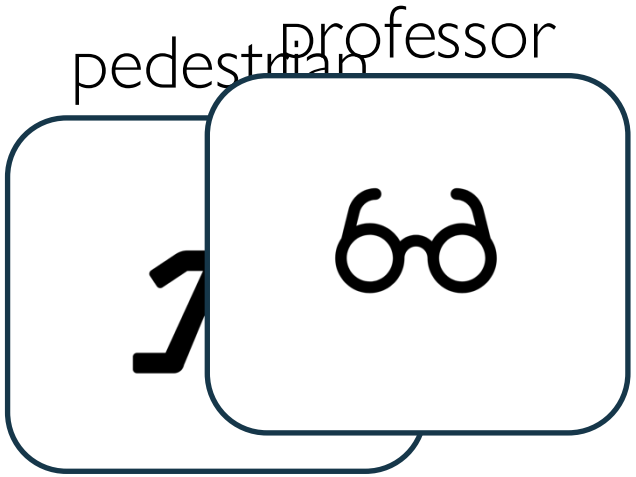
Language as a signal of inductive potential

how well a category
supports generalization

low ⟷ high

categories seen as minimal,
not supportive of generalization

categories seen as coherent/meaningful,
supportive of generalization



Language as a signal of inductive potential

how well a category
supports generalization

low — high

categories seen as minimal,
not supportive of generalization

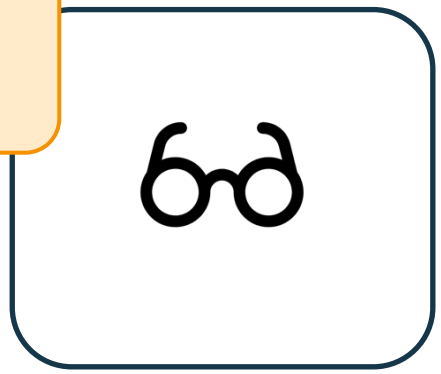
categories seen as coherent/meaningful,
supportive of generalization

more generics

pedestrian

"**Professors** are absent-minded."

professor



Language as a signal of inductive potential

how well a category
supports generalization

low — high

categories seen as minimal,
not supportive of generalization

categories seen as coherent/meaningful,
supportive of generalization

more specifics

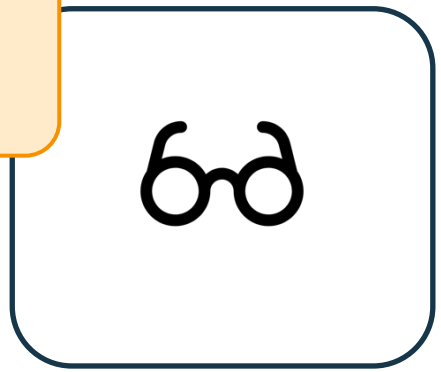
more generics

pedestrian

"That pedestrian
is absent-minded."

"Professors are
absent-minded."

professor



Language can signal what categories are thought to support generalization.

Learning about the inductive potential of categories from language

Learning from generic vs specific language

- Computational model
- **Exp 1a:** generics vs specifics
- **Exp 1b**: mix of generics and specifics

Learning to generalize features based on feature similarity

- Extending the model
- **Exp 1b**: different types of features

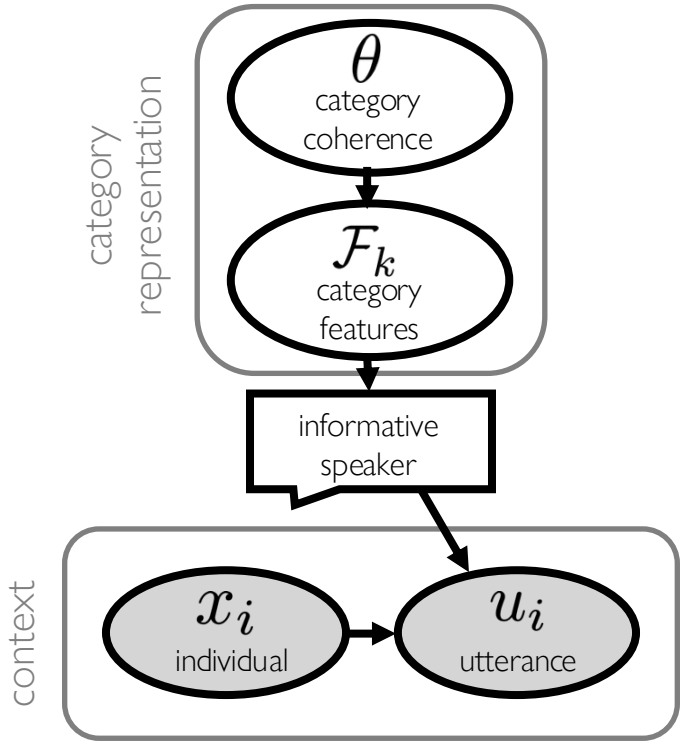
Computational model
for learning inductive potential
from language

Computational model

for learning inductive potential from language

Learning a category representation
via hierarchical Bayesian reasoning
(Kemp et al., 2007; A. Gelman, 1995)

Reasoning about what a speaker meant
via rational speech acts framework
(Goodman & Frank, 2016)

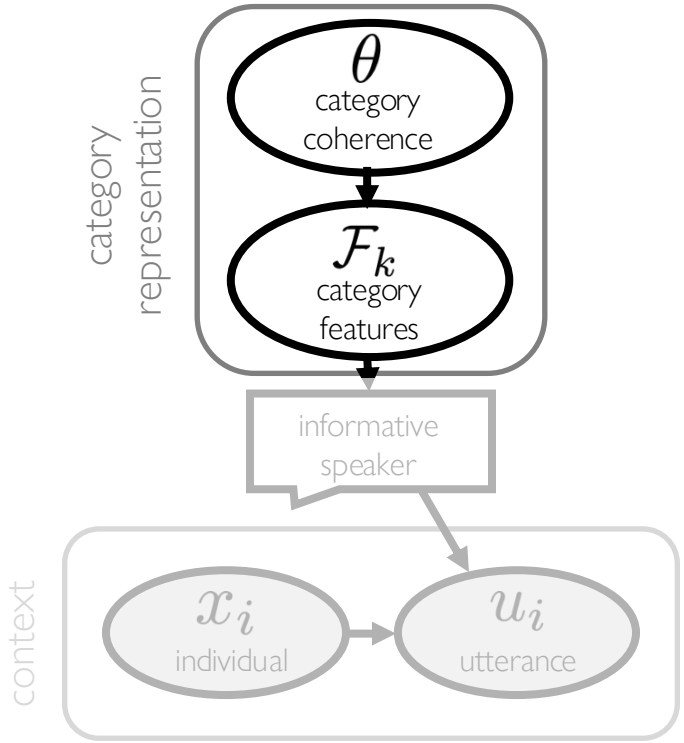


Computational model

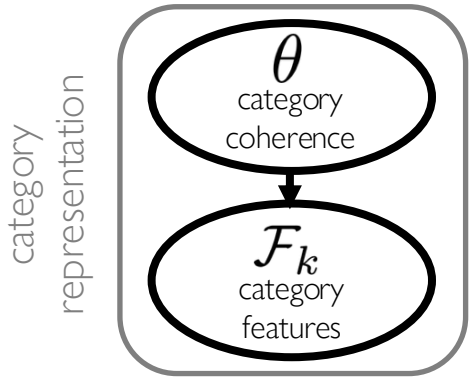
for learning inductive potential from language

Learning a category representation
via hierarchical Bayesian reasoning
(Kemp et al., 2007; A. Gelman, 1995)

Reasoning about what a speaker meant
via rational speech acts framework
(Goodman & Frank, 2016)



Learning a category representation
via hierarchical Bayesian reasoning

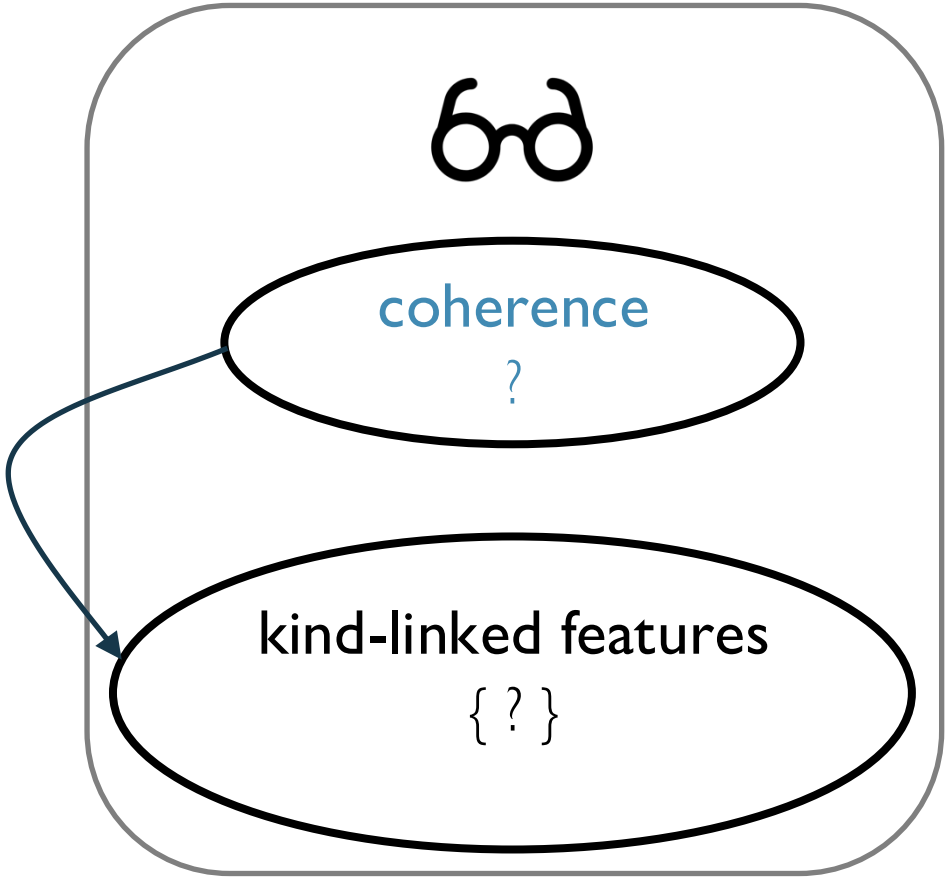


individual professors

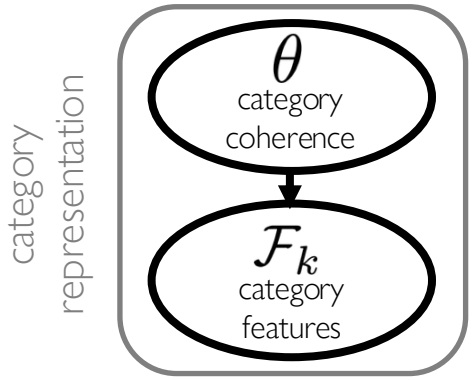
smart | funny | absent-minded

features under consideration

professor



Learning a category representation
via hierarchical Bayesian reasoning

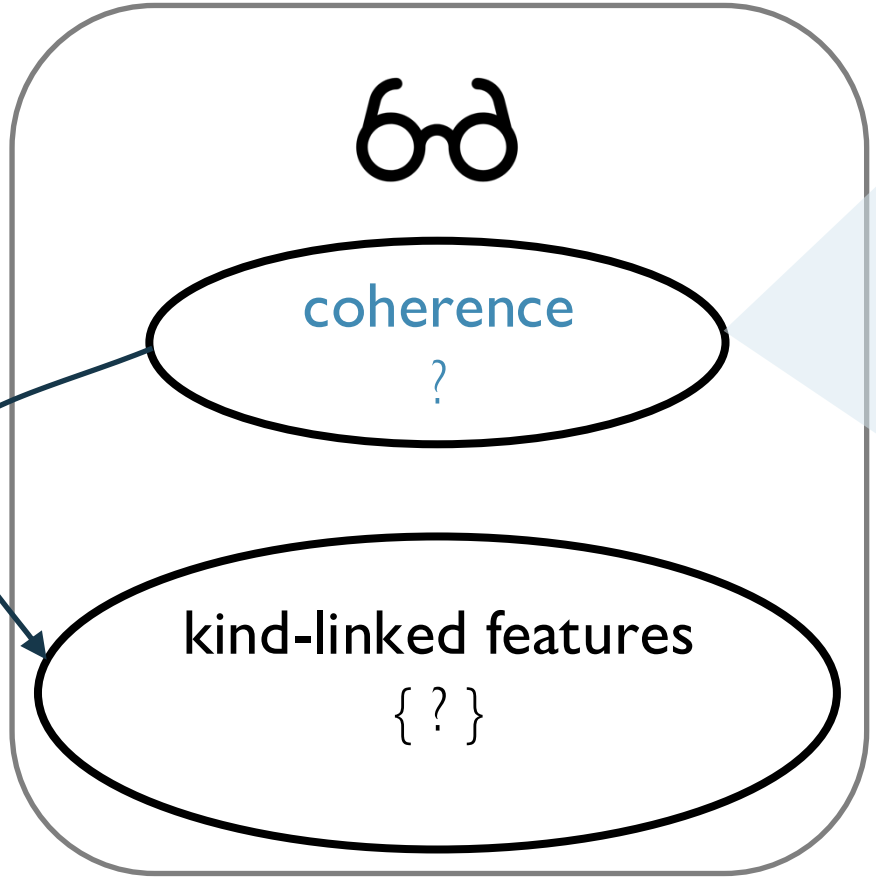


individual professors

smart | funny | absent-minded

features under consideration

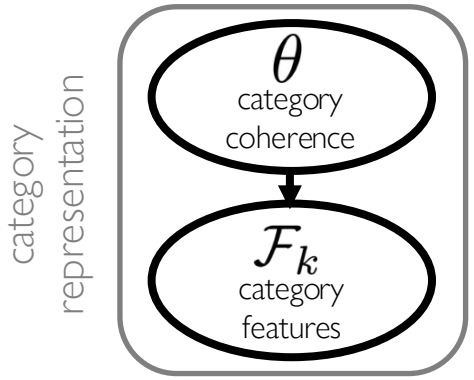
professor



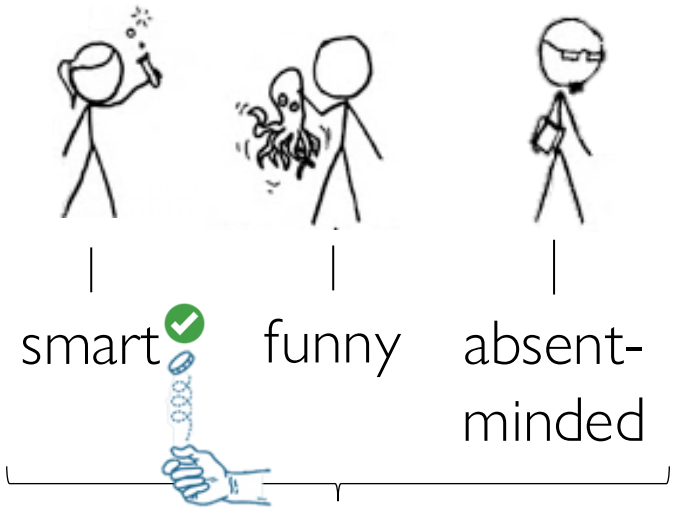
coherence

an **overhypothesis** about the probability that any feature of an individual will be kind-linked

Learning a category representation
via hierarchical Bayesian reasoning

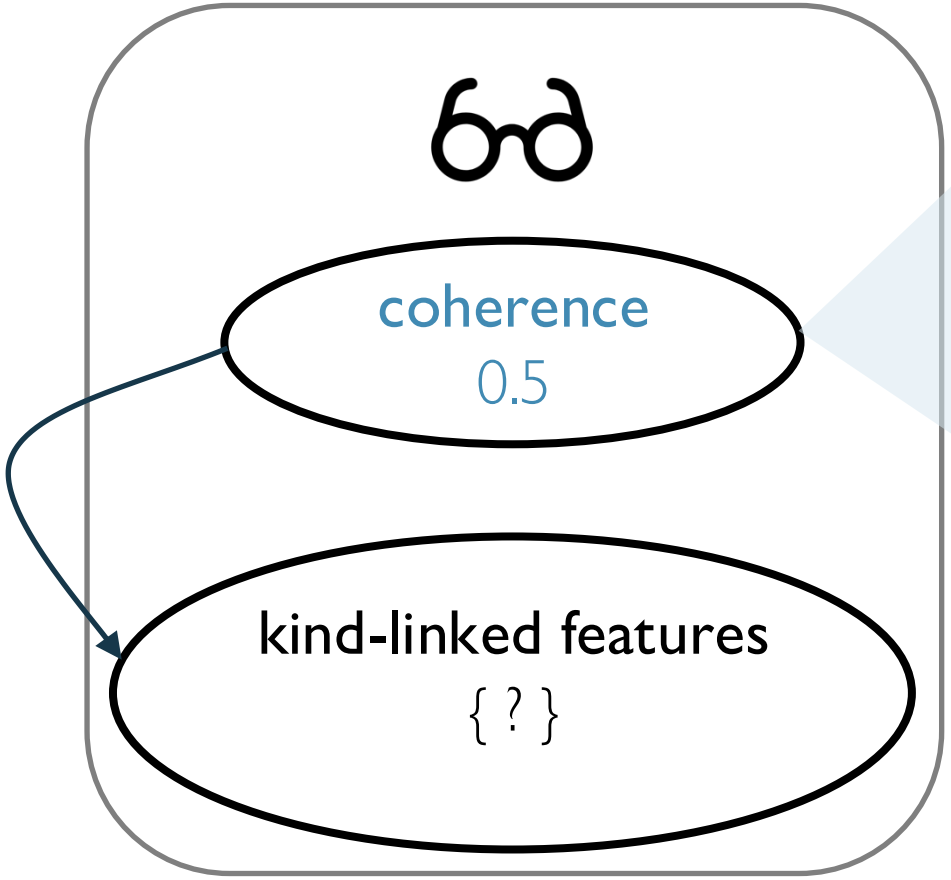


individual professors



features under consideration

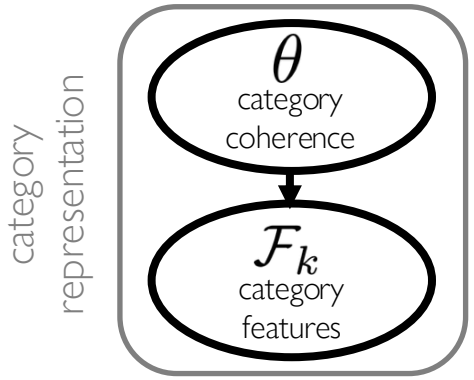
professor



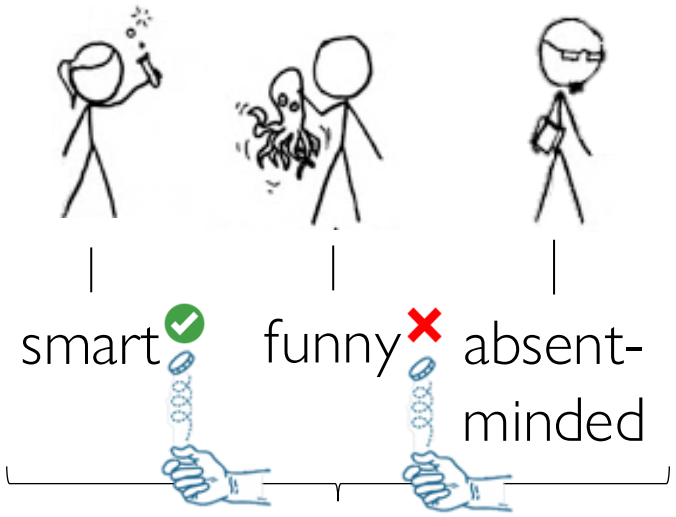
coherence

an **overhypothesis** about the probability that any feature of an individual will be kind-linked

Learning a category representation
via hierarchical Bayesian reasoning

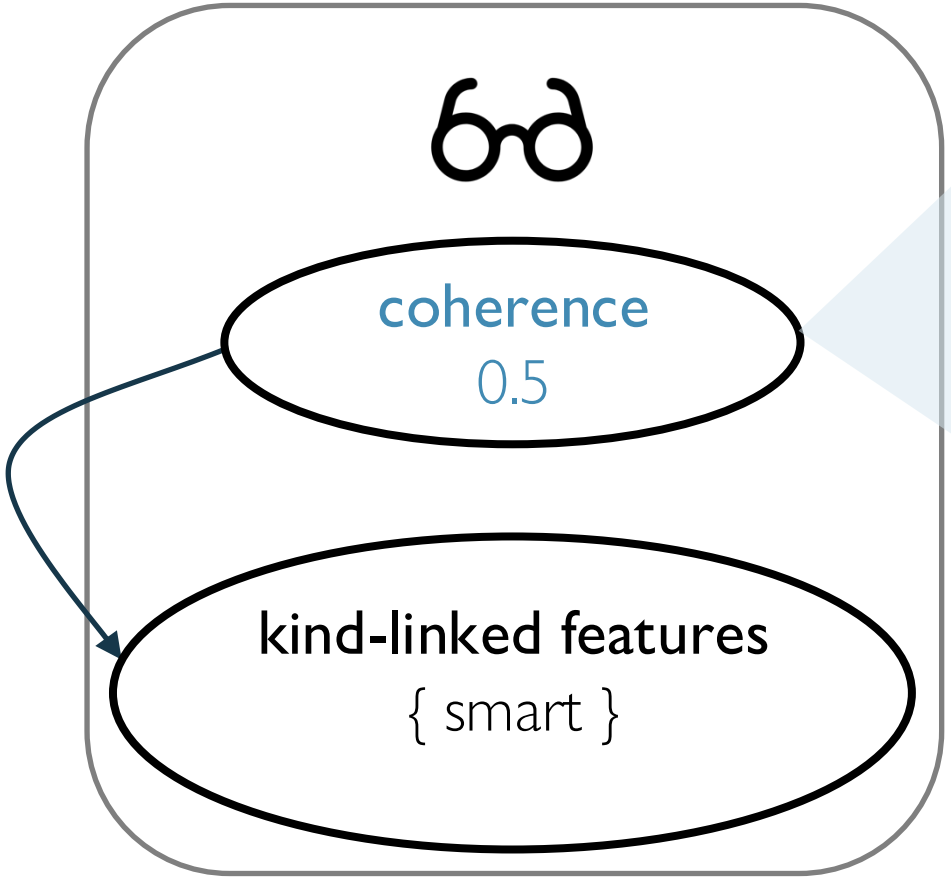


individual professors



features under consideration

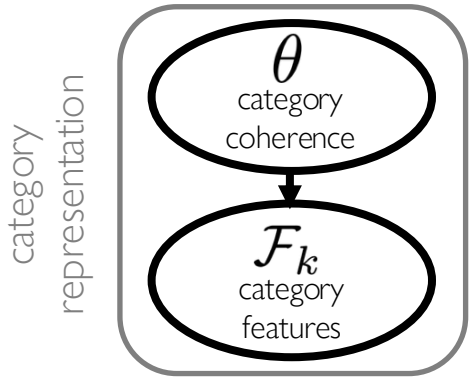
professor



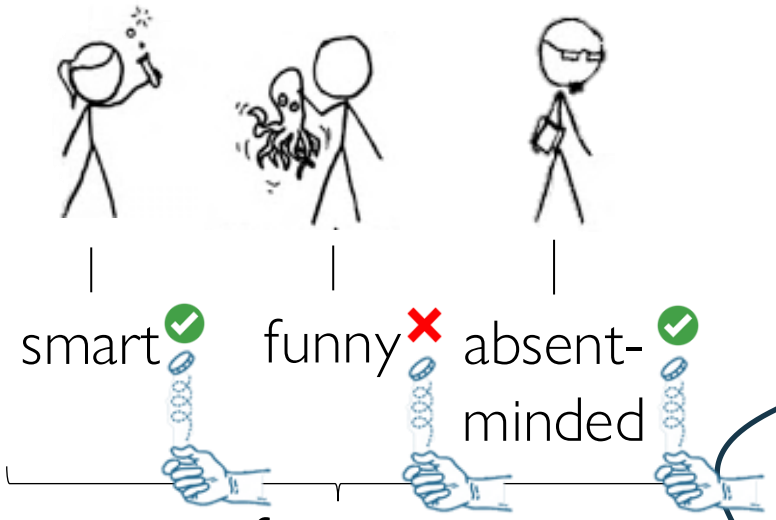
coherence

an **overhypothesis** about the probability that any feature of an individual will be kind-linked

Learning a category representation
via hierarchical Bayesian reasoning

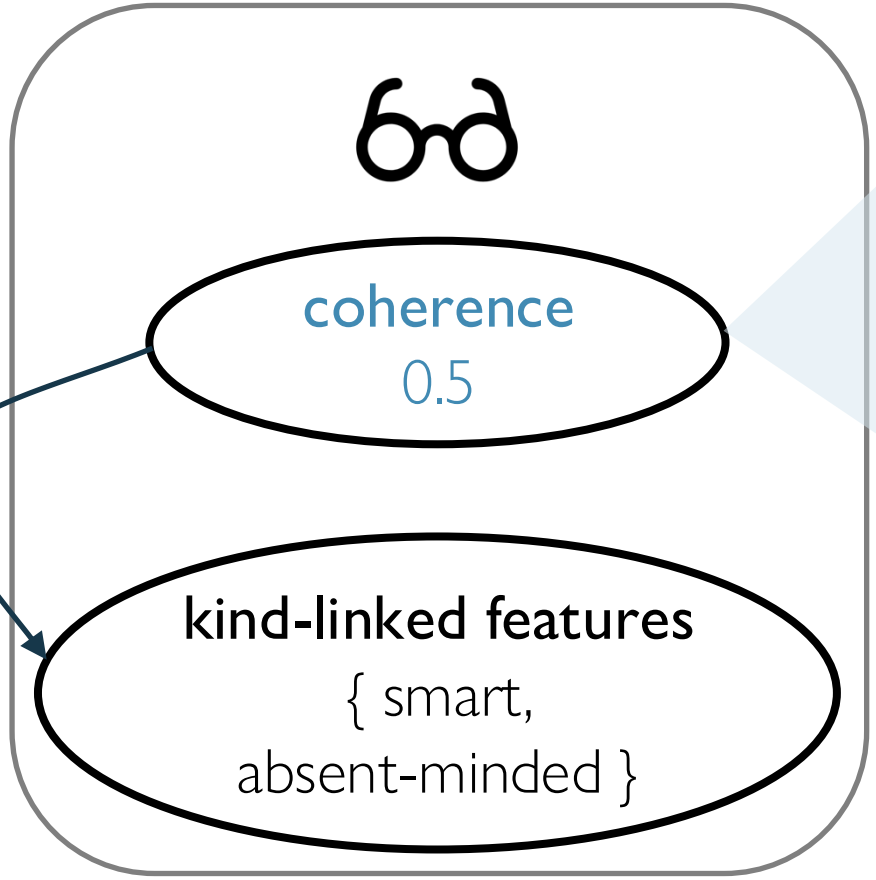


individual professors



features under consideration

professor



coherence

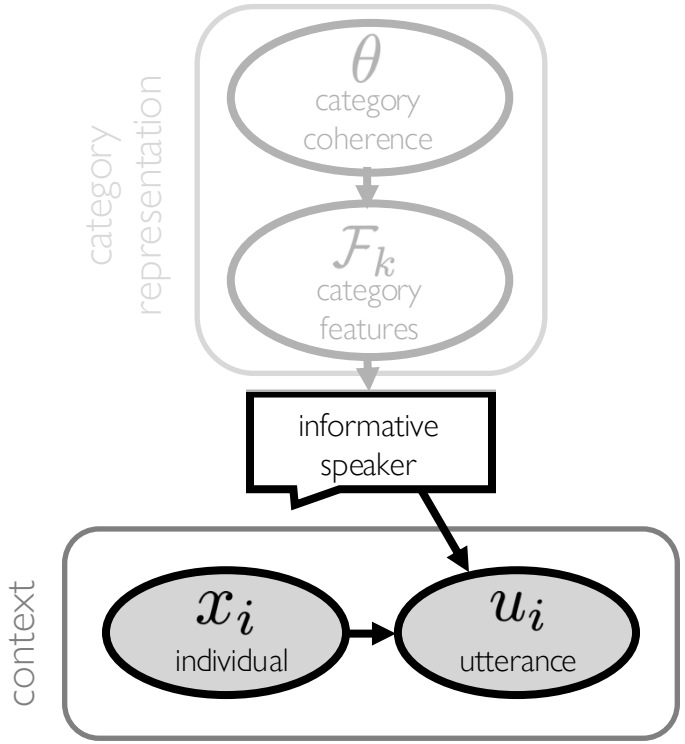
an **overhypothesis** about the probability that any feature of an individual will be kind-linked

Computational model

for learning inductive potential from language

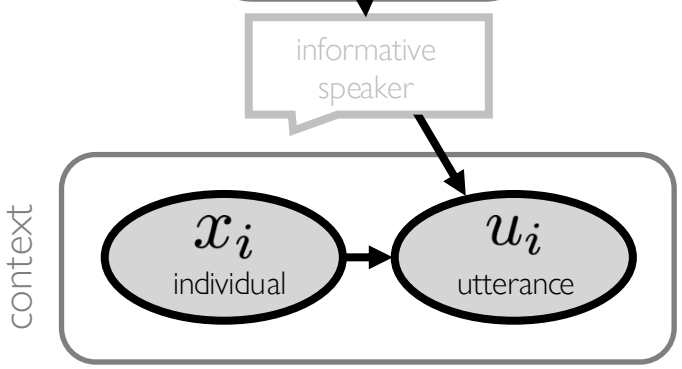
Learning a category representation
via hierarchical Bayesian reasoning
(Kemp et al., 2007; A. Gelman, 1995)

Reasoning about what a speaker meant
via rational speech acts framework
(Goodman & Frank, 2016)



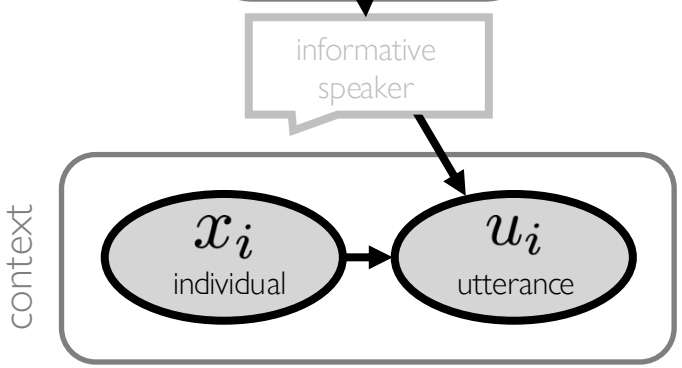
Reasoning about what a speaker meant

via rational speech acts framework



Reasoning about what a speaker meant

via rational speech acts framework



literal listener

infers coherence & kind-linked features from **literal meaning** of what was said

kind-linked features

coherence

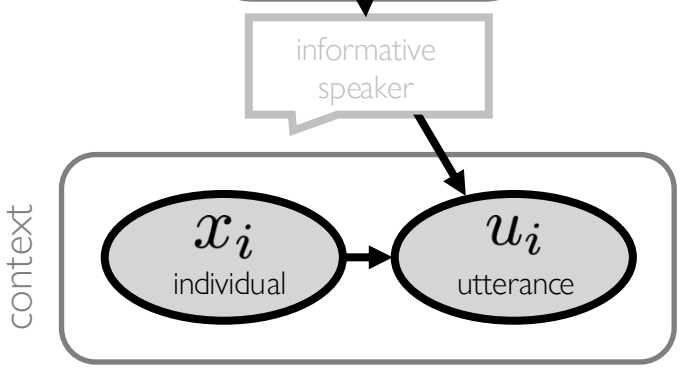
$$\mathrm{Lit}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u})$$

observed individual

utterance

Reasoning about what a speaker meant

via rational speech acts framework



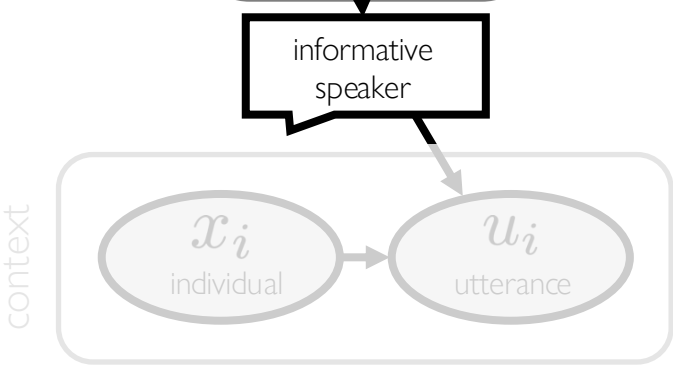
literal listener

infers coherence & kind-linked features from **literal meaning** of what was said

generics are true if feature is kind-linked
specifics are true is individual has feature

$$\mathrm{Lit}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto \underbrace{P(\theta)}_{\text{coherence prior}} \; \underbrace{P(\mathcal{F}_k \mid \theta)}_{\text{likelihood of kind-linked features, given coherence}} \; \underbrace{\prod_i [\![u_i]\!](\mathcal{F}_k, x_i)}_{\text{literal meaning of what was said}}$$

Reasoning about what a speaker meant
via rational speech acts framework



literal listener

infers coherence & kind-linked features from **literal meaning** of what was said

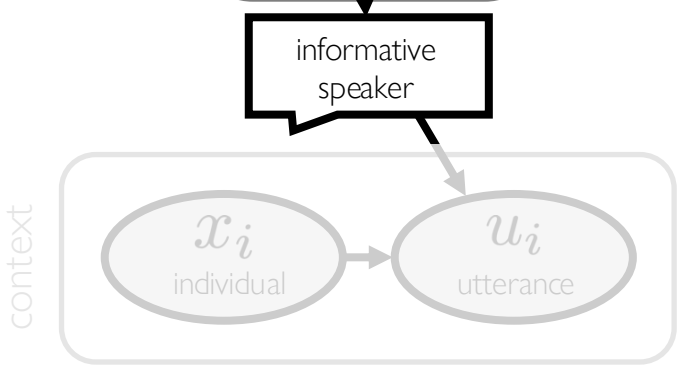
generics are true if feature is kind-linked
specifics are true is individual has feature

$$\mathrm{Lit}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i [\![u_i]\!](\mathcal{F}_k, x_i)$$

speaker

says a generic or specific to inform the **literal listener** which features of the individual are kind-linked

Reasoning about what a speaker meant
via rational speech acts framework



literal listener

infers coherence & kind-linked features from **literal meaning** of what was said

generics are true if feature is kind-linked
specifics are true is individual has feature

$$\mathrm{Lit}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i [\![u_i]\!](\mathcal{F}_k, x_i)$$

speaker

says a generic or specific to inform the **literal listener** which features of the individual are kind-linked

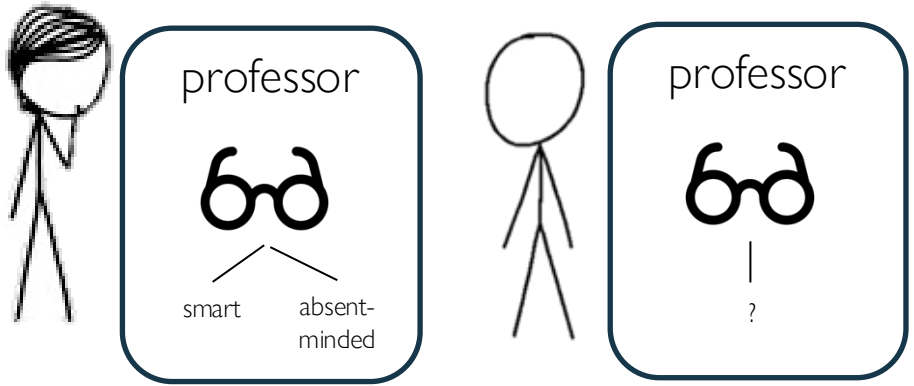
$$\mathrm{Utility}(u_i, x_i, \mathcal{F}_k^*)$$

$$= \sum_{\mathcal{F}_k} \mathrm{Lit}(\mathcal{F}_k \mid x_i, u_i) \cdot \mathrm{Similarity}(\mathcal{F}_k^* \cap x_i, \mathcal{F}_k \cap x_i)$$

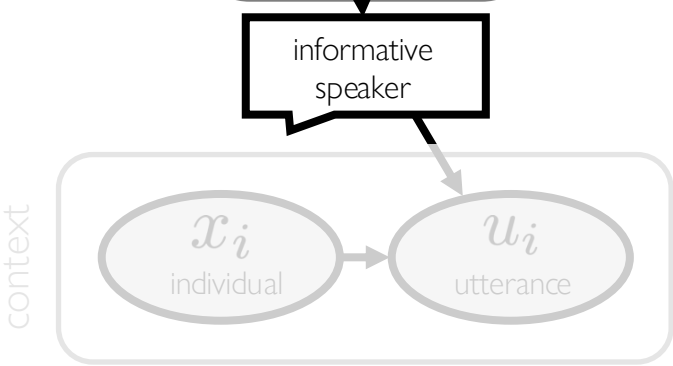
Jaccard similarity between sets: [0, 1]

observed features the **speaker** knows are kind-linked

observed features the **literal listener** thinks are kind-linked



Reasoning about what a speaker meant
via rational speech acts framework



literal listener

infers coherence & kind-linked features from **literal meaning** of what was said

generics are true if feature is kind-linked
specifics are true is individual has feature

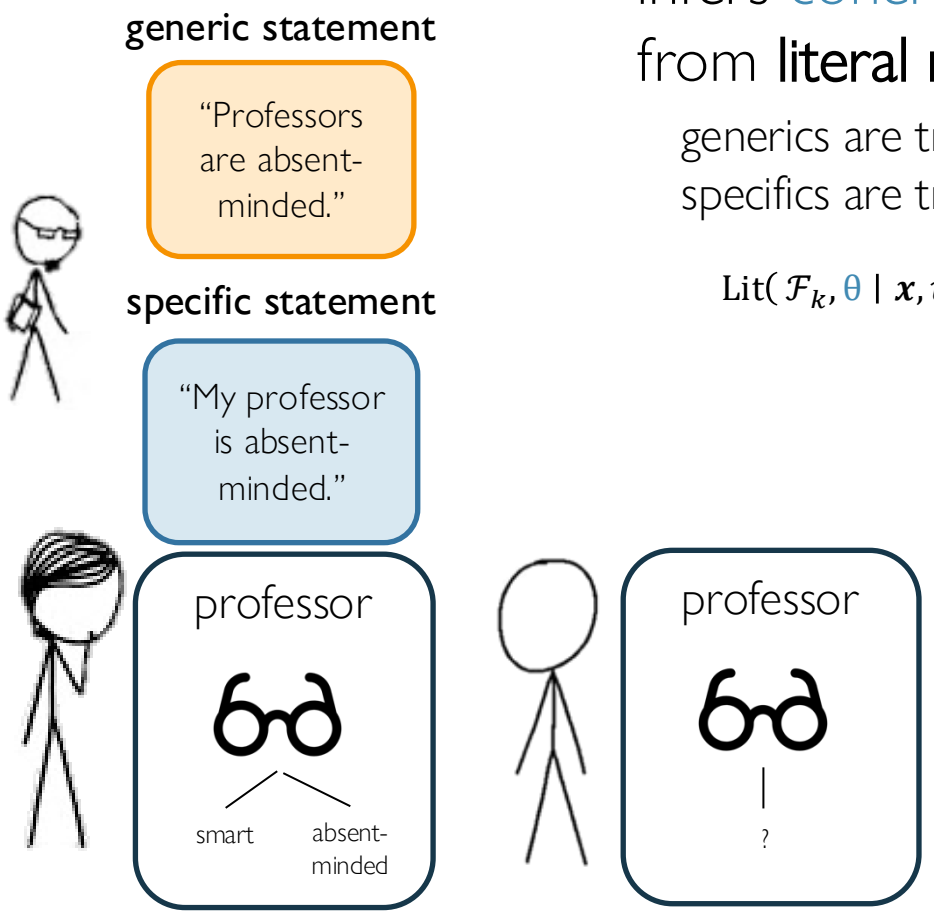
$$\mathrm{Lit}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i [\![u_i]\!](\mathcal{F}_k, x_i)$$

speaker

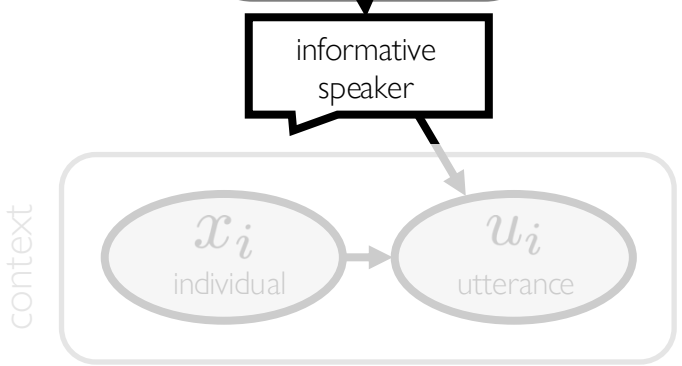
says a generic or specific to inform the **literal listener** which features of the individual are kind-linked

$$\mathrm{Sp}(u_i \mid \mathcal{F}_k^*, x_i) \propto \exp\{\beta \cdot \mathrm{Utility}(u_i, x_i, \mathcal{F}_k^*)\}$$

$$\mathrm{Utility}(u_i, x_i, \mathcal{F}_k^*) = \sum_{\mathcal{F}_k} \mathrm{Lit}(\mathcal{F}_k \mid x_i, u_i) \cdot \mathrm{Similarity}(\mathcal{F}_k^* \cap x_i, \mathcal{F}_k \cap x_i)$$



Reasoning about what a speaker meant
via rational speech acts framework



literal listener

infers coherence & kind-linked features from **literal meaning** of what was said

generics are true if feature is kind-linked
specifics are true is individual has feature

$$\mathrm{Lit}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i [\![u_i]\!](\mathcal{F}_k, x_i)$$

speaker

says a generic or specific to inform the **literal listener** which features of the individual are kind-linked

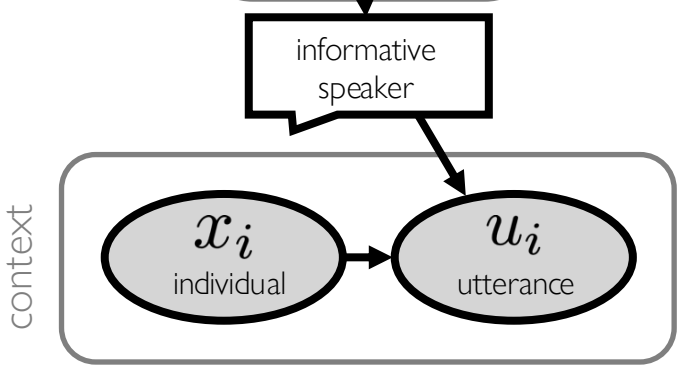
$$\mathrm{Sp}(u_i \mid \mathcal{F}_k^*, x_i) \propto \exp\{\beta \cdot \mathrm{Utility}(u_i, x_i, \mathcal{F}_k^*)\}$$

$$\mathrm{Utility}(u_i, x_i, \mathcal{F}_k^*)$$

$$= \sum_{\mathcal{F}_k} \mathrm{Lit}(\mathcal{F}_k \mid x_i, u_i) \cdot \mathrm{Similarity}(\mathcal{F}_k^* \cap x_i, \mathcal{F}_k \cap x_i)$$

Reasoning about what a speaker meant

via rational speech acts framework



literal listener

infers coherence & kind-linked features from **literal meaning** of what was said

generics are true if feature is kind-linked
specifics are true is individual has feature

$$\mathrm{Lit}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i [\![u_i]\!](\mathcal{F}_k, x_i)$$

speaker

says a generic or specific to inform the **literal listener** which features of the individual are kind-linked

$$\mathrm{Sp}(u_i \mid \mathcal{F}_k^*, x_i) \propto \exp\{\beta \cdot \mathrm{Utility}(u_i, x_i, \mathcal{F}_k^*)\}$$

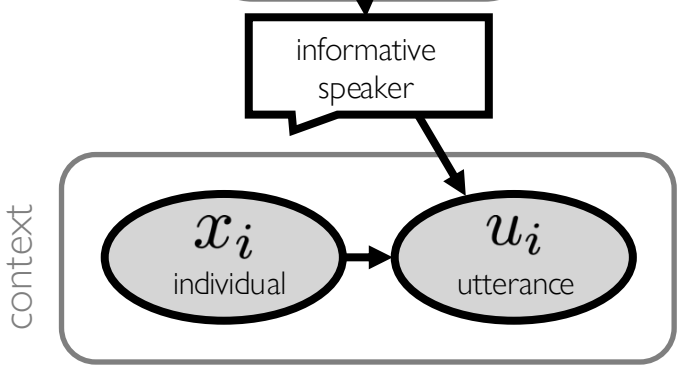
$$\mathrm{Utility}(u_i, x_i, \mathcal{F}_k^*) = \sum_{\mathcal{F}_k} \mathrm{Lit}(\mathcal{F}_k \mid x_i, u_i) \cdot \mathrm{Similarity}(\mathcal{F}_k^* \cap x_i, \mathcal{F}_k \cap x_i)$$

pragmatic listener

infers coherence & kind-linked features by reasoning about what the **speaker** meant

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

Reasoning about what a speaker meant
via rational speech acts framework



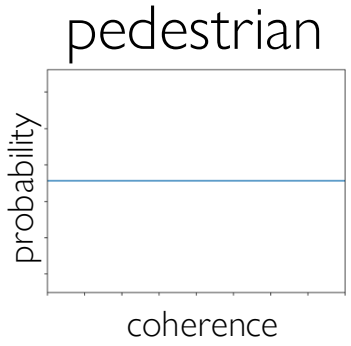
specific statement

"That pedestrian is is absent-minded."

literal listener

infers category's coherence & features from **literal meaning** of what was said

- generics are true
  - if mentioned feature is a category feature
- specifics are true
  - if individual has the mentioned feature

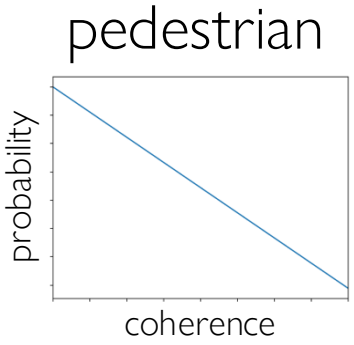


? coherence

pragmatic listener

infers category's coherence & features by reasoning about what the **speaker** meant

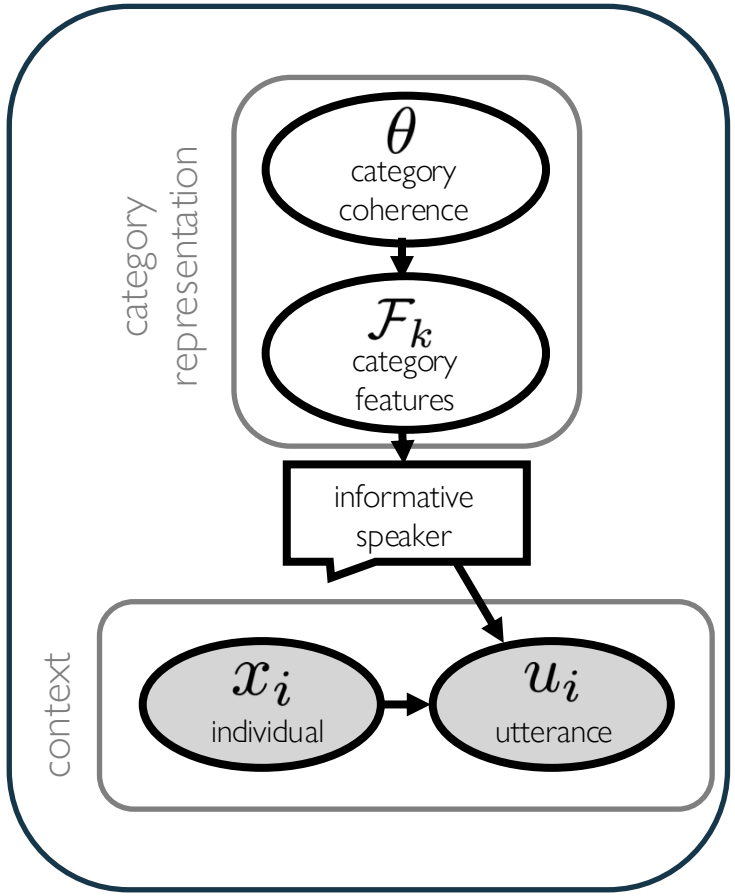
- speaker wants to communicate about
  - whether this feature is a category feature
- speaker can say either generic or specific



low coherence

Computational model for learning about categories from language

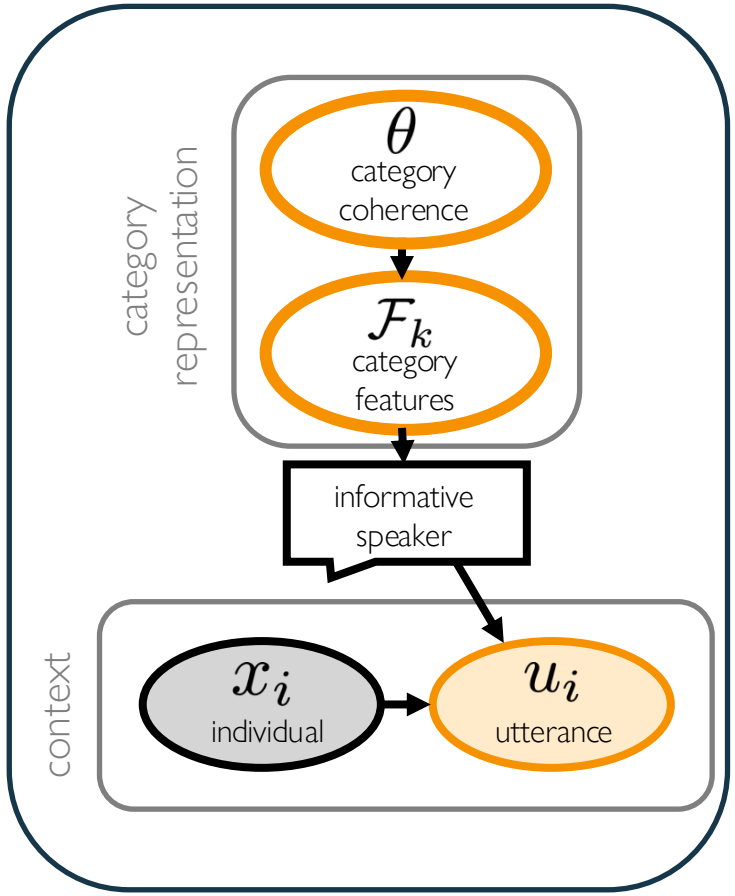
Our hierarchical Bayesian + rational speech acts model predicts that:

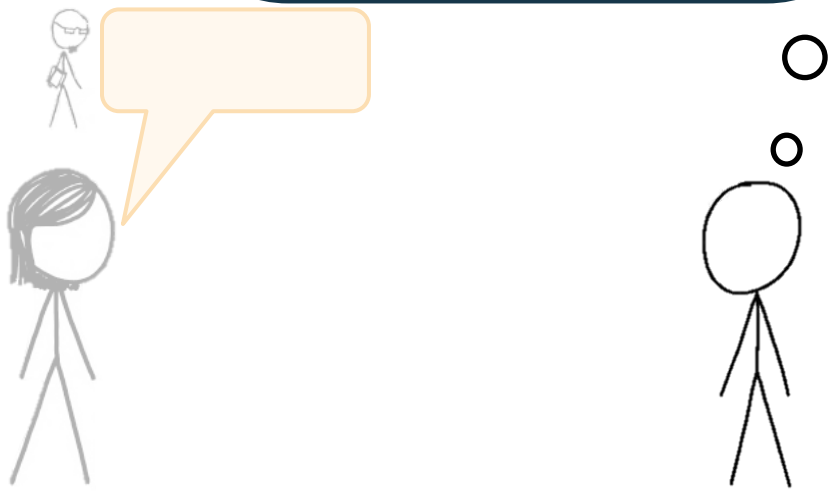


Computational model for learning about categories from language

Our hierarchical Bayesian + rational speech acts model predicts that:

1. Hearing generic statements about a category (e.g., "Professors are absent-minded.") will increase generalization across the category.



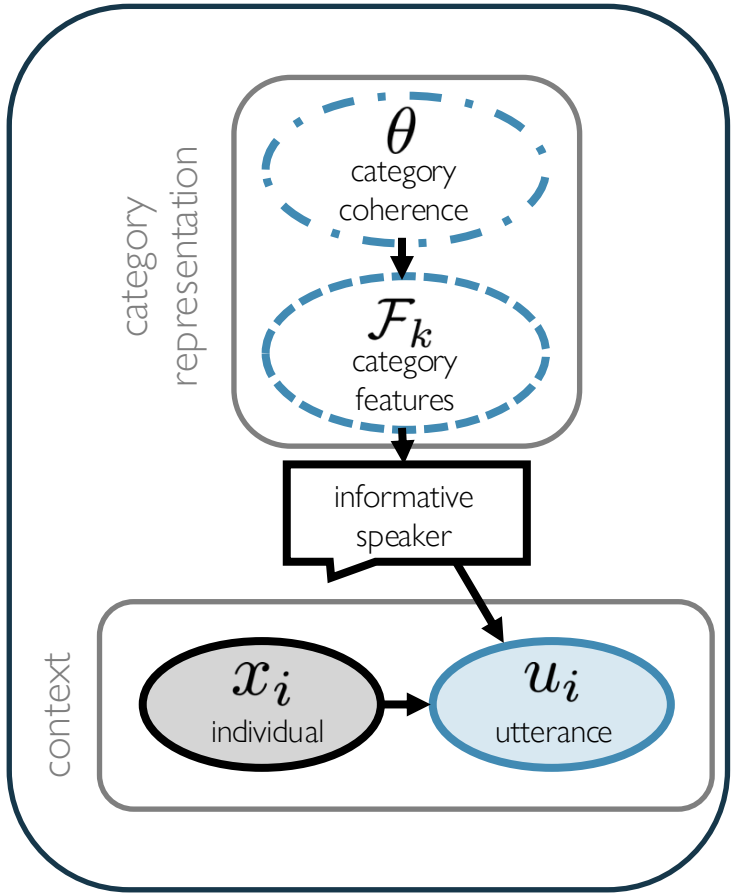


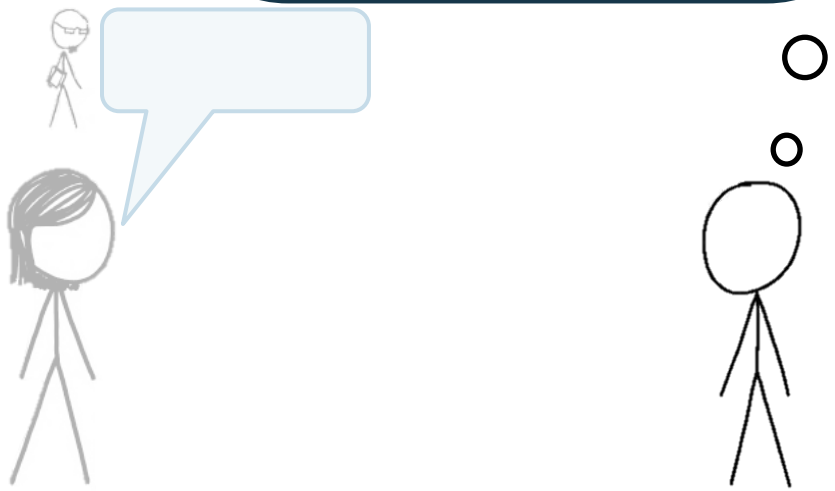
Computational model for learning about categories from language

Our hierarchical Bayesian + rational speech acts model predicts that:

1. Hearing generic statements about a category (e.g., "Professors are absent-minded.") will increase generalization across the category.

2. Hearing specific statements about individuals (e.g., "That pedestrian is absent-minded.") will decrease generalization across the category.





Experiment 1a

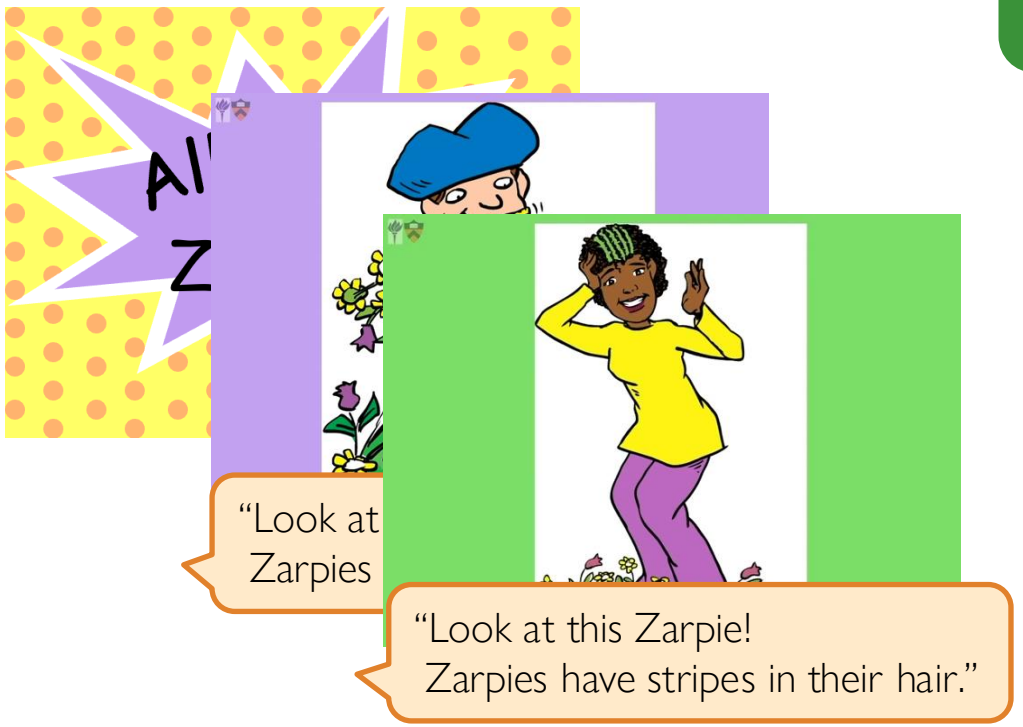
generic vs specific language

Exp 1a: design

adults learned about a novel social group, Zarpies

(US, Prolific, n=284, n=90-99/condition)

generic condition

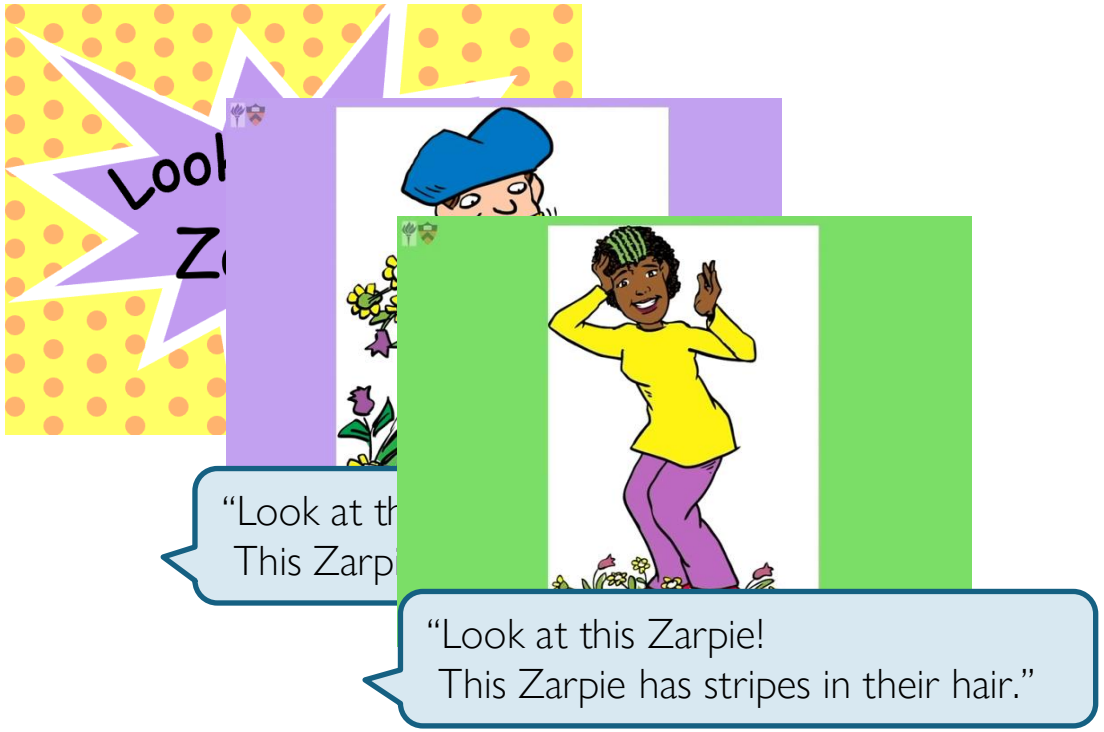


x16 statements

baseline condition

no info

specific condition



x16 statements

Exp 1a: design

adults learned about a novel social group, Zarpies

(US, Prolific, n=284, n=90-99/condition)

generic condition

16 generics about Zarpies

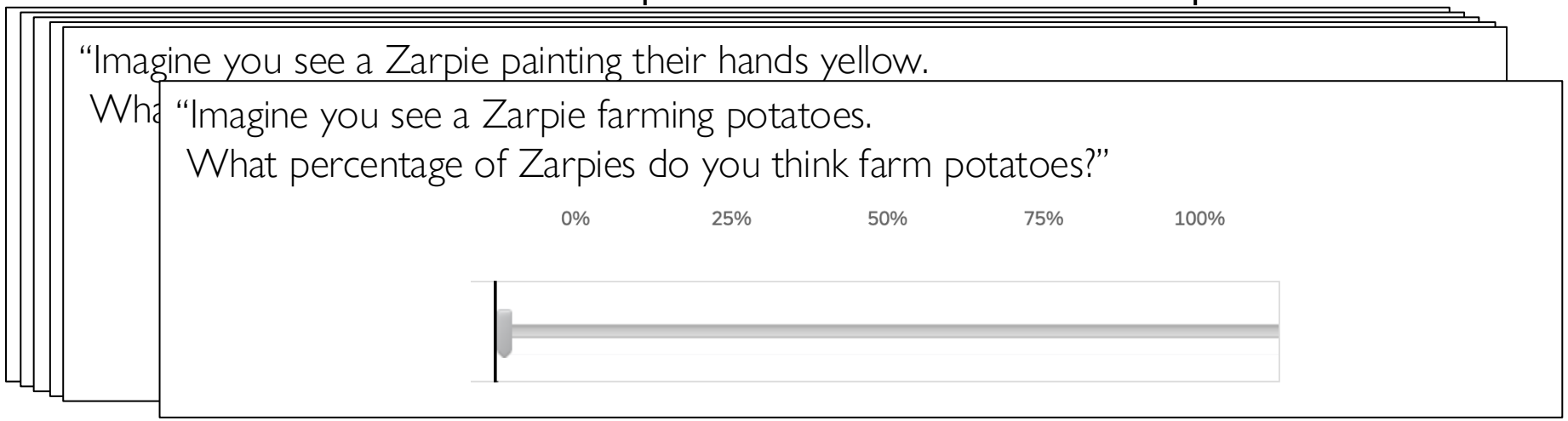
baseline condition

no info

specific condition

16 specifics about individual Zarpies

…then completed an inductive potential task



x16 test features, order randomized

From coherence to feature prevalence

pragmatic listener

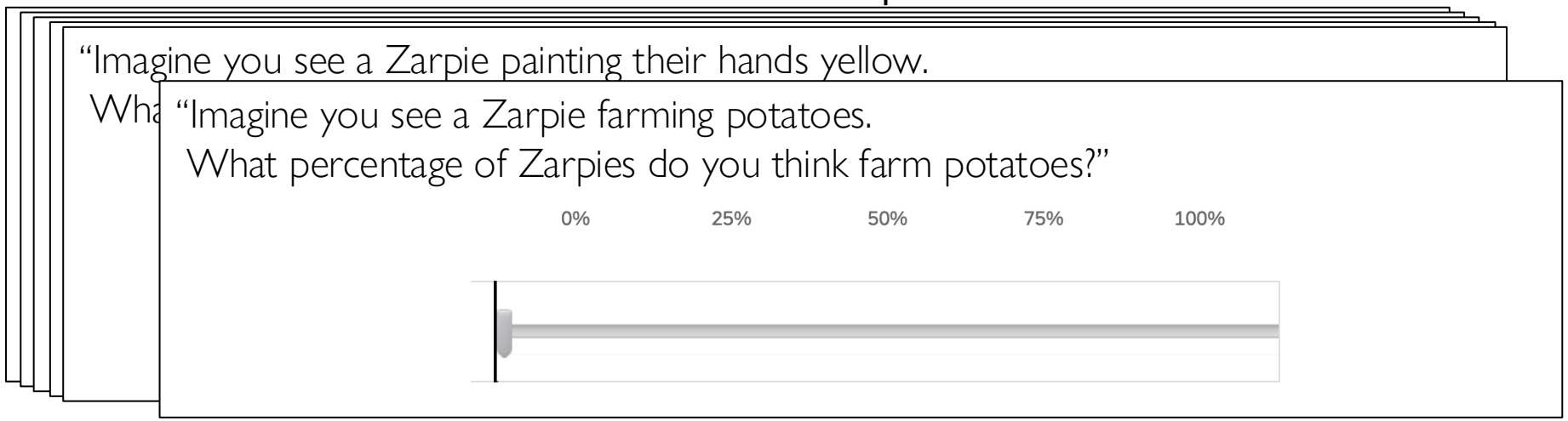
infers coherence & kind-linked features by reasoning about what the **speaker** meant

→ a single category-wide estimate of **coherence** (θ)

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

?

inductive potential task



→ feature-specific estimates of **prevalence**

From coherence to feature prevalence

pragmatic listener

infers coherence & kind-linked features by reasoning about what the **speaker** meant

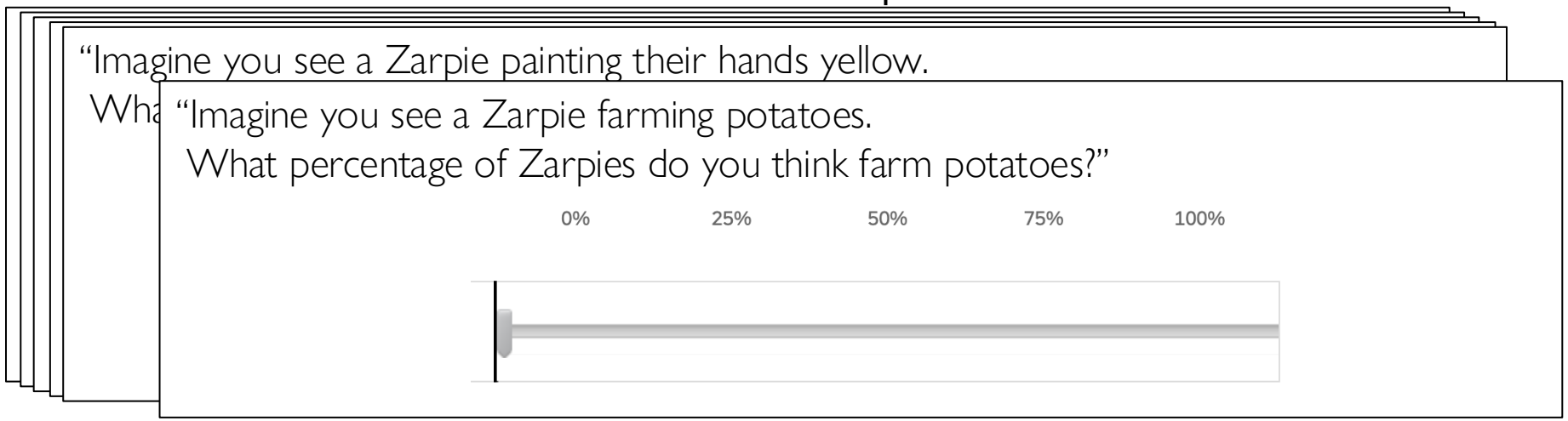
→ a single category-wide estimate of **coherence** (θ)

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

✨ **linking function** ✨

(see Tessler & Goodman, 2019)

inductive potential task



→ feature-specific estimates of **prevalence**

From coherence to feature prevalence

pragmatic listener

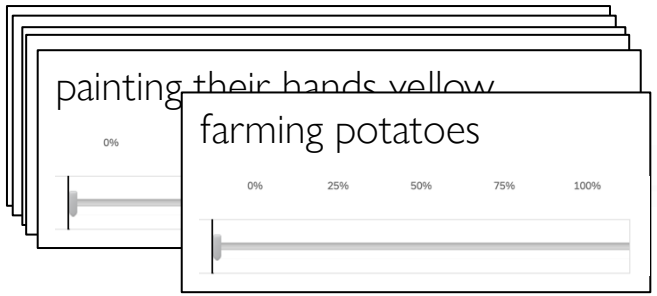
infers coherence & kind-linked features by reasoning about what the **speaker** meant

→ a single category-wide estimate of **coherence** (θ)

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

linking function

for each test feature,
use prevalence ratings to fit:



→ feature-specific estimates of **prevalence**

From coherence to feature prevalence

pragmatic listener

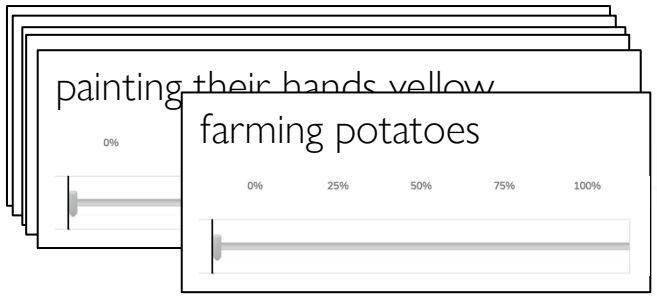
infers coherence & kind-linked features by reasoning about what the **speaker** meant

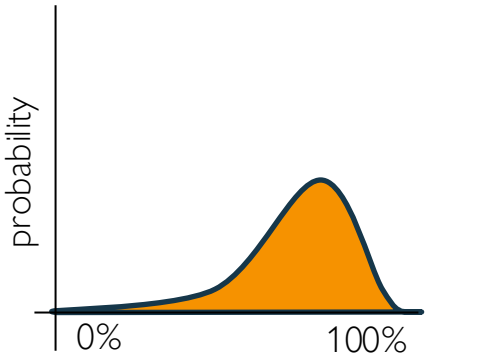
→ a single category-wide estimate of **coherence** (θ)

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

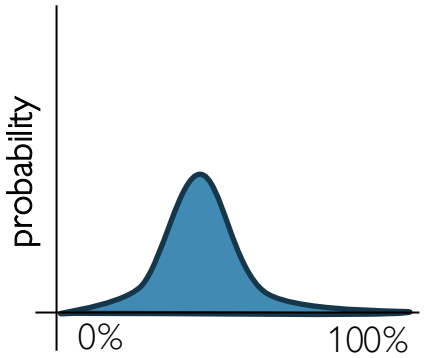
linking function

for each test feature,
use prevalence ratings to fit:





prevalence when kind-linked



prevalence when not kind-linked

→ feature-specific estimates of **prevalence**

From coherence to feature prevalence

pragmatic listener

infers coherence & kind-linked features by reasoning about what the **speaker** meant

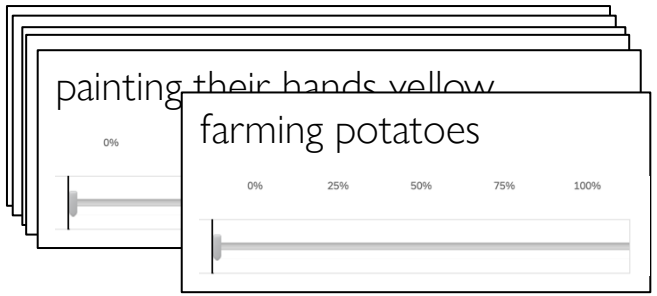
→ a single category-wide estimate of <u>coherence</u> (θ)

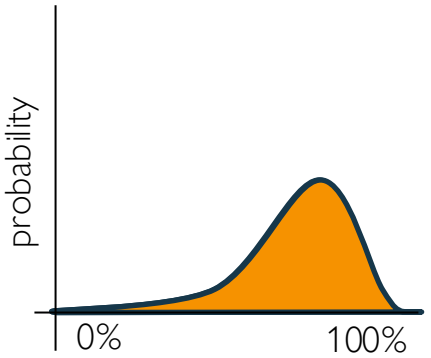
the probability any feature of an individual will be kind-linked

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

linking function

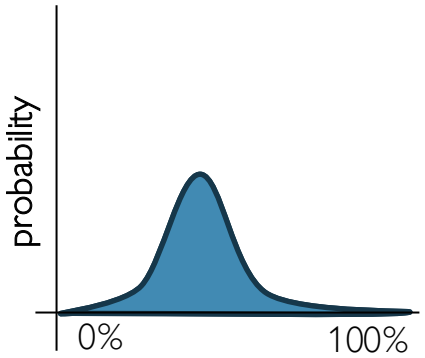
for each test feature, use prevalence ratings to fit:





prevalence when kind-linked

× coherence (θ)



prevalence when not kind-linked

× (1 – θ)

→ feature-specific estimates of prevalence

From coherence to feature prevalence

pragmatic listener

infers coherence & kind-linked features by reasoning about what the **speaker** meant

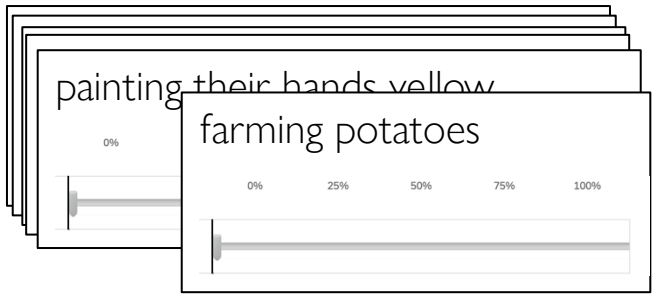
→ a single category-wide estimate of <u>coherence</u> (θ)

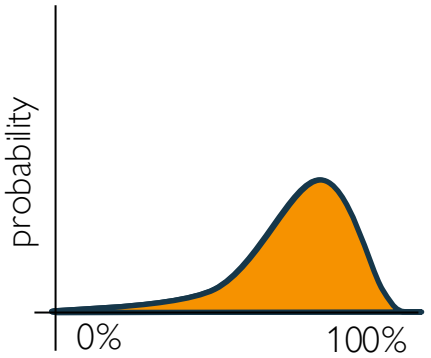
the probability any feature of an individual will be kind-linked

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

linking function

for each test feature, use prevalence ratings to fit:

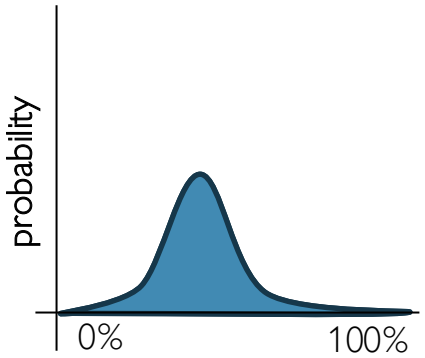




prevalence when kind-linked

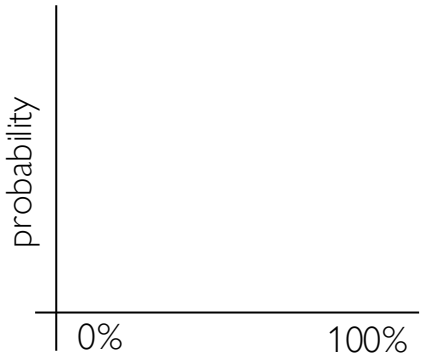
× coherence (θ)

+



prevalence when not kind-linked

× (1 – θ)



→ feature-specific estimates of **prevalence**

From coherence to feature prevalence

pragmatic listener

infers coherence & kind-linked features by reasoning about what the **speaker** meant

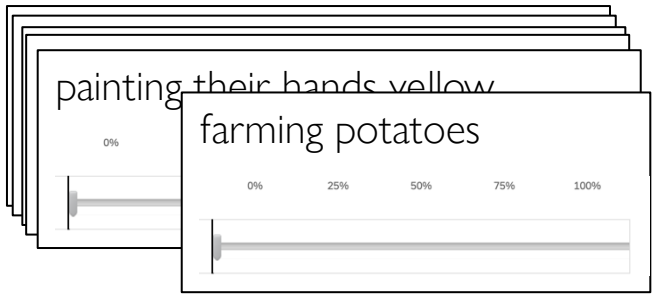
→ a single category-wide estimate of <u>coherence</u> (θ)

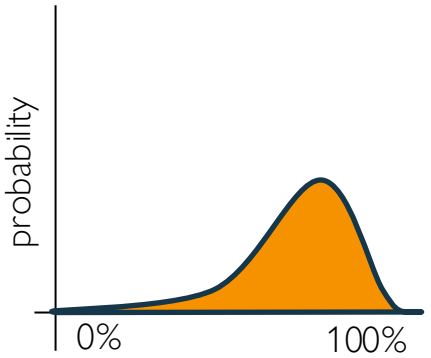
the probability any feature of an individual will be kind-linked

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

linking function

for each test feature, use prevalence ratings to fit:

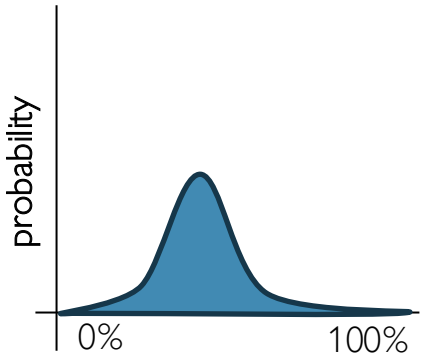




prevalence when kind-linked

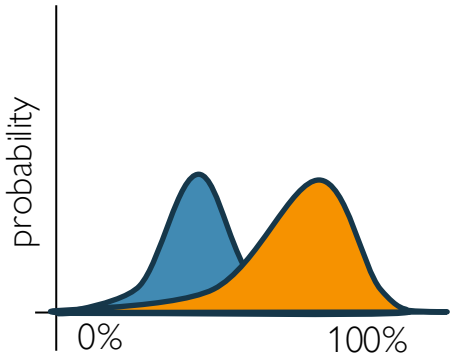
× coherence (θ)

+



prevalence when not kind-linked

× (1 – θ)



→ feature-specific estimates of **prevalence**

From coherence to feature prevalence

pragmatic listener

infers coherence & kind-linked features by reasoning about what the **speaker** meant

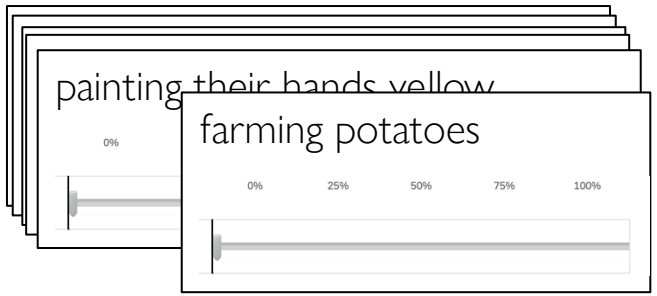
→ a single category-wide estimate of <u>coherence</u> (θ)

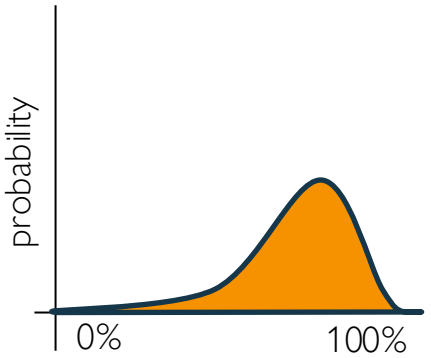
the probability any feature of an individual will be kind-linked

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

linking function

for each test feature, use prevalence ratings to fit:

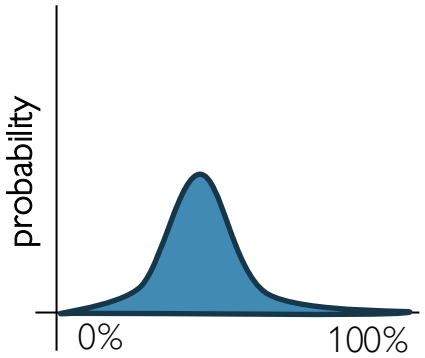




prevalence when kind-linked

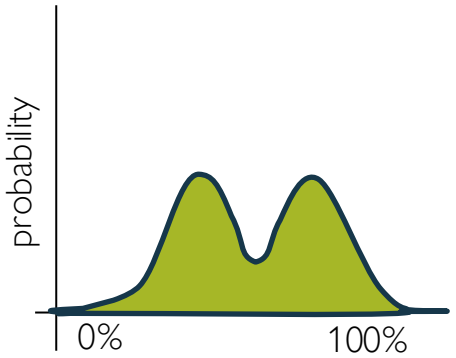
× coherence (θ)

+



prevalence when not kind-linked

× (1 – θ)



→ feature-specific estimates of **prevalence**

From coherence to feature prevalence

pragmatic listener

infers coherence & kind-linked features by reasoning about what the **speaker** meant

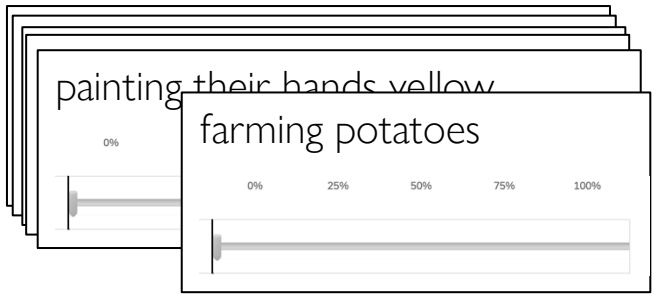
→ a single category-wide estimate of <u>coherence</u> (θ)

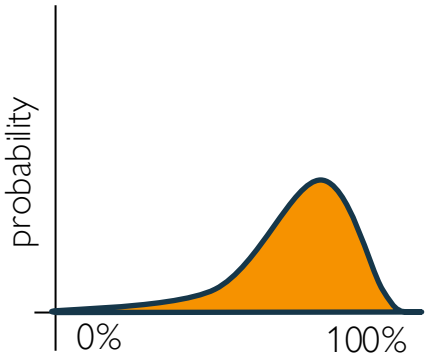
the probability any feature of an individual will be kind-linked

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

linking function

for each test feature, use prevalence ratings to fit:

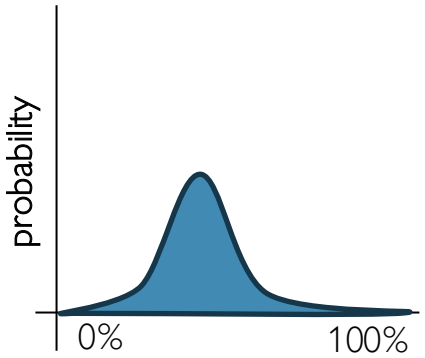




prevalence when kind-linked

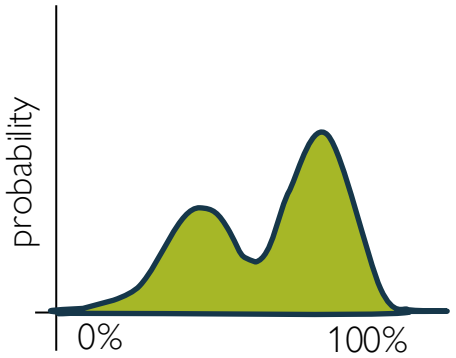
× coherence (0.8)

+



prevalence when not kind-linked

× (1 – 0.8)



→ feature-specific estimates of **prevalence**

From coherence to feature prevalence

pragmatic listener

infers coherence & kind-linked features by reasoning about what the **speaker** meant

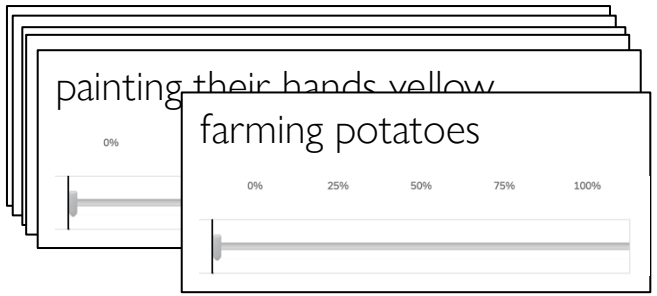
→ a single category-wide estimate of <u>coherence</u> (θ)

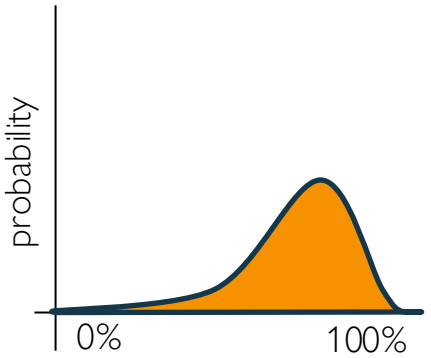
the probability any feature of an individual will be kind-linked

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

linking function

for each test feature, use prevalence ratings to fit:

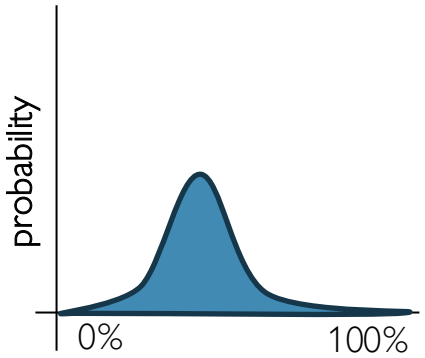




prevalence when kind-linked

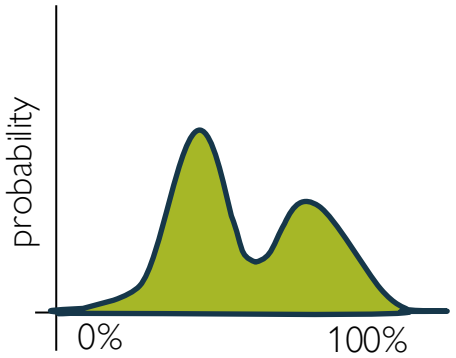
× coherence (0.2)

+



prevalence when not kind-linked

× (1 – 0.2)



→ feature-specific estimates of **prevalence**

From coherence to feature prevalence

pragmatic listener

infers coherence & kind-linked features by reasoning about what the **speaker** meant

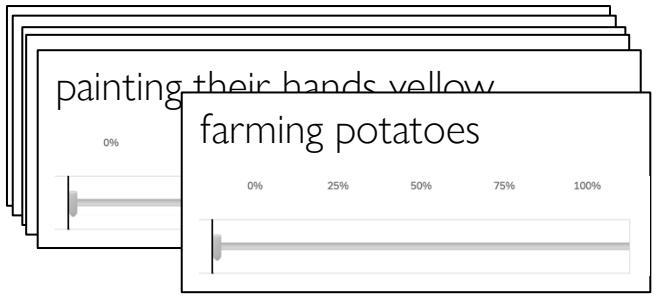
→ a single category-wide estimate of <u>coherence</u> (θ)

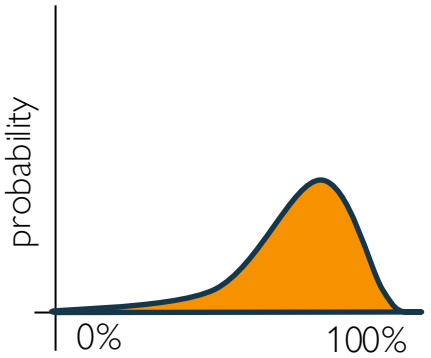
the probability any feature of an individual will be kind-linked

$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

linking function

for each test feature, use prevalence ratings to fit:

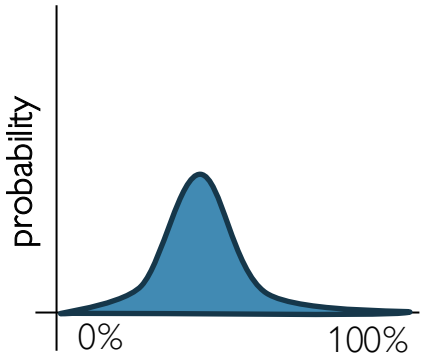




prevalence when kind-linked

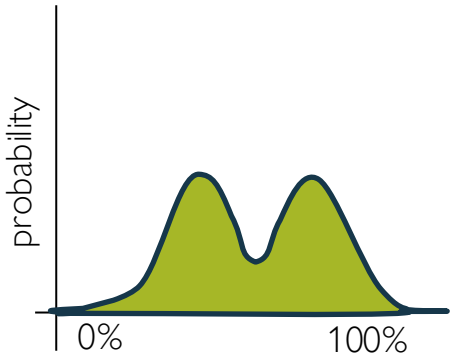
× coherence (θ)

+



prevalence when not kind-linked

× (1 – θ)



→ feature-specific estimates of **prevalence**

From coherence to feature prevalence

pragmatic listener

infers coherence & kind-linked features by reasoning about what the **speaker** meant

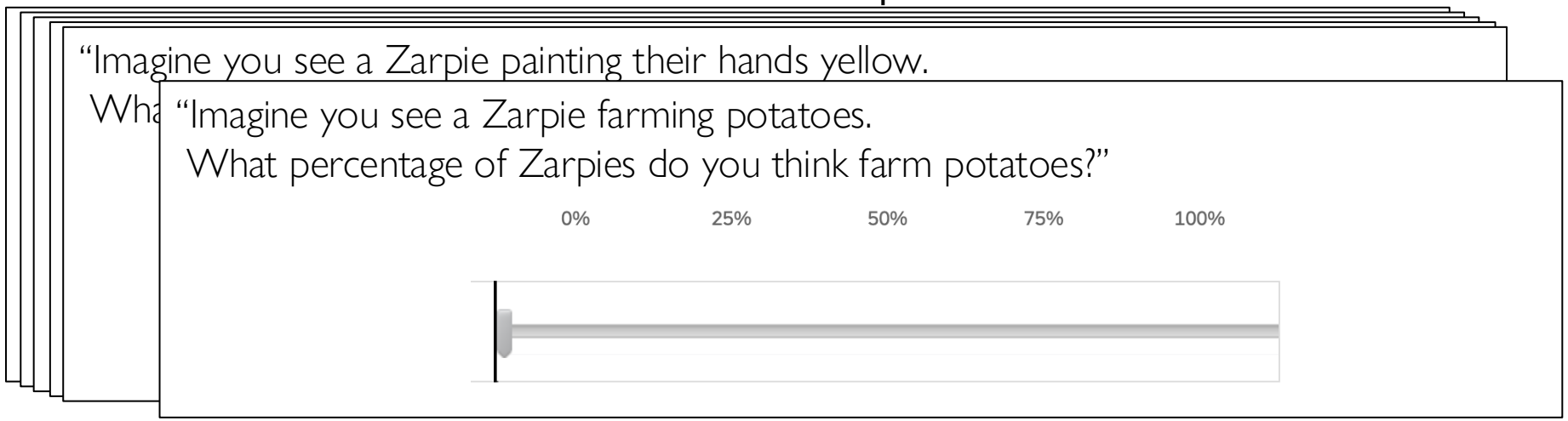
→ a single category-wide estimate of **coherence** (θ)

✨ **linking function** ✨

→ feature-specific estimates of **prevalence**

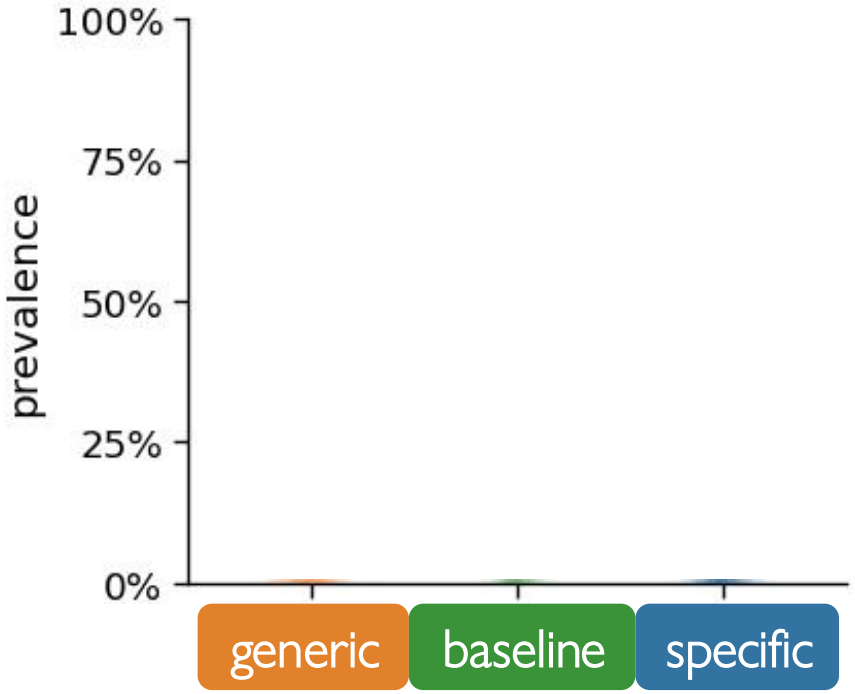
$$\mathrm{Prag}(\mathcal{F}_k, \theta \mid \boldsymbol{x}, \boldsymbol{u}) \propto P(\theta) P(\mathcal{F}_k \mid \theta) \prod_i \mathrm{Sp}(u_i \mid \mathcal{F}_k, x_i)$$

inductive potential task



→ feature-specific estimates of **prevalence**

Exp 1a results: human data

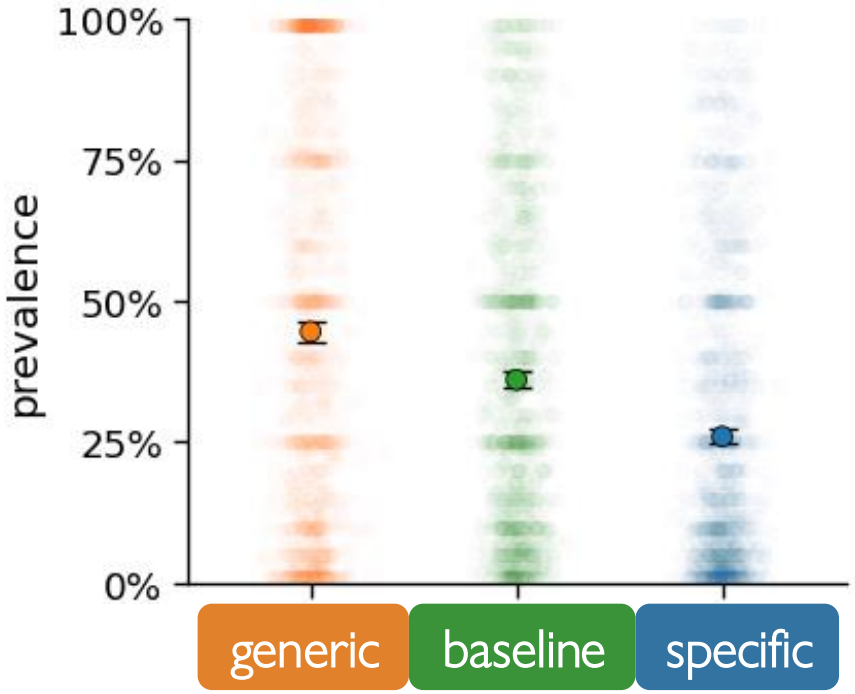


large dots with bars = means with 95% CIs
small dots = individual responses to each trial

Exp 1a results: human data

As predicted by our model,

1. Hearing generic statements about a category increased generalization across the category.
2. Hearing specific statements about individuals decreased generalization across the category.

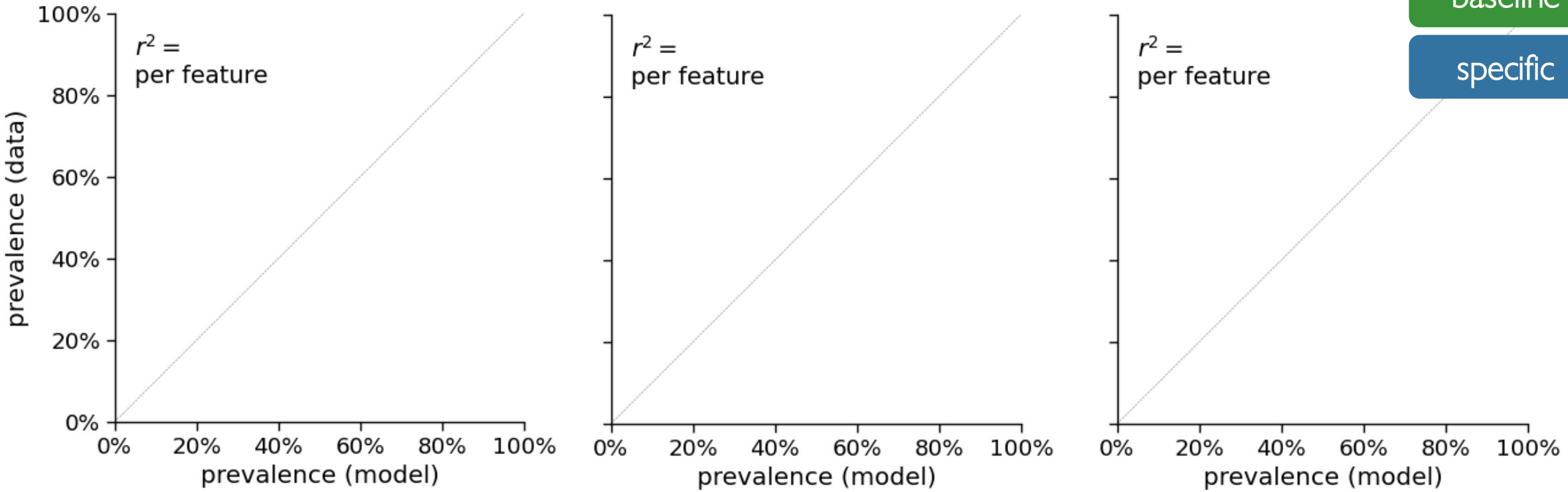


large dots with bars = means with 95% CIs
small dots = individual responses to each trial

Exp 1a results: model vs data

conditions

- generic
- baseline
- specific



null model

reasoning about individuals and features only

literal listener

reasoning about **categories** and literal **meaning** of generics

pragmatic listener

reasoning about categories and what the **speaker** meant

n = 284 adults vs 300 simulated responses in each model
dots = mean per test feature per condition; large dots = mean per condition

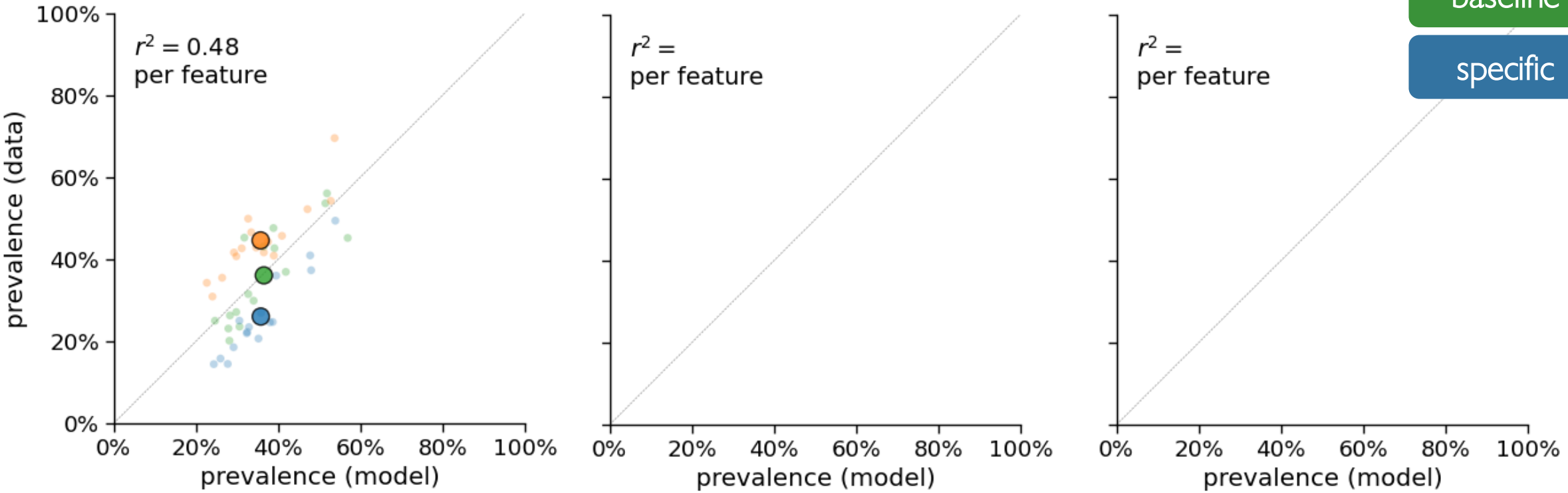
Exp 1a results: model vs data

conditions

generic

baseline

specific



null model

reasoning about individuals and features only

literal listener

reasoning about **categories** and literal **meaning** of generics

pragmatic listener

reasoning about categories and what the **speaker** meant

n = 284 adults vs 300 simulated responses in each model

dots = mean per feature per condition; large dots = mean per condition

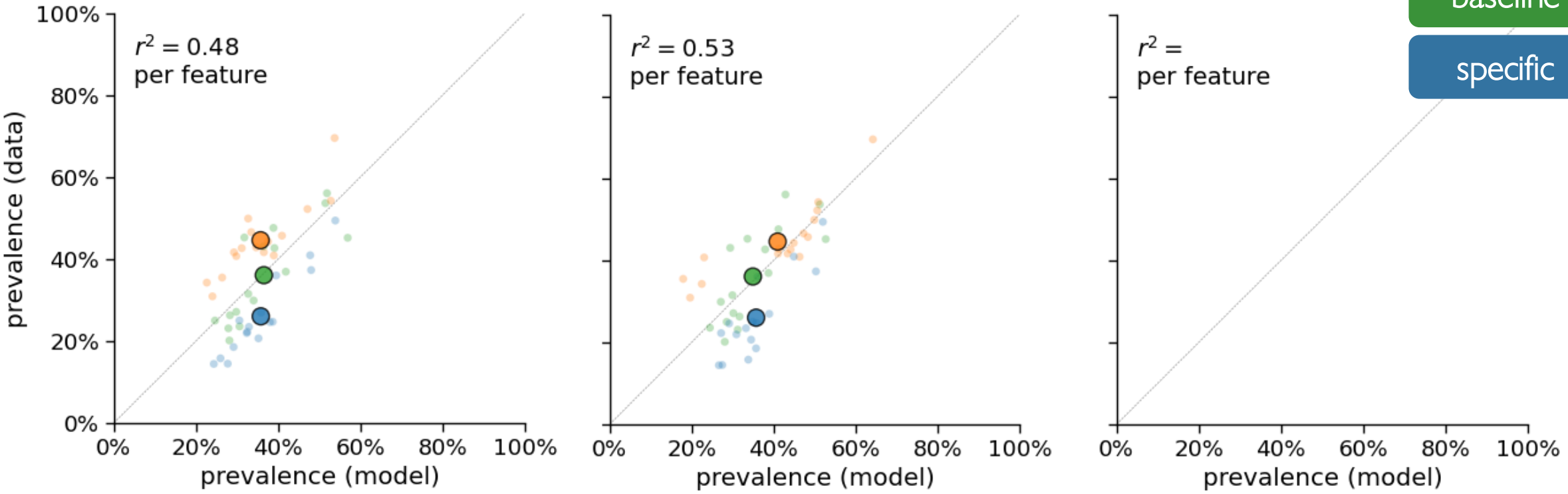
Exp 1a results: model vs data

conditions

generic

baseline

specific



null model

reasoning about individuals and features only

literal listener

reasoning about **categories** and literal **meaning** of generics

pragmatic listener

reasoning about categories and what the **speaker** meant

generics

data: n = 284; model: 300 simulations

dots = individual features within condition; large dots = mean per condition

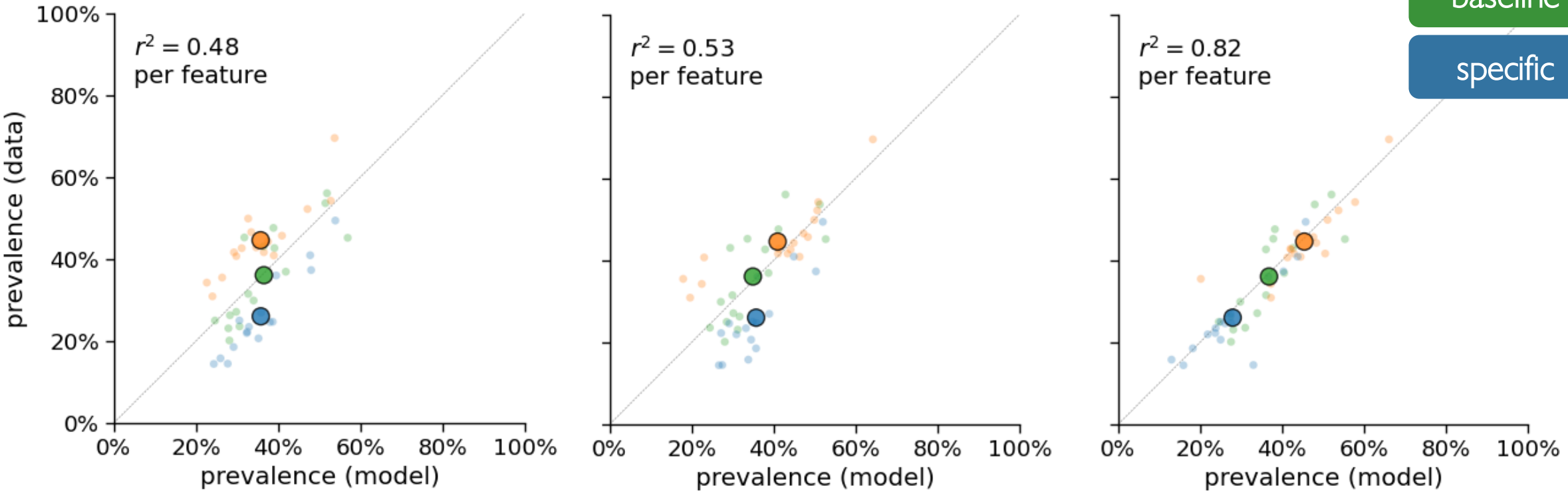
Exp 1a results: model vs data

conditions

generic

baseline

specific



null model

reasoning about individuals and features only

literal listener

reasoning about **categories** and literal **meaning** of generics

pragmatic listener

reasoning about categories and what the **speaker** meant

generics

specifics

data: n = 284; model: 300 simulations

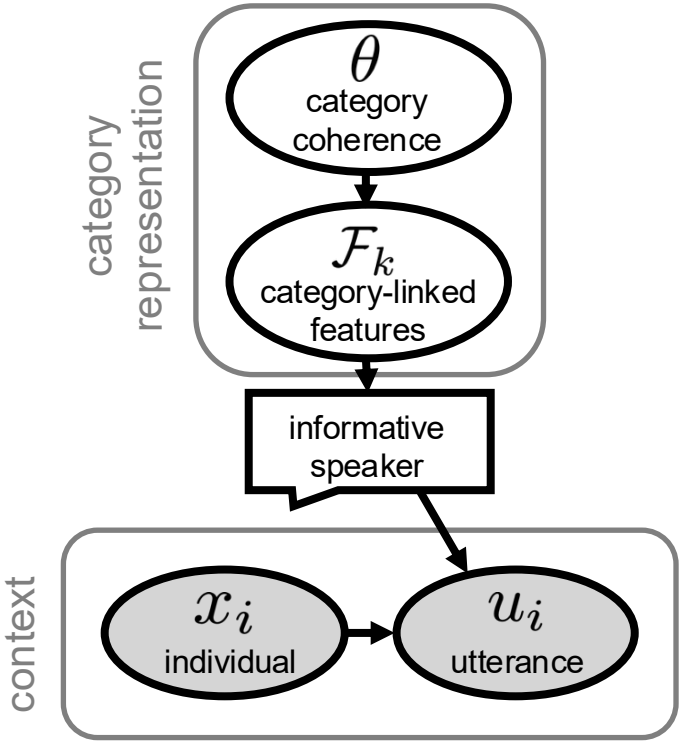
dots = individual features within condition; large dots = mean per condition

Exp 1a: generics vs specifics

A hierarchical Bayesian + rational speech acts model explains why:

1. Hearing generic statements about a category **increases** generalization across the category.
   → due to **reasoning about semantic meaning** and the **inductive potential** of the category

2. Hearing specific statements about category members **decreases** the inferred prevalence of more features.
   → due to **pragmatic reasoning** about what the speaker could have but chose not to say



Experiment 1b

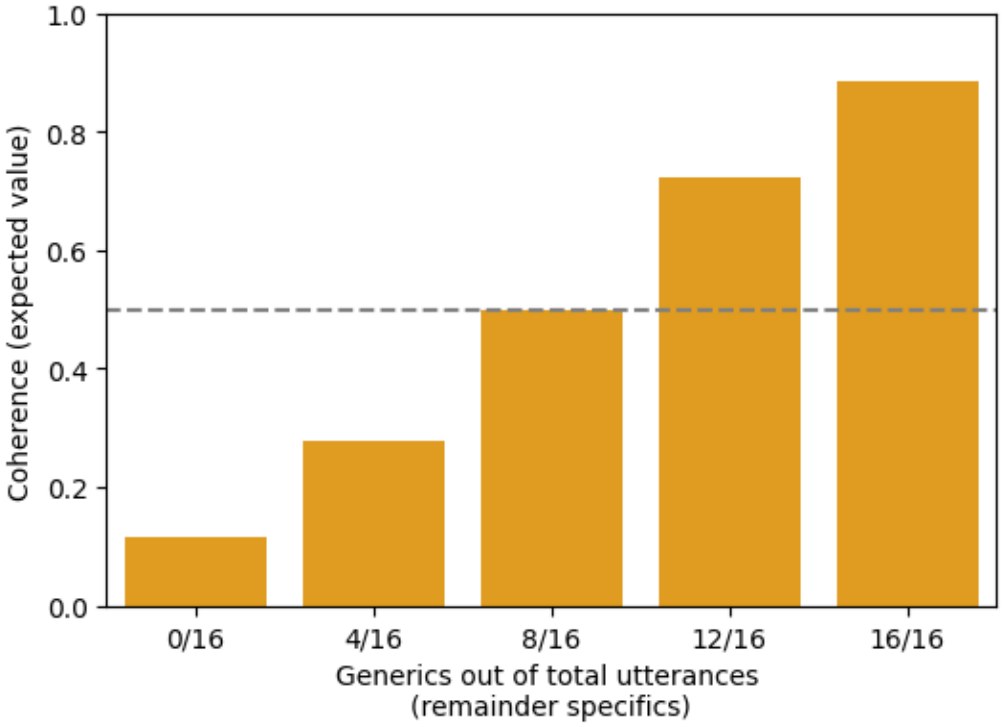
inference from a mix of generic and specific statements

Exp 1b: model coherence

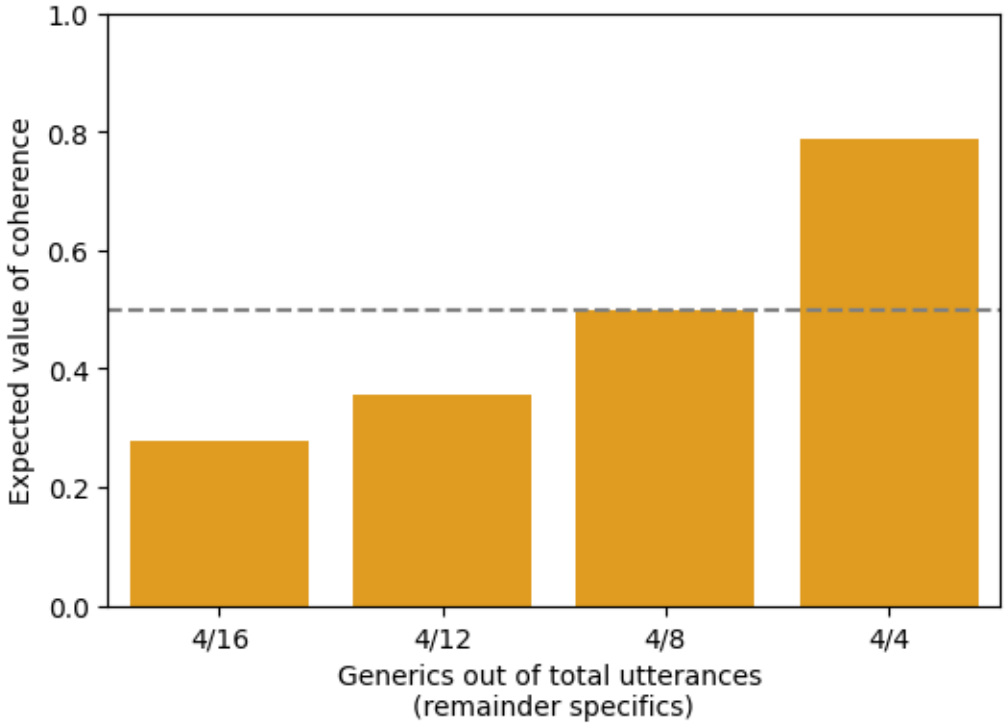
pragmatic listener

reasoning about categories and what the **speaker** meant

number of generics matters



proportion of generics matters



inverse temperature: set to 20

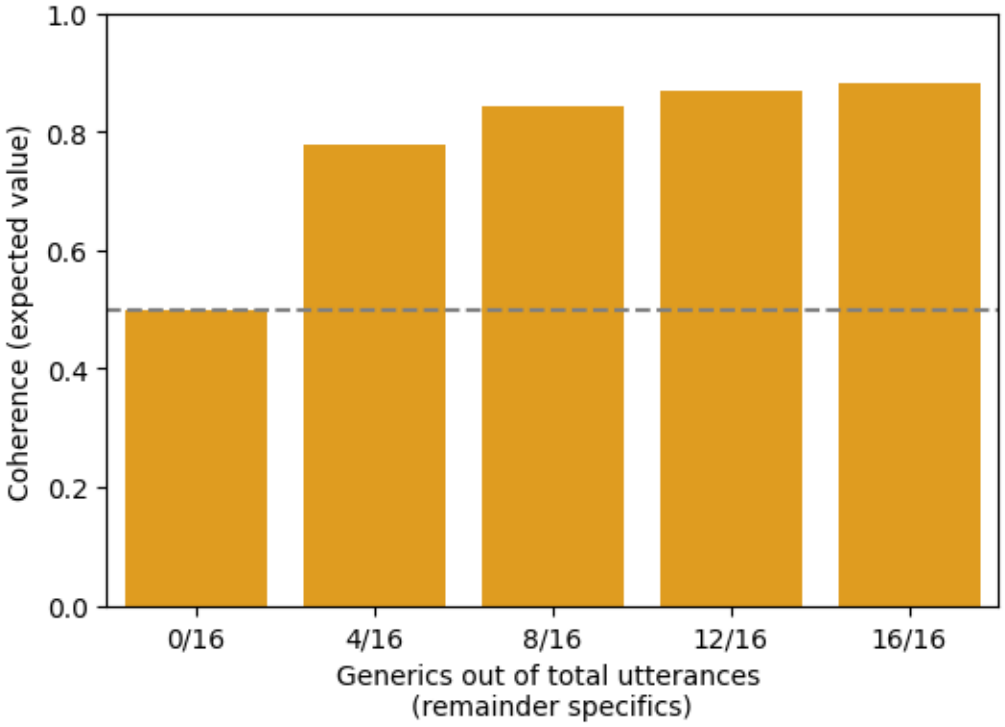
dashed line = prior on coherence = uniform over [0, 1]

Exp 1b: model coherence

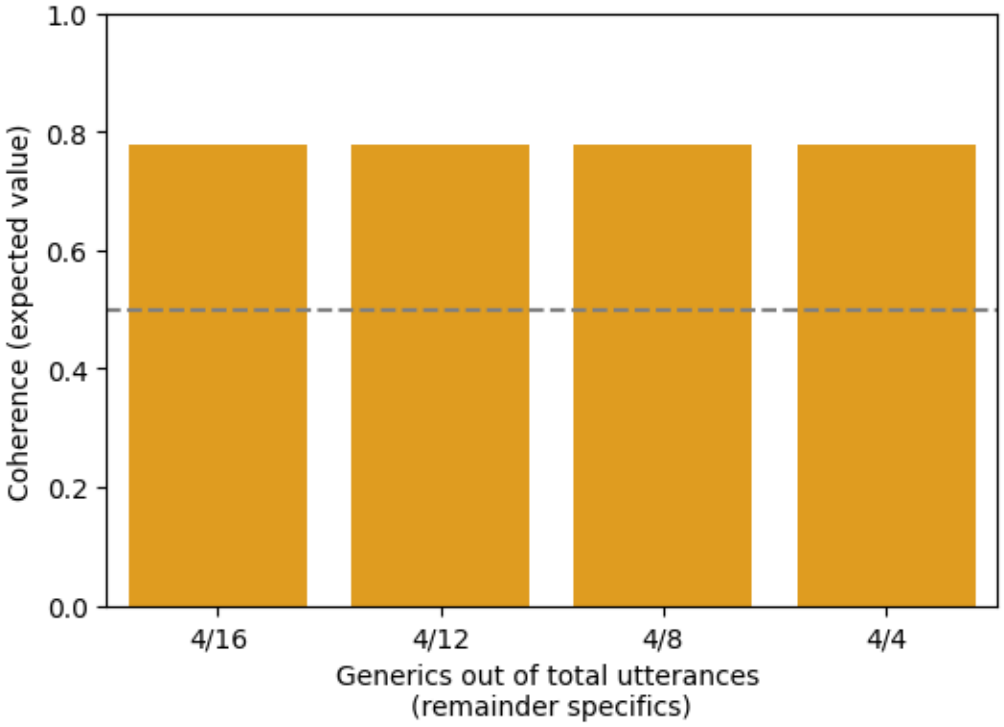
literal listener

reasoning about **categories** and literal **meaning** of generics

number of generics matters



if same number of generics,
proportion of generics doesn't matter



dashed line = prior on coherence = uniform over [0, 1]

Exp 1b: design

adults learned about a novel social group, Zarpies

(US, Prolific, *n* = 524, *n* = 32-25/condition)

16/16 generics

baseline

0/16 generics

Exp 1b: design

adults learned about a novel social group, Zarpies

(US, Prolific, *n* = 524, *n* = 32-25/condition)

number of generics

				16/16 generics
			12/12	12/16 generics
		8/8	8/12	8/16 generics
	4/4	4/8	4/12	4/16 generics
baseline	0/4	0/8	0/12	0/16 generics

number of utterances

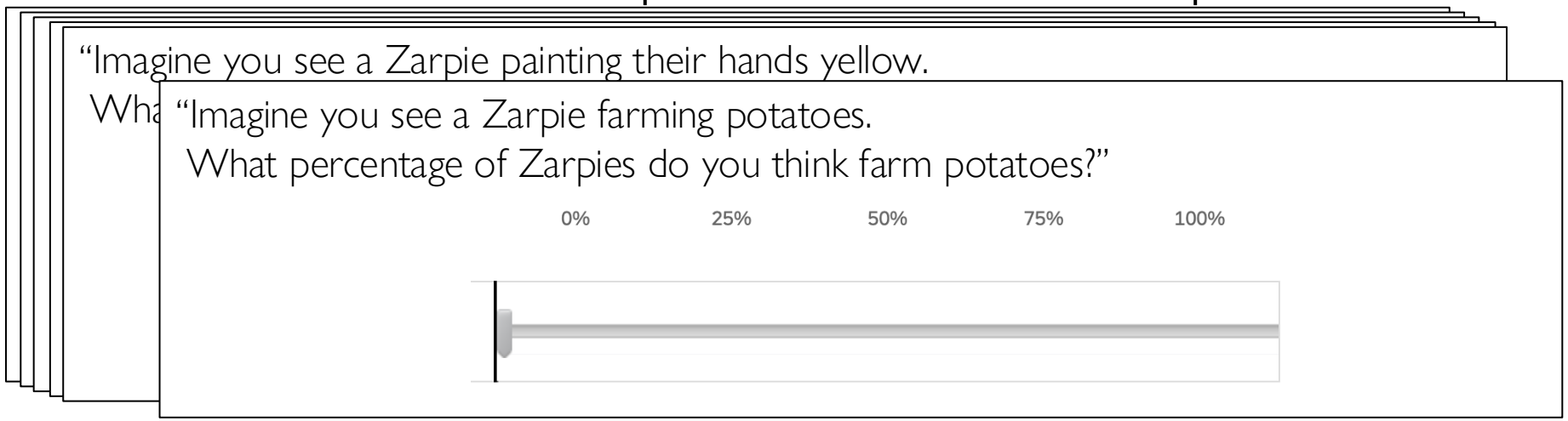
(rest specifics)

Exp 1b: design

adults learned about a novel social group, Zarpies

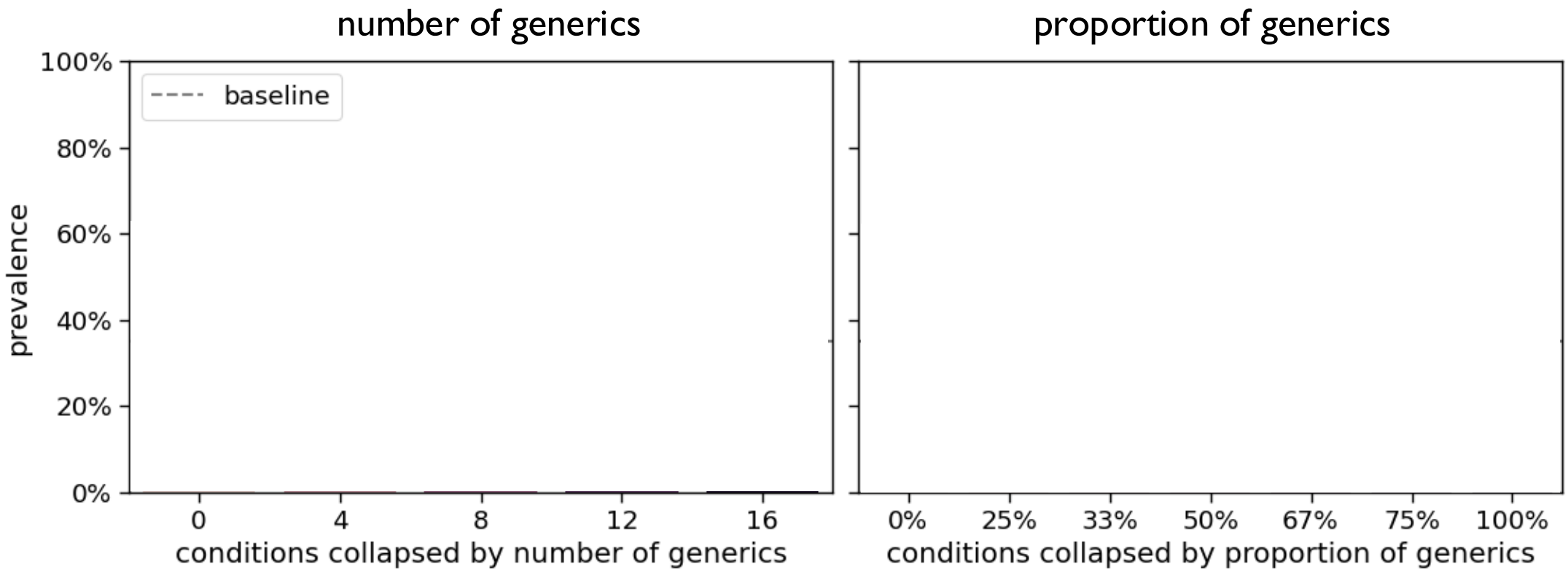
(US, Prolific, *n* = 524, *n* = 32-25/condition)

…then completed an inductive potential task



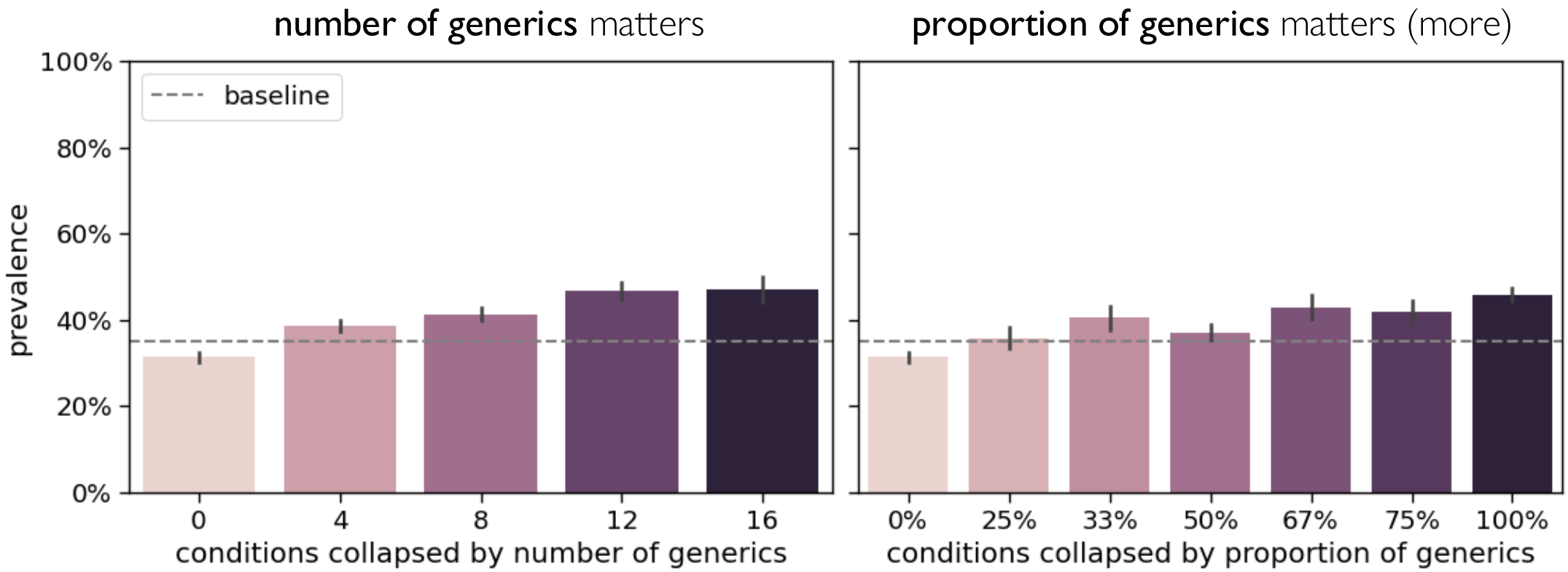
x16 test features,
order randomized

Exp 1b: data



dots w/ bars = means w/ 95% CIs
from n = 524 adults (n = 32-35/condition)

Exp 1b: data



mixed effects logistic regression
w/ random intercepts per participant, test feature:

on their own, main effect of **number of generics** ($p < .001$), main effect of **proportion of generics** ($p < .001$)

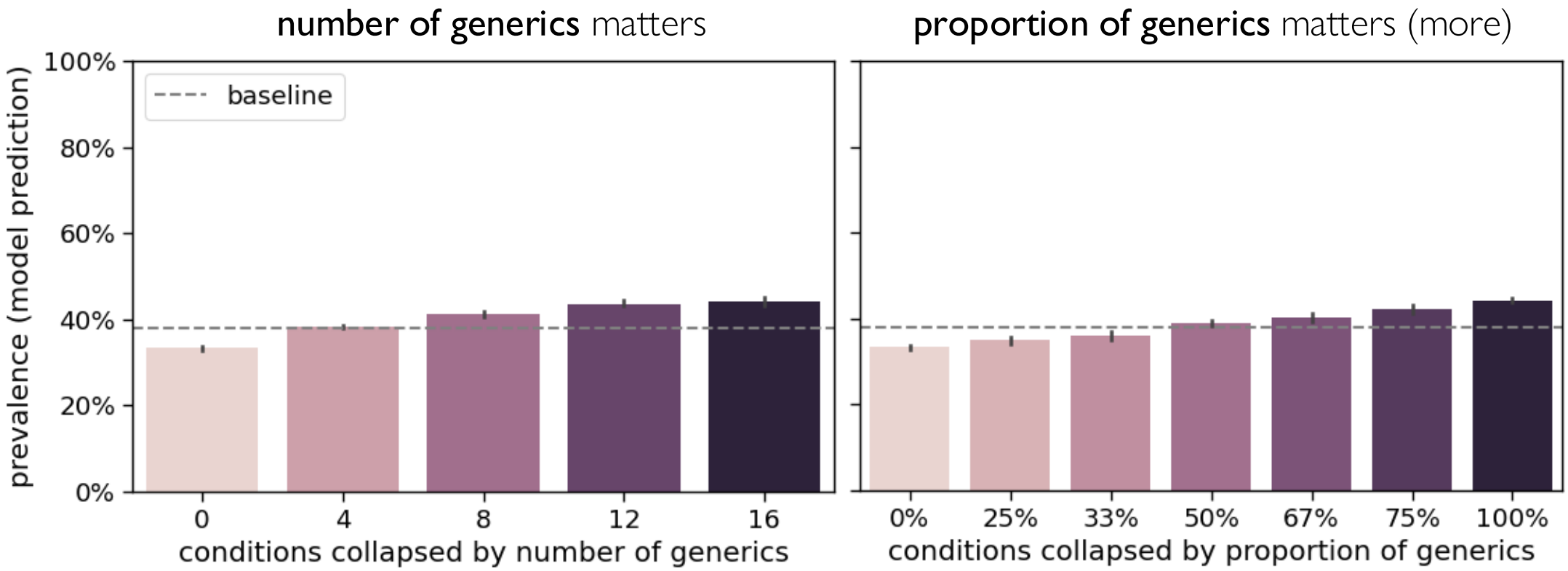
simultaneously, **proportion of generics** remains significant ($p = .018$), **number of generics** does not ($p = .32$)

dots w/ bars = means w/ 95% CIs
from $n = 524$ adults ($n = 32$-35/condition)

Exp 1b: model fit

pragmatic listener

reasoning about categories and what the **speaker** meant



inverse temperature = 20

dots w/ bars = means w/ 95% CIs

from 300 simulated responses from model's linking function

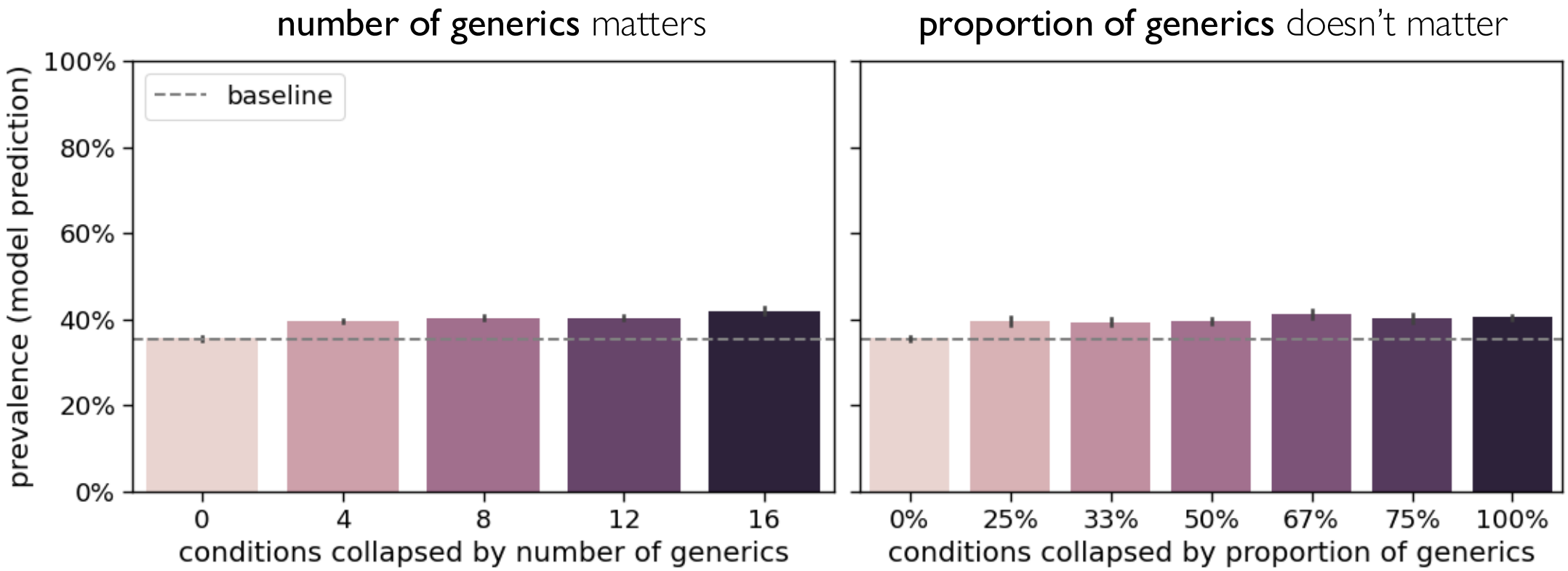
(fit for each feature to prevalence ratings, weighted by coherence)

R^2 with data = 0.67

Exp 1b: model fit

literal listener

reasoning about **categories** and literal **meaning** of generics



dots w/ bars = means w/ 95% CIs
from 300 simulated responses from model's linking function
(fit for each feature to prevalence ratings, weighted by coherence)

R^2 with data = 0.55

Exp 1b: conclusions

A hierarchical Bayesian + rational speech acts model explains why:

Hearing **proportionally more** generic statements about a category vs specific statements about individual category members **increases** the inferred inductive potential of the category.

→ due to **pragmatic reasoning** about what the speaker could have but chose not to say

Learning about the inductive potential of categories from language

Part I

- **Computational model**
- **Exp 1a:** learning from generic versus specific language
- **Exp 1b**: learning from a mix of generics and specifics

Part II

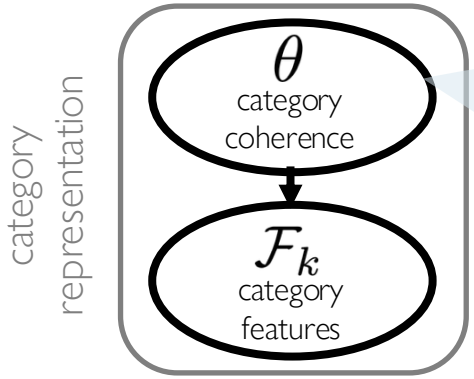
- **Extending the model**: feature-based generalization
- **Exp 2**: learning to generalize over features

extending the model

feature-based generalization

Computational model

for modeling feature-based generalization



coherence

the probability
that a feature will be
kind-linked
for this category

earlier model

coherence

a **single fixed value**
for all features

feature-specific model

coherence function

a **function** over **feature space**
providing a feature-specific coherence

Computational model

for modeling feature-based generalization

generic statements

"Zarpies have freckles on their feet."
"Zarpies tell bad jokes."
"Zarpies like to chug maple syrup."
…

grouped into a priori feature types:

- physical
- appearance
- abilities
- hobbies_behaviors
- habitat
- diet_preferences
- beliefs_culture
- personality

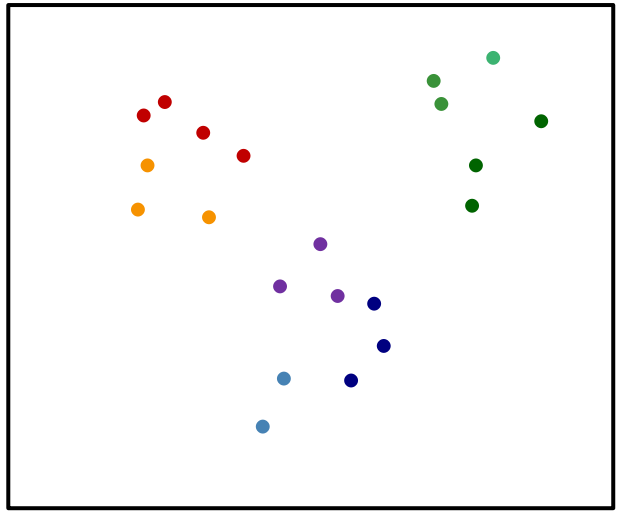
sentence transformer (MiniLM-L6-v2) →

embeddings in high-dimensional feature space

[…, …, …, …]
[…, …, …, …]
[…, …, …, …]
…

PCA →

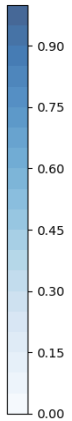
embeddings in 2D feature space



similar features are **closer** in feature space

Computational model

for modeling feature-based generalization



coherence

the probability
that a feature
will be kind-linked
for this category

earlier model

coherence

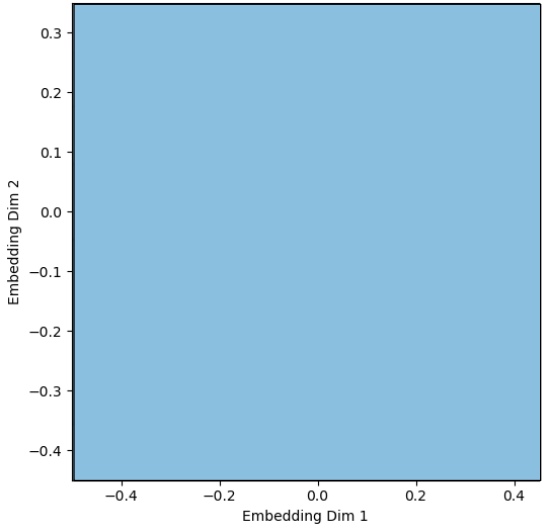
a **single fixed value**
for all features

feature-specific model

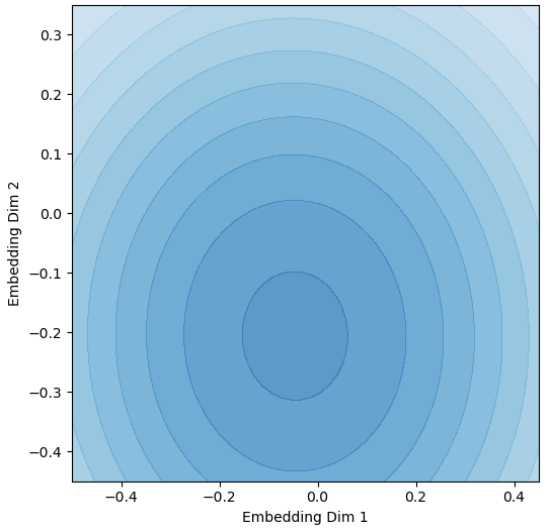
coherence function

a **function** over **feature space**
returning a feature-specific value

2D feature space



2D feature space



Computational model

for modeling feature-based generalization

coherence

the probability that a feature will be kind-linked for this category

earlier model

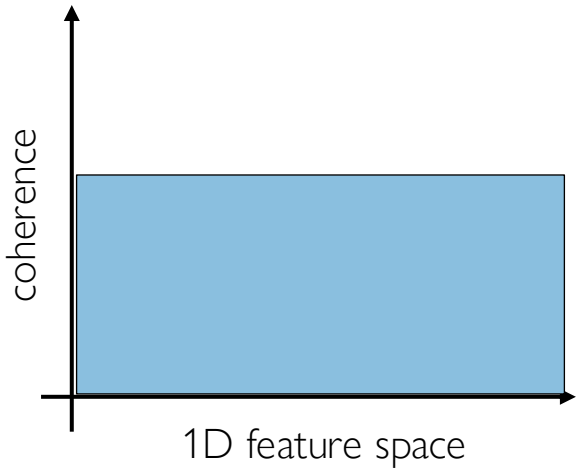
coherence

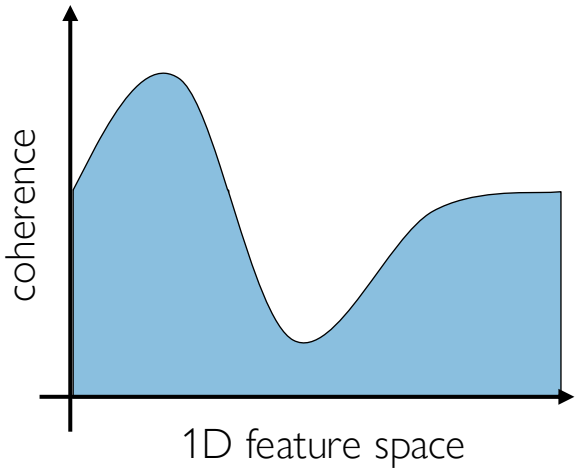
a **single fixed value** for all features

feature-specific model

coherence function

a **function** over **feature space** returning a feature-specific value



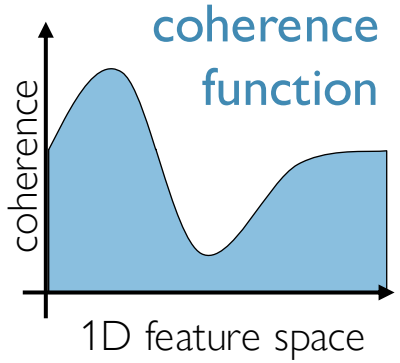


Gaussian process represents a distribution over functions

w/ Maya Malaviya!

Gaussian process (GP)

a distribution over functions



a GP takes in **observed data points** and returns a **probability distribution over possible functions**

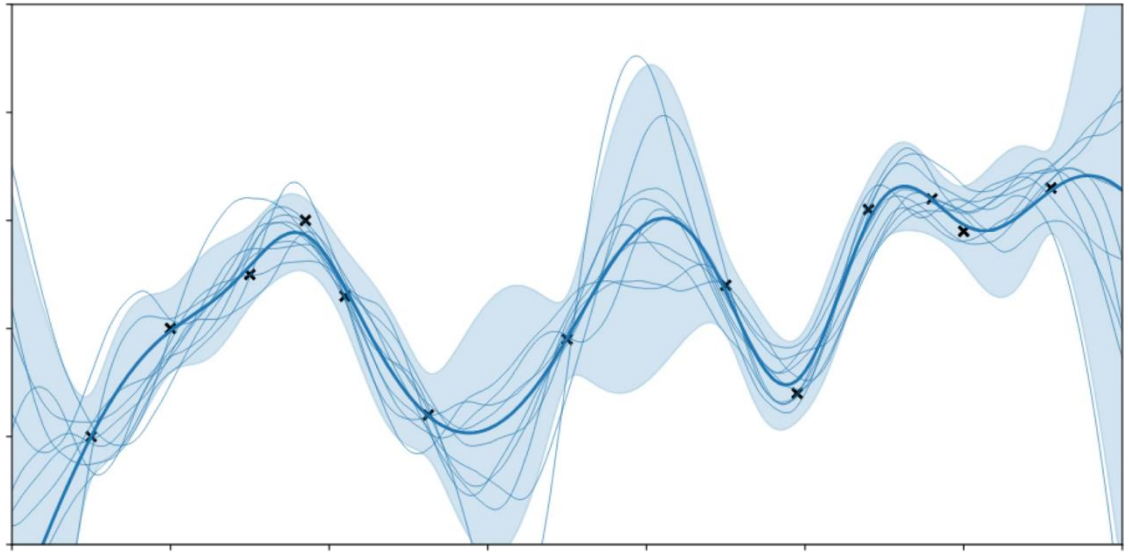
$$f \sim \mathcal{GP}(\mu, k)$$

*GPs are **parameterized** by:*

- μ *(mean function)*
- k *(kernel function)*

Typical GP

takes in continuous data points



x = training point

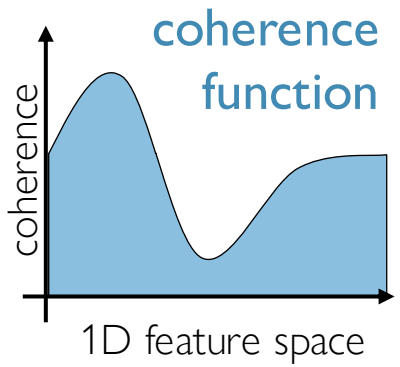
slide from Maya Malaviya

visualization from Newman 2022 blog

Williams & Rasmussen 1995, Griffiths et al. 2008, Duvenaud 2014, Lucas et al. 2015, Schulz et al. 2017

Gaussian process (GP)

a distribution over functions



a GP takes in **observed data points** and returns a **probability distribution over possible functions**

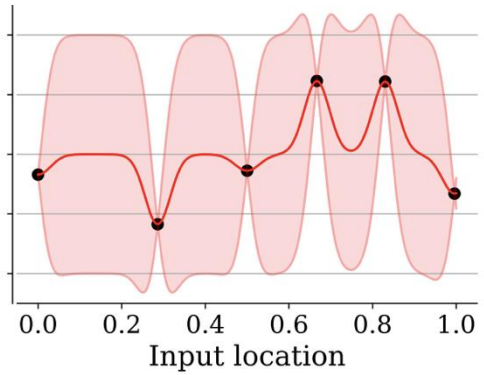
$$f \sim \mathcal{GP}(\mu, k)$$

*GPs are **parameterized** by:*

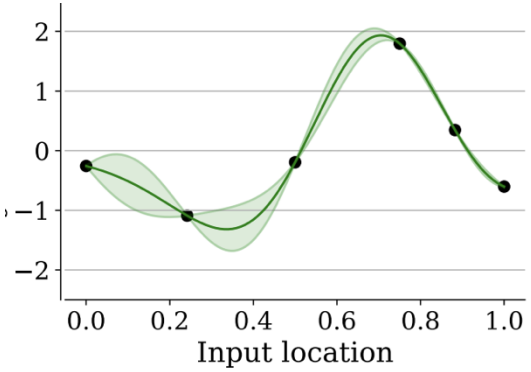
- μ *(mean function)*
- k *(kernel function)*

$$k_{sim}(x, x') = \sigma^2 \exp\left(-\frac{(x - x')}{\boxed{\ell}^2}\right)$$

short length scale (l)



long length scale (l)

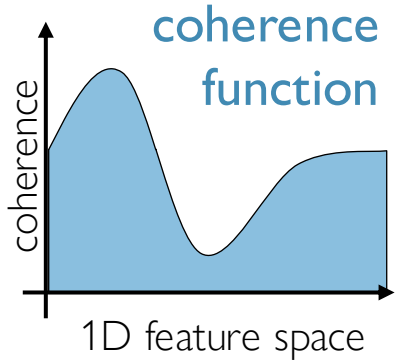


slide from Maya Malaviya
visualization from Gondu 2024

Williams & Rasmussen 1995, Griffiths et al. 2008, Duvenaud 2014, Lucas et al. 2015, Schulz et al. 2017

Gaussian process (GP)

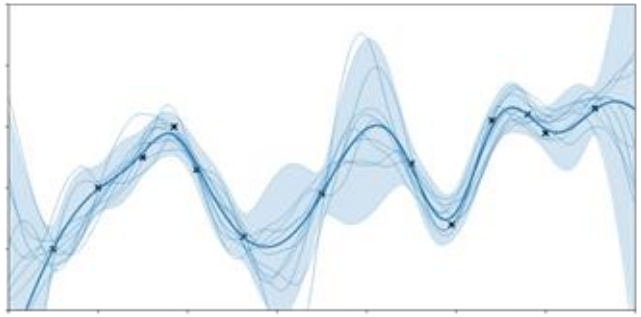
a distribution over functions



Typical GP

takes in continuous data points

$$P(y|f) = \mathcal{N}(f(x), \sigma^2)$$



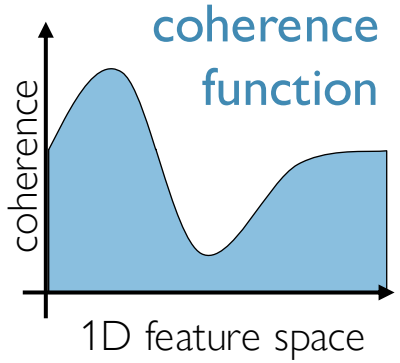
Pragmatic listener + GP

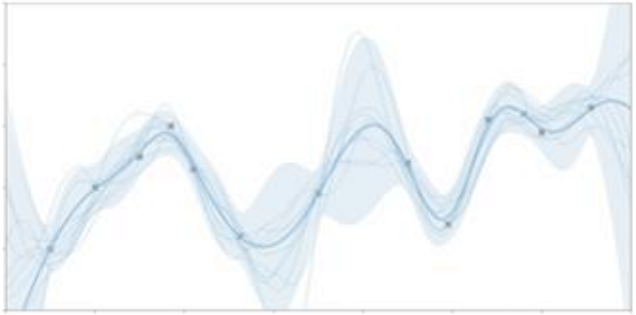
takes in utterances

returns a distribution over coherence functions

Gaussian process (GP)

a distribution over functions





Typical GP

takes in continuous data points

$$P(y|f) = \mathcal{N}(f(x), \sigma^2)$$

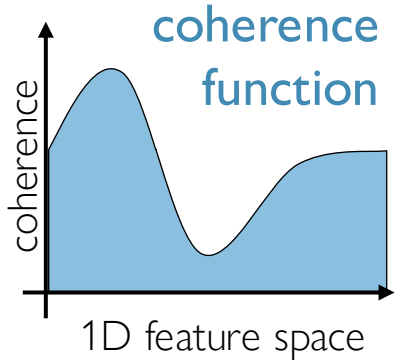
Bernoulli GP

takes in binary data points

(e.g., training features' embeddings & whether each is kind-linked or not)

Gaussian process (GP)

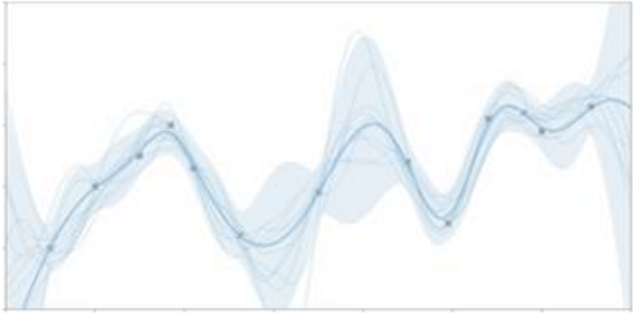
a distribution over functions



Typical GP

takes in continuous data points

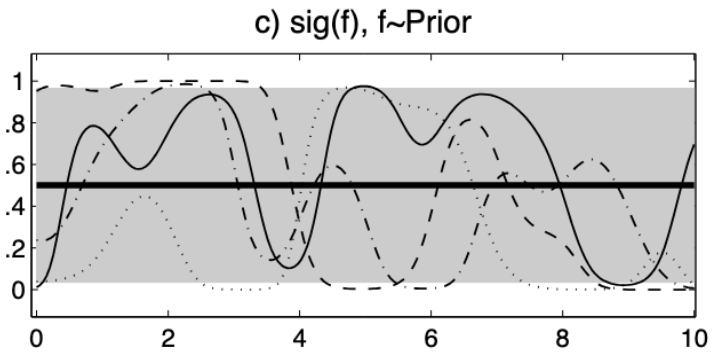
$$P(y|f) = \mathcal{N}(f(x), \sigma^2)$$



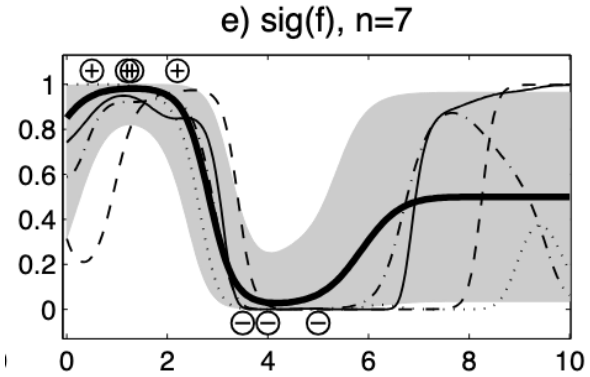
Bernoulli GP

takes in binary data points

$$P(y|f) = sigmoid\big(f(x)\big)^{y} \cdot \Big(1 - sigmoid\big(f(x)\big)\Big)^{1-y}$$



many possible functions
can be sampled from the prior

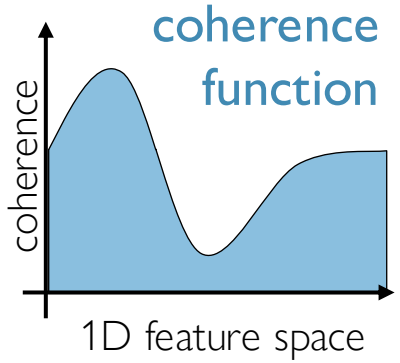


⊕ = positive training point
⊖ = negative training point

Nickisch & Rasmussen (2008), Fig 2c/e

Gaussian process (GP)

a distribution over functions



Bernoulli GP

takes in binary data points

(e.g., training features' embeddings & whether each is kind-linked or not)

$$P(z'|x,z,x',l) = \sum_f P(f|x,z)\ P(z'|f,x',l)$$

test feature: kind-linked or not?

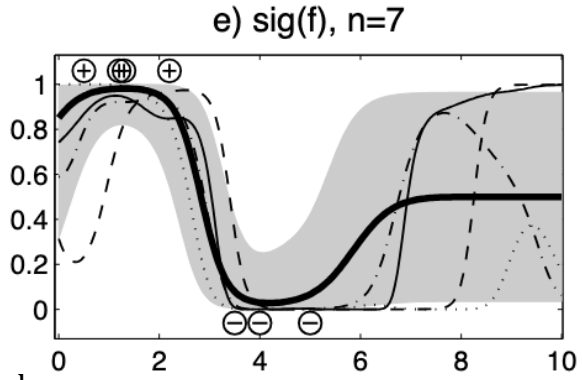
training features embeddings & kind-linked or not

test feature embedding

length scale for kernel

coherence function

$$sigmoid(f(x))^y \quad \text{if } z' = \text{kind-linked}$$
$$\left(1 - sigmoid(f(x))\right)^{1-y} \quad \text{if } z' = \text{not kind-linked}$$



⊕ = positive training point

⊖ = negative training point

Nickisch & Rasmussen (2008), Fig 2c/e

Experiment 2

feature-based variability

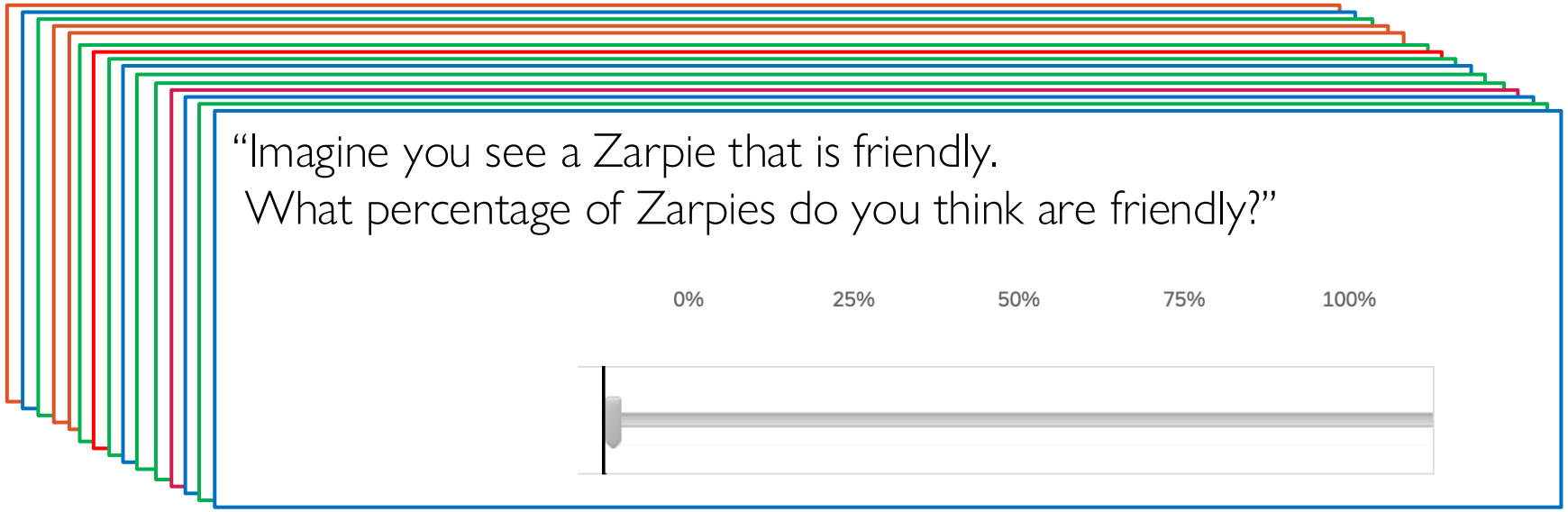
Exp 2: design

adults heard 15 generics about a novel social group, Zarpies

(US, Prolific, n=397, n=98-100/condition)

physical condition	diet condition	personality condition	heterogeneous condition

…then completed an inductive potential task



15 novel test features:

- 5 physical
- 5 diet
- 5 personality

test feature order randomized

training features

physical condition

have freckles on their feet
have a flat belly button
tan easily
can touch their toes
have orange earwax
have tattoos
can see really far
have a loud voice
have wide feet
have bony elbows
paint their toenails
draw circles on their knees
have short nails
wear tall socks
have stripes in their hair

diet condition

eat rice
like to eat broccoli
like to eat herbs
really don't like soup
make bread
drink milk
eat nuts
eat meat
eat cheese
like to eat sweets
like to drink maple syrup
like to drink tea
like to drink fizzy water
really don't like ice cream
farm potatoes

personality condition

are friendly
are angry
are brave
like to ride bikes
tell bad jokes
are forgetful
are curious
are silly
are smart
are thoughtful
teach children
live with their families
give good gifts
take care of others
can do cartwheels

heterogeneous condition

inductive potential test features

snore when they sleep
can snap with their toes
have cold hands and feet
can wiggle their ears
can roll their tongue

can eat spicy food
eat breakfast very late
like juice with pulp
eat five meals a day
put pepper on all their foods

like to collect rocks
like to give high-fives
like to read books
cry easily
like to dance

Exp 2: feature space

heterogeneous

- training feature

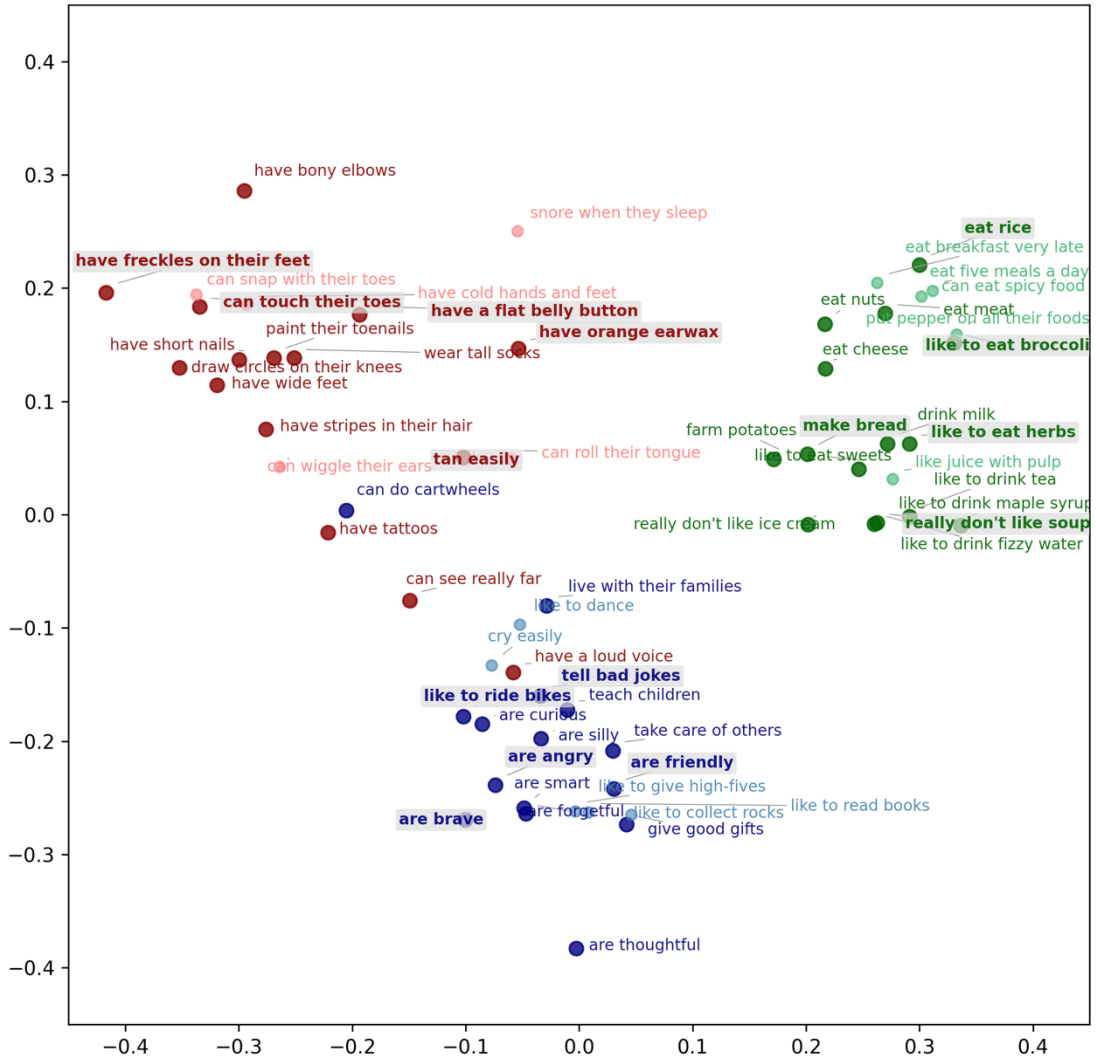
physical

- training feature
- inductive potential test feature

personality

- training feature
- inductive potential test feature

2D feature space



diet

- training feature
- inductive potential test feature

Exp 2 behavioral predictions

1. Hearing generic statements about some features should increase generalization of other similar features.
   (e.g., generics about Zarpies' physical features → generalize novel physical features more)

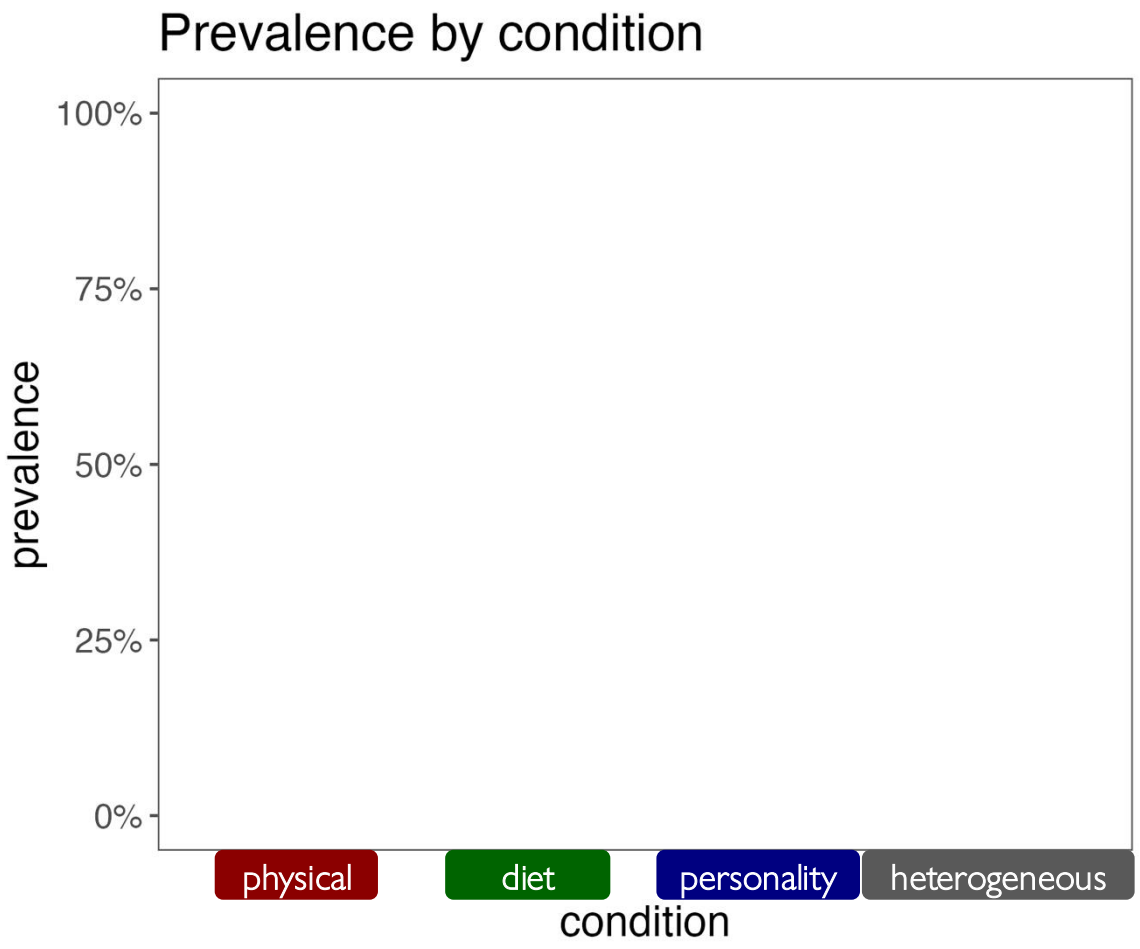
2. Hearing generic statements about heterogeneous features should result in the most increase in generalization overall, across features.

Exp 2 behavioral predictions

1. Hearing generic statements about some features should increase generalization of other similar features.
   (e.g., generics about Zarpies' physical features → generalize novel physical features more)

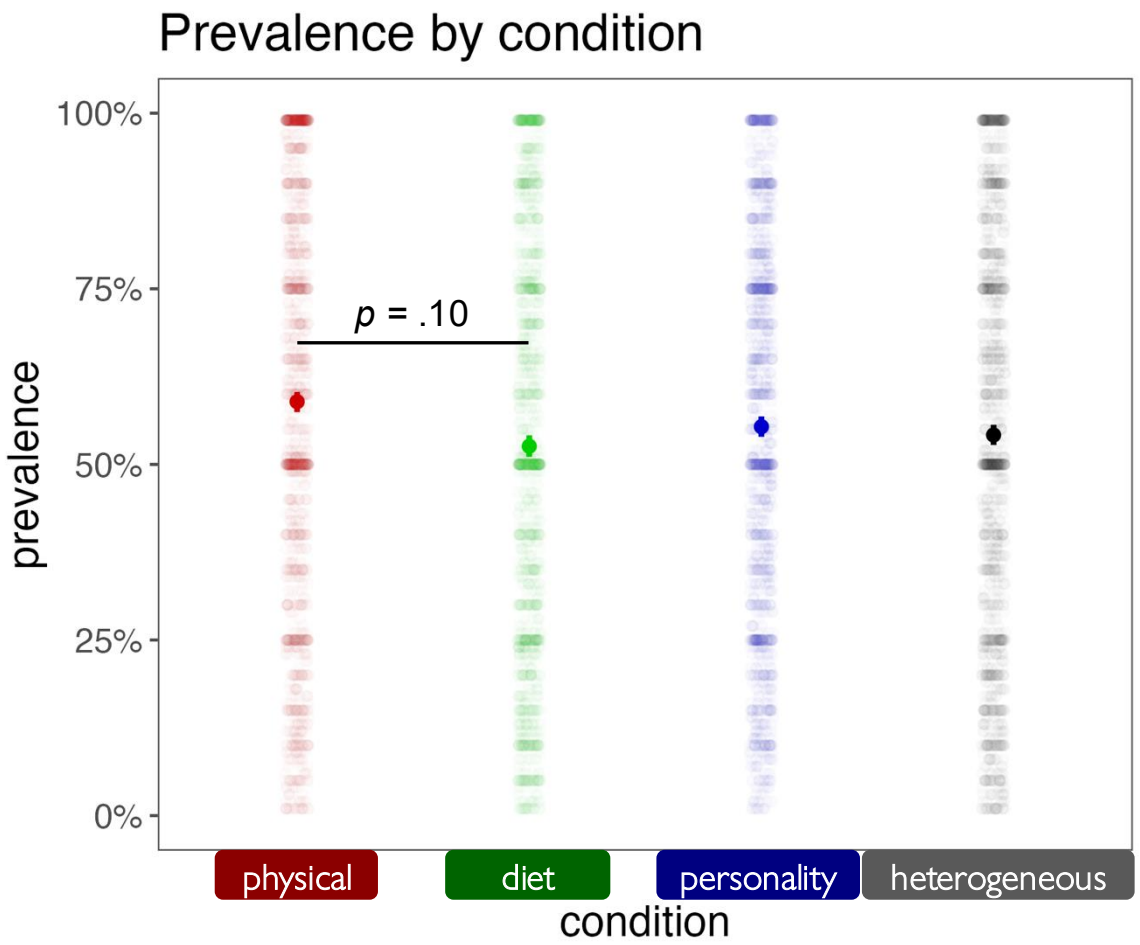
2. Hearing generic statements about heterogeneous features should result in the most increase in generalization overall, across features.

Exp 2 results: human data



small dots = individual responses (participant * test feature type)
large dots w/ bars = means w/ 95% CIs
all pairwise comparisons with FDR correction

Exp 2 results: human data



mixed effects logistic regression
w/ random intercepts per participant, test feature:
marginal effect of condition ($\chi(3) = 6.48$, $p = .091$)

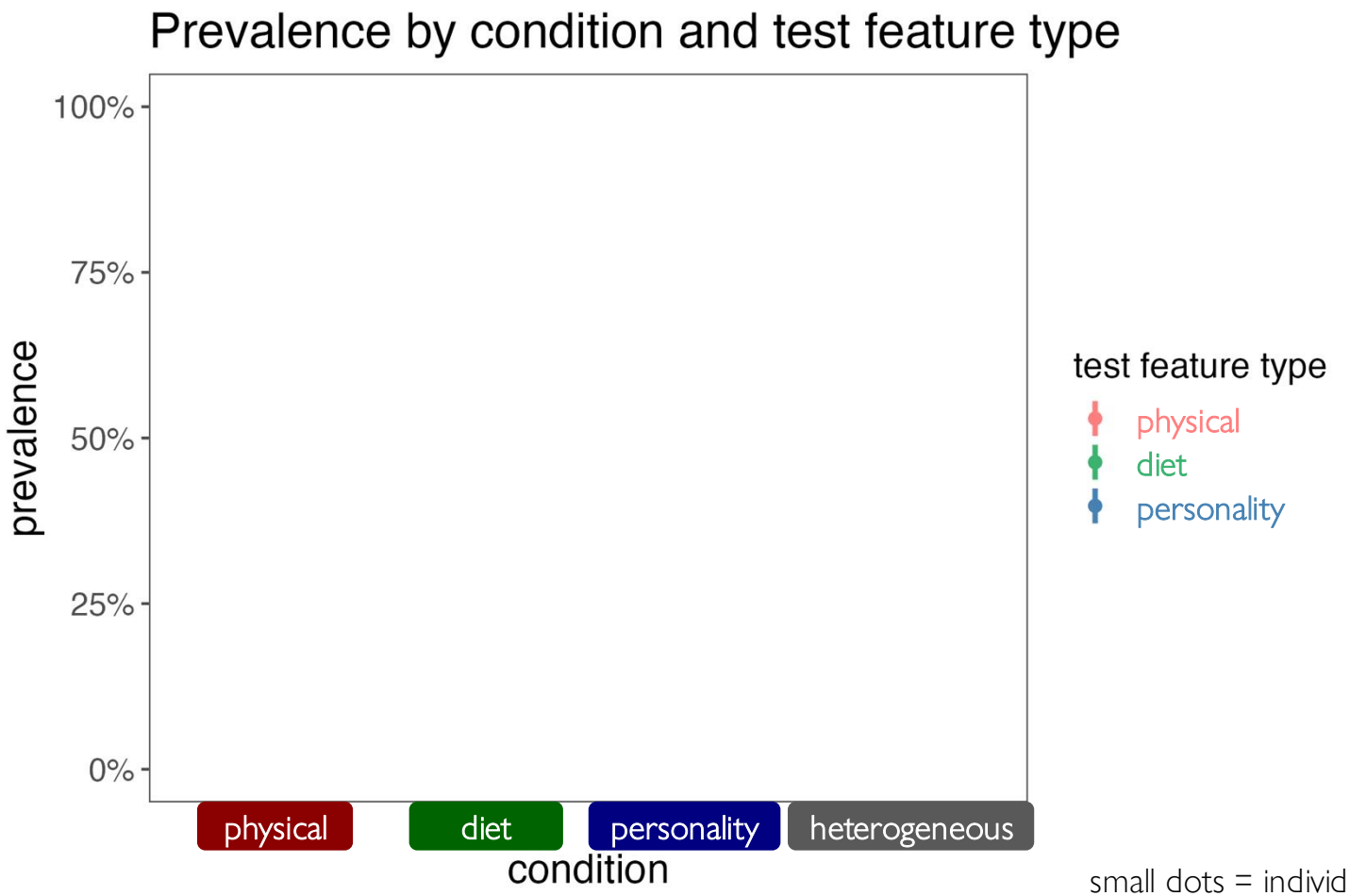
small dots = individual responses (participant * test feature type)
large dots w/ bars = means w/ 95% CIs
all pairwise comparisons with FDR correction

Exp 2 behavioral predictions

1. Hearing generic statements about some features should increase generalization of other similar features.
   (e.g., generics about Zarpies' physical features → generalize novel physical features more)

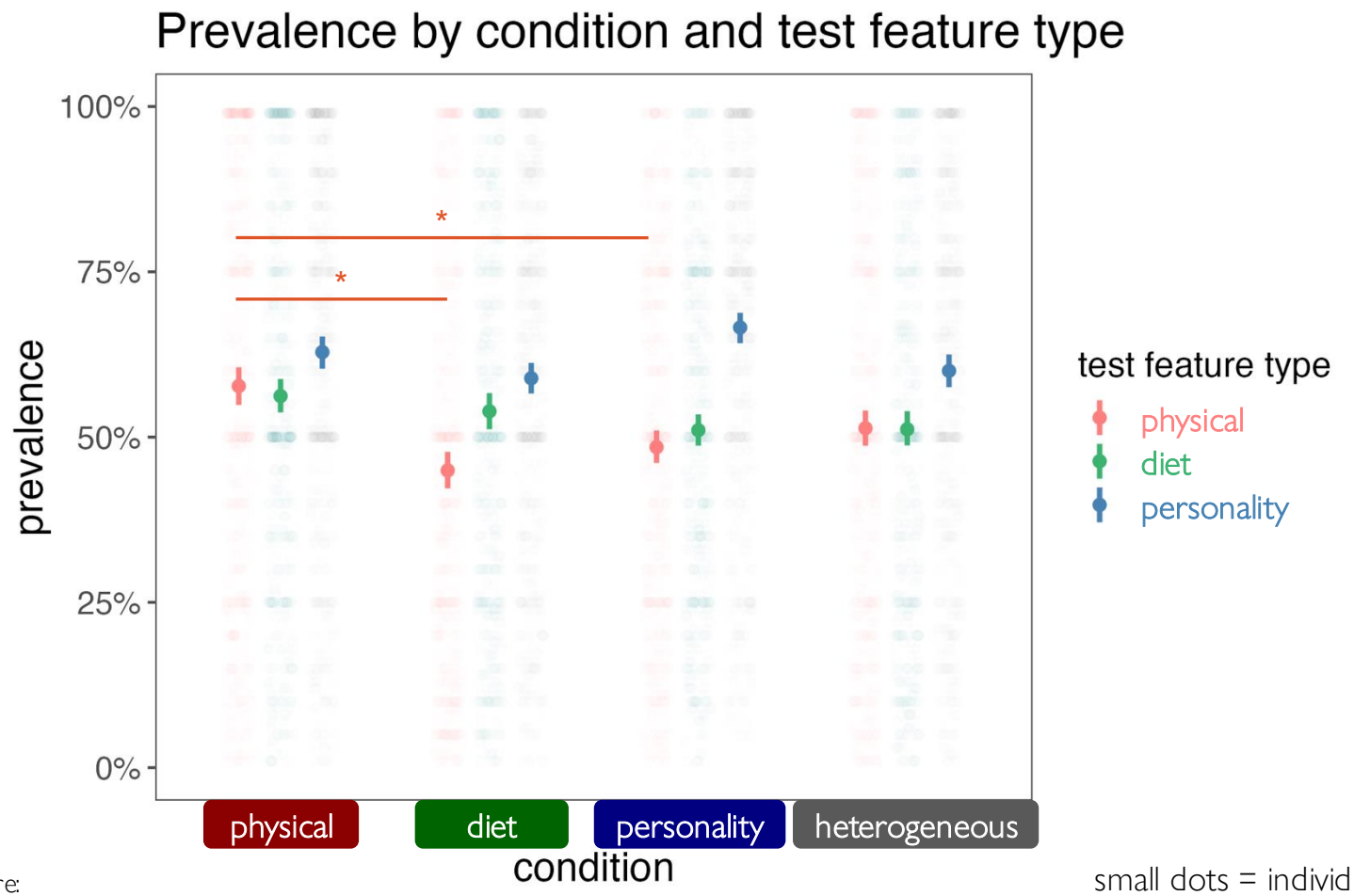
2. Hearing generic statements about heterogeneous features should result in the most increase in generalization overall, across features.

Exp 2 results: human data



small dots = individual responses (participant * test feature type)
large dots w/ bars = means w/ 95% CIs
all pairwise comparisons with FDR correction

Exp 2 results: human data



mixed effects logistic regression
w/ random intercepts per participant, test feature:

effect of test feature type ($\chi(2) = 6.62$, $p = .037$)

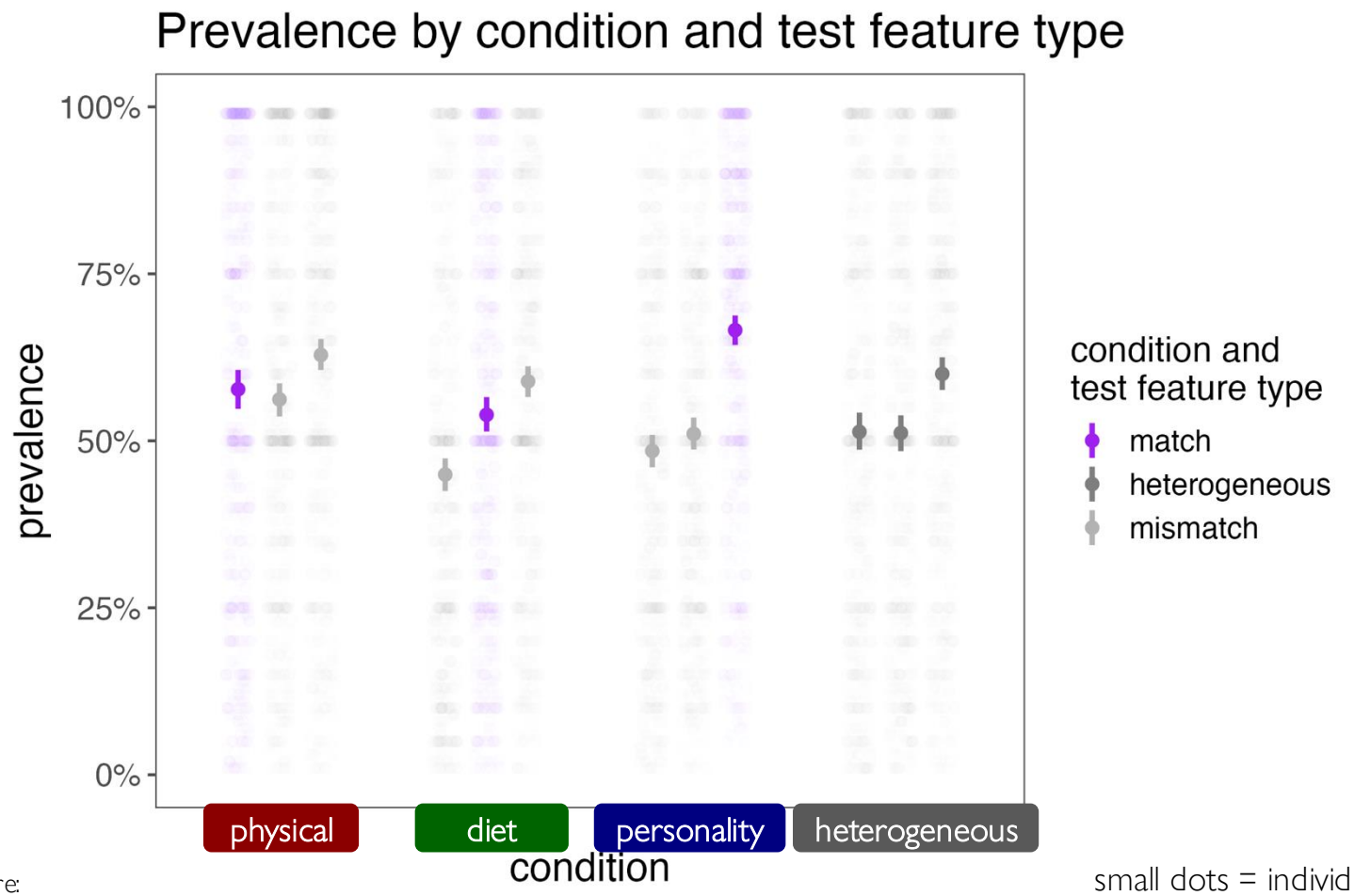
effect of condition * test feature type ($\chi(6) = 86.17$, $p < .001$)

small dots = individual responses (participant * test feature type)

large dots w/ bars = means w/ 95% CIs

all pairwise comparisons with FDR correction

Exp 2 results: human data



mixed effects logistic regression
w/ random intercepts per participant, test feature:

effect of test feature type ($\chi(2) = 6.62$, $p = .037$)

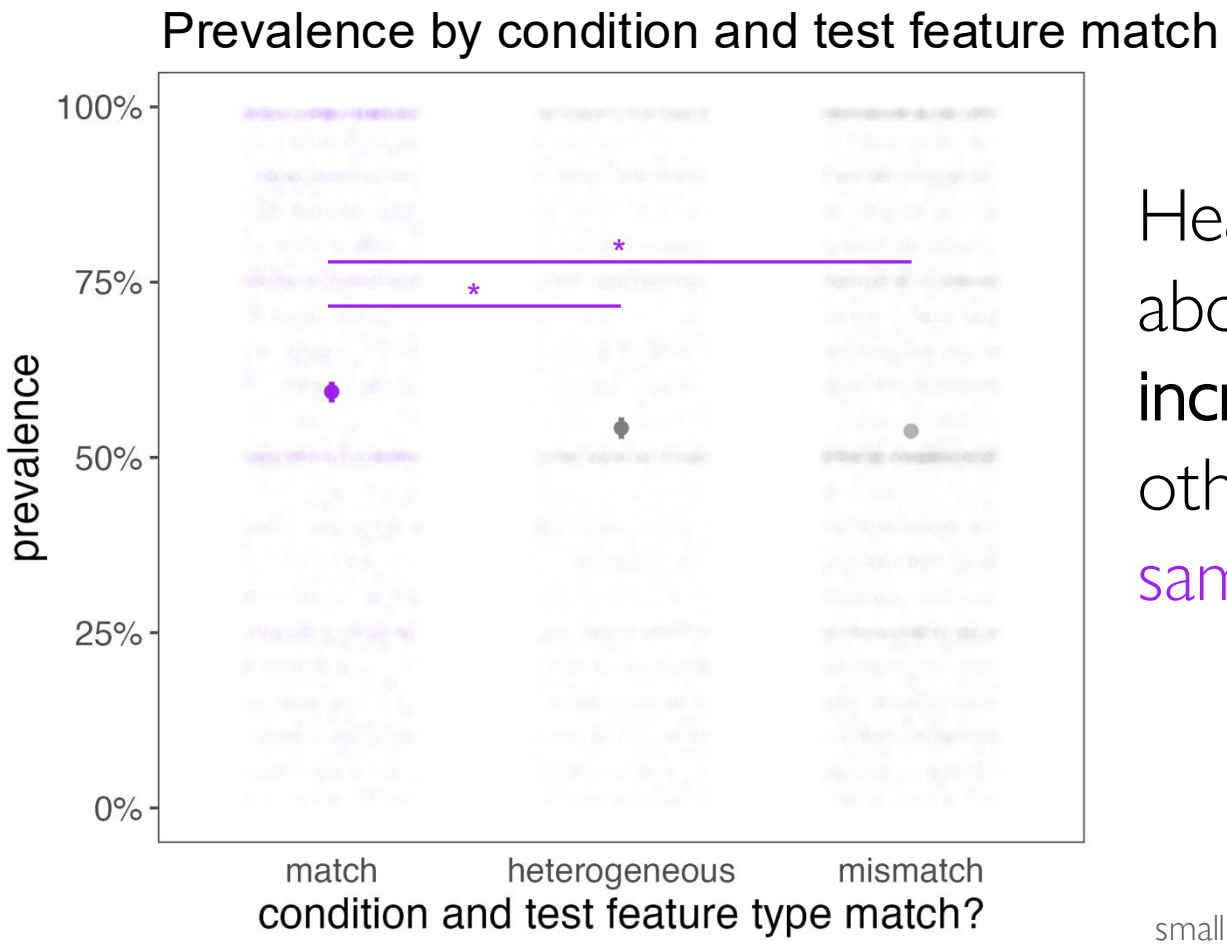
effect of condition * test feature type ($\chi(6) = 86.17$, $p < .001$)

small dots = individual responses (participant * test feature type)

large dots w/ bars = means w/ 95% CIs

all pairwise comparisons with FDR correction

Exp 2 results: human data



Hearing generic statements about one type of feature **increased** generalization of other features of the same type.

small dots = individual responses (participant * test feature type)
large dots w/ bars = means w/ 95% CIs
all pairwise comparisons with FDR correction

main effect of match ($\chi(2) = 74.03$, $p < .001$)

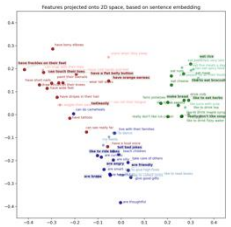
Exp 2: model

Bernoulli GP

takes in binary data points
(generic = feature is kind-linked,
specific = feature is not kind-linked)

returns distribution over possible
coherence functions

posterior over coherence
plotted in 2D feature space



physical

diet

personality

heterogeneous

Exp 2: model

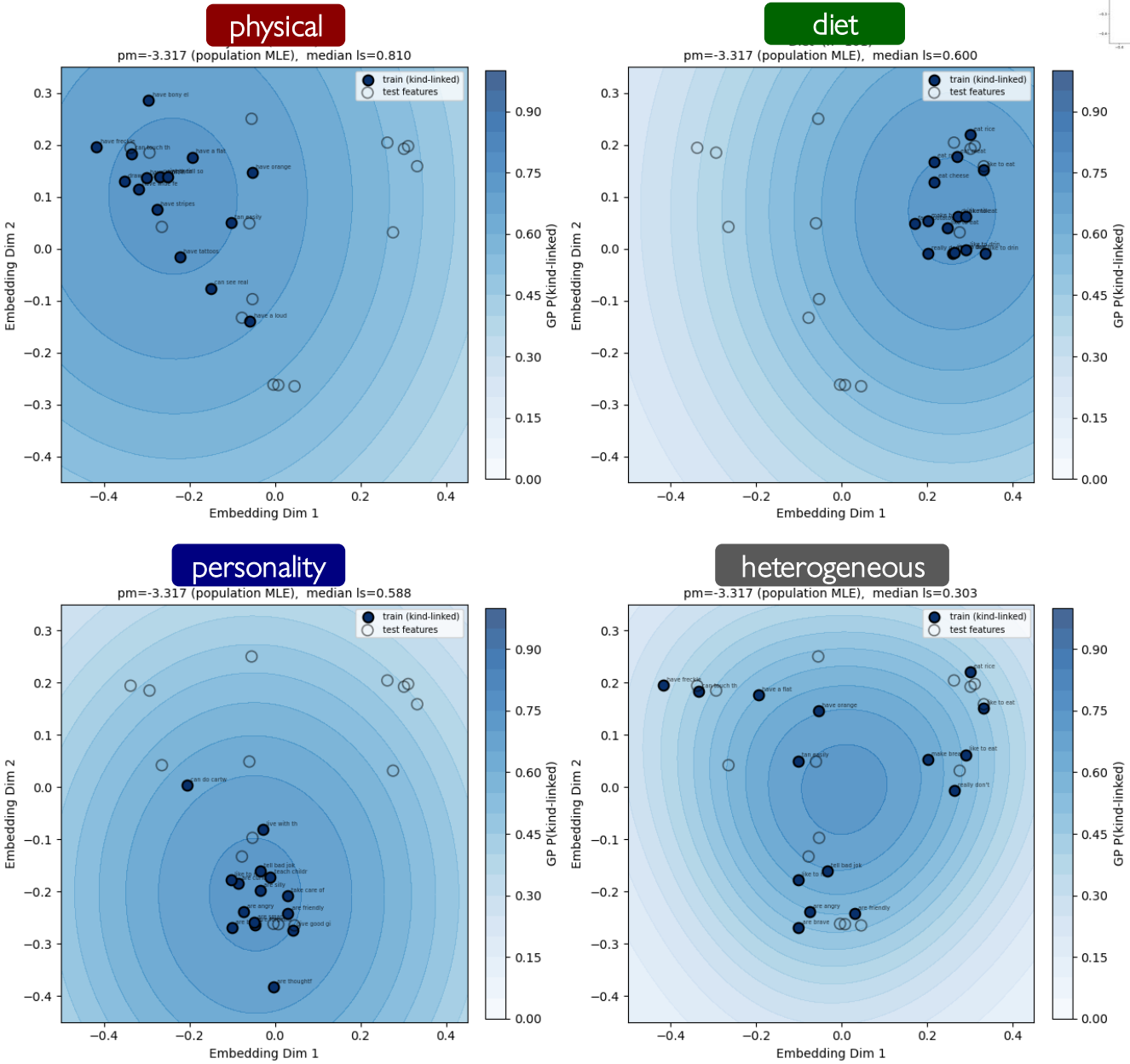
Bernoulli GP

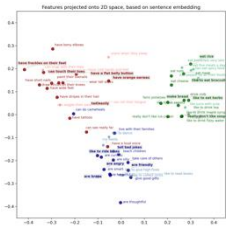
takes in binary data points
(generic = feature is kind-linked,
specific = feature is not kind-linked)

returns distribution over possible coherence functions

→ coherence is **higher** over the area of feature space mentioned in training generics

posterior over coherence
plotted in 2D feature space





plot made by Maya Malaviya!

Exp 2 results: model

Bernoulli GP

takes in binary data points
(generic = feature is kind-linked,
specific = feature is not kind-linked)

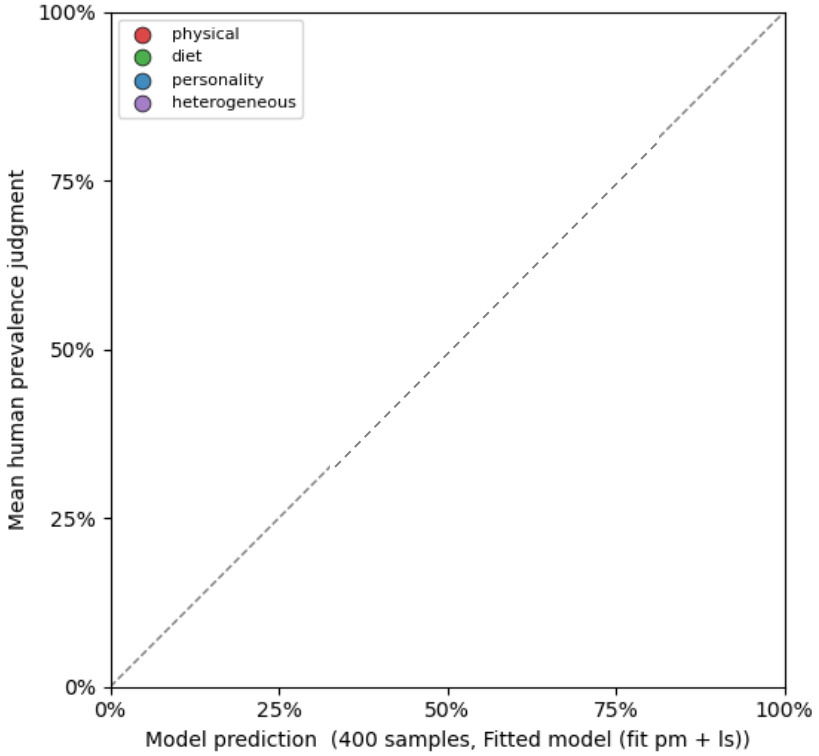
returns distribution over possible
coherence functions

linking function

translates coherence
into prevalence judgment

Full Bernoulli GP

fitted population-level prior mean
fitted participant-level length scale



n = 284 adults vs 400 simulated responses from linking function
dots = mean per test feature per condition

plot made by Maya Malaviya!

Exp 2 results: model

Bernoulli GP

takes in binary data points
(generic = feature is kind-linked,
specific = feature is not kind-linked)

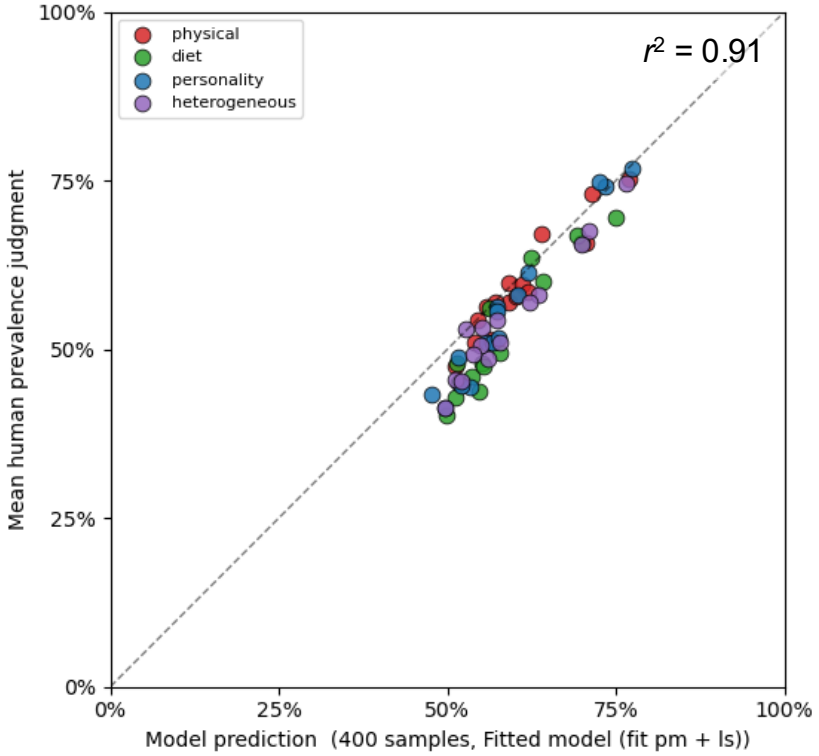
returns distribution over possible
coherence functions

linking function

translates coherence
into prevalence judgment

Full Bernoulli GP

fitted population-level prior mean
fitted participant-level length scale



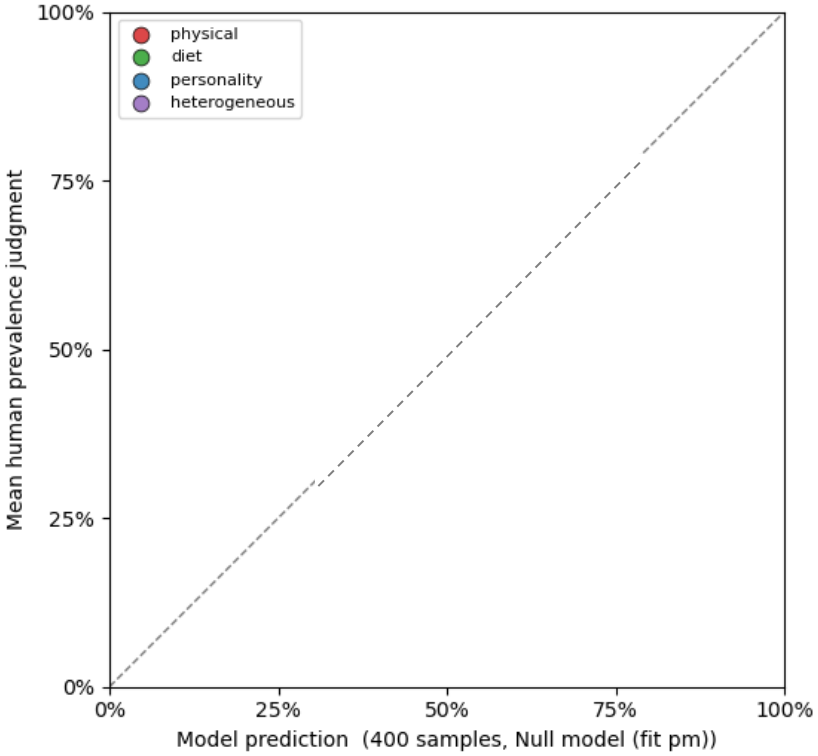
n = 284 adults vs 400 simulated responses from linking function
dots = mean per test feature per condition

plot made by Maya Malaviya!

Exp 2 results: model

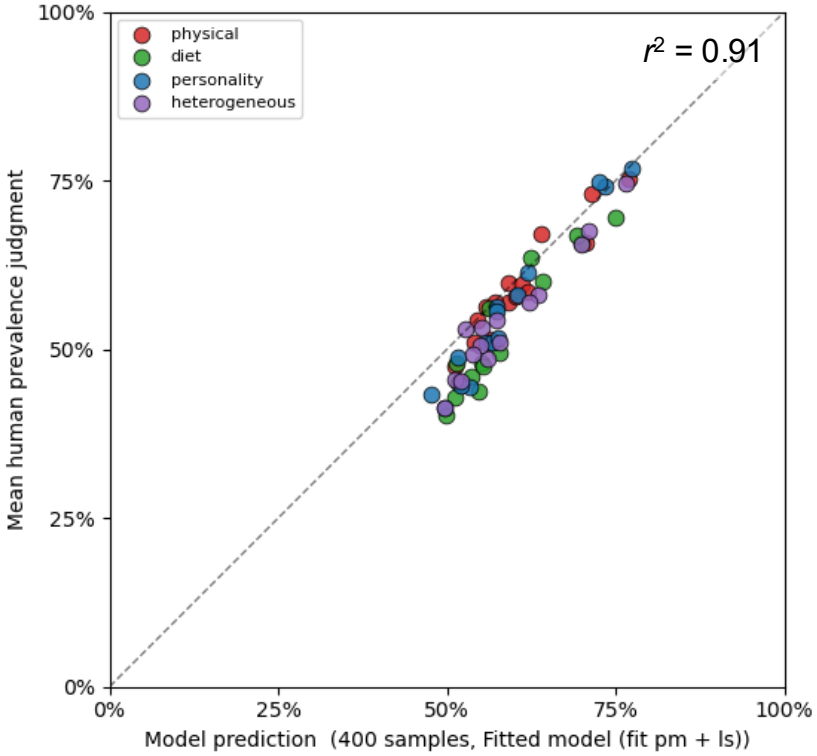
Null Bernoulli GP

fitted population-level prior mean
fixed long length scale



Full Bernoulli GP

fitted population-level prior mean
fitted participant-level length scale



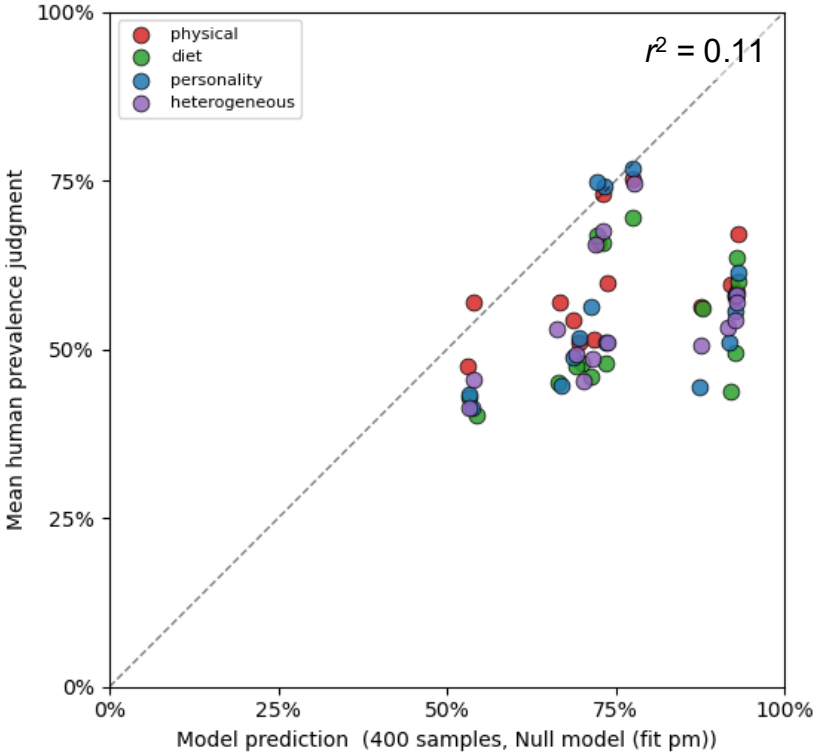
n = 284 adults vs 400 simulated responses from linking function
dots = mean per test feature per condition

plot made by Maya Malaviya!

Exp 2 results: model

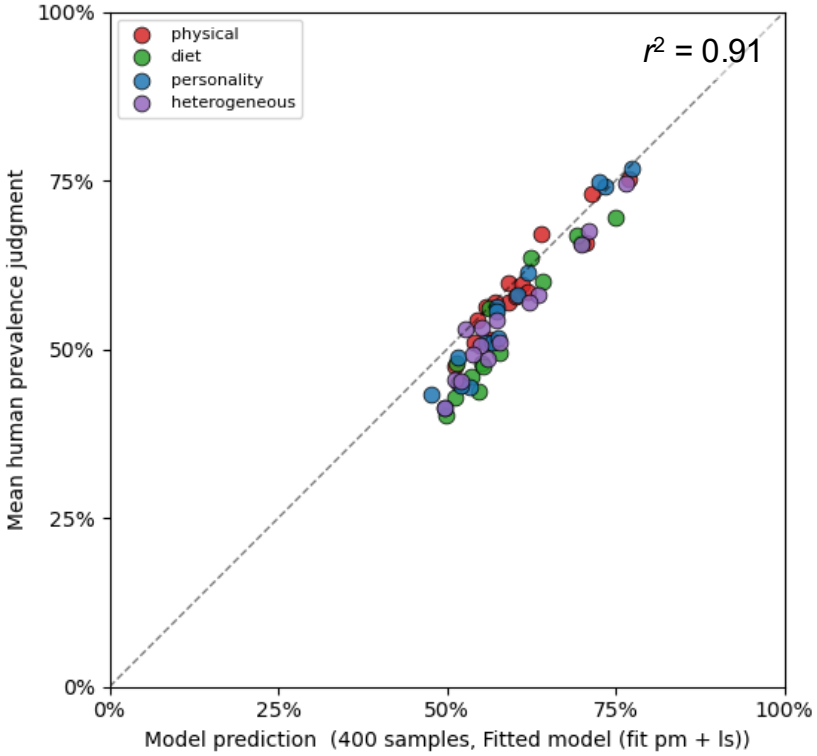
Null Bernoulli GP

fitted population-level prior mean
fixed long length scale



Full Bernoulli GP

fitted population-level prior mean
fitted participant-level length scale



χ^2 LLR(402) = 7539.63, p < .001

n = 284 adults vs 400 simulated responses from linking function
dots = mean per test feature per condition

Exp 2 conclusion

A Gaussian process model captures how:

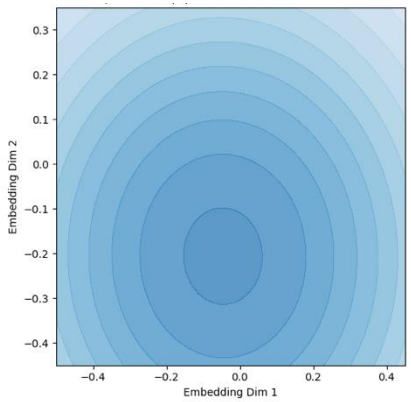
Hearing generic statements especially increases the generalization of features similar to those mentioned.

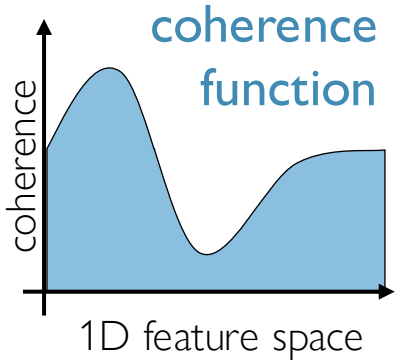
Pragmatic listener + GP

takes in utterances

returns a distribution over coherence functions

2D feature space





Learning about the inductive potential of categories from language

Learning from generic vs specific language

- Hierarchical Bayesian and RSA model
- **Exp 1a:** generalize more after generics, less after specifics
- **Exp 1b**: graded generalizations to proportion of generics to specifics

Learning to generalize features based on feature similarity

- Gaussian process and RSA model
- **Exp 2**: generalize more to features similar to those heard in generics

Language can signal what categories support generalization.

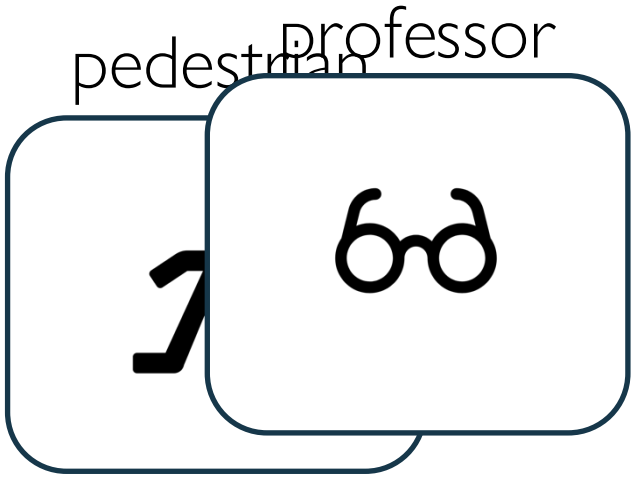
Language as a signal of inductive potential

how well a category
supports generalization

low ⟷ high

categories seen as minimal,
not supportive of generalization

categories seen as coherent/meaningful,
supportive of generalization



Language as a signal of inductive potential

how well a category
supports generalization

low — high

categories seen as minimal,
not supportive of generalization

categories seen as coherent/meaningful,
supportive of generalization

more specifics

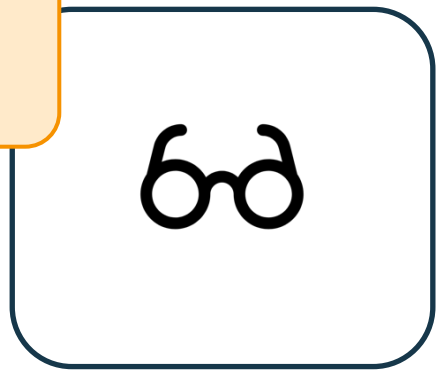
more generics

pedestrian

"That pedestrian
is absent-minded."

"Professors are
absent-minded."

professor



Future directions

Do **children** learn categories through these mechanisms?

→ similar experiments with children over pragmatic reasoning development

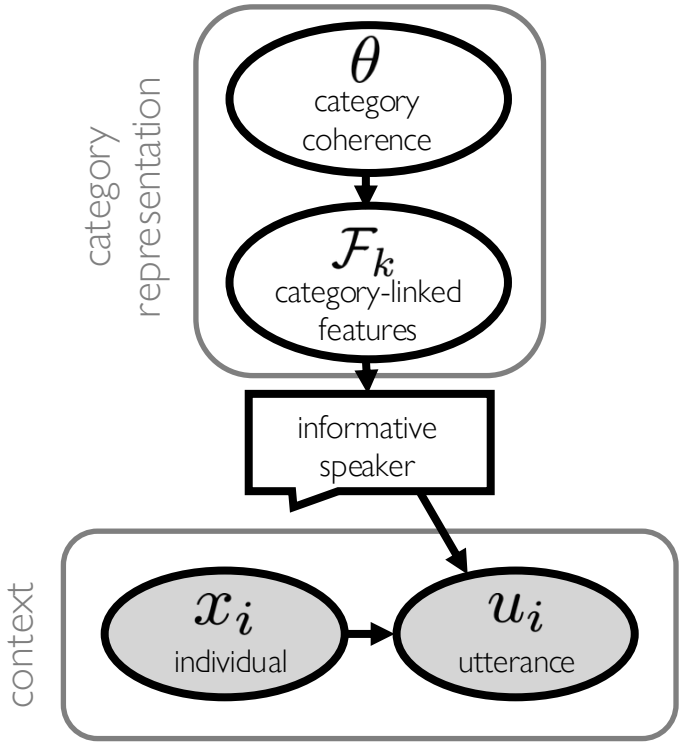
How can an understanding of the **causal relationships between features** guide generalization?

(Rehder & Burnett, 2005)

→ e.g., modeling the causal centrality of physical features vs other features

How can higher levels of understanding about types of **categories** and **domains** be learned?

→ over-overhypotheses about e.g., occupational groups, or about social groups



Thank you!

github.com/mariannazhang/compgenerics

CDSC

Computation and Decision-Making Lab

Marjorie Rhodes

Maya Malaviya

Mark Ho

project support
Jess Stephenson
Julia Johnson

funding
NYU Postdoctoral Trailblazer Award
NSF SBE Postdoctoral Research Fellowship
(formerly)

contact
mariannazhang.github.io
marianna.zhang@nyu.edu
@mariannazhang.bsky.social